

SYMBIOT

PARASITE IN SYMBIOSIS WITH LIVING ORGANISM



APPENDIX

ADS5

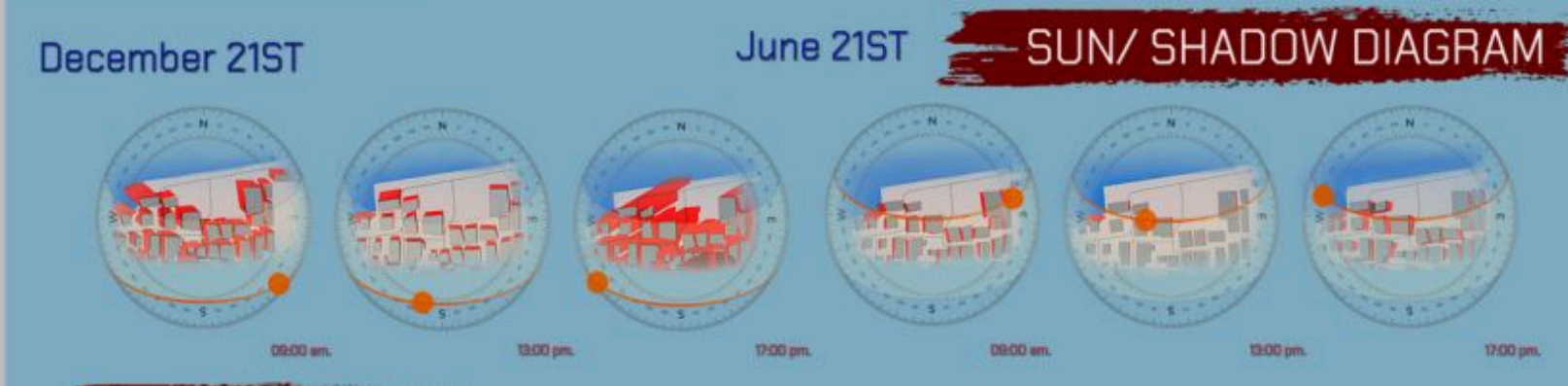
COLLAGE FELT/LIVED
POSTER 1
POSTER 2
HISTORY/ CLIMATE LITERACY
MICRO SITE ANALYSIS
MACRO SITE ANALYSIS
PRECEDENCE STUDIES
ZONING
POD DEVELOPMENT
FLOOR PLANS
SECTIONS
UNIT PLAN
RENDER VIEWS
PHYSICAL MODELS
FINAL PHYSICAL MODEL
GROUP PROJECT

FELT AND LIVED ABSTRACT ARTISTIC COLLAGE OF AL FHAHID HISTORICAL NEINGHOURHOOD



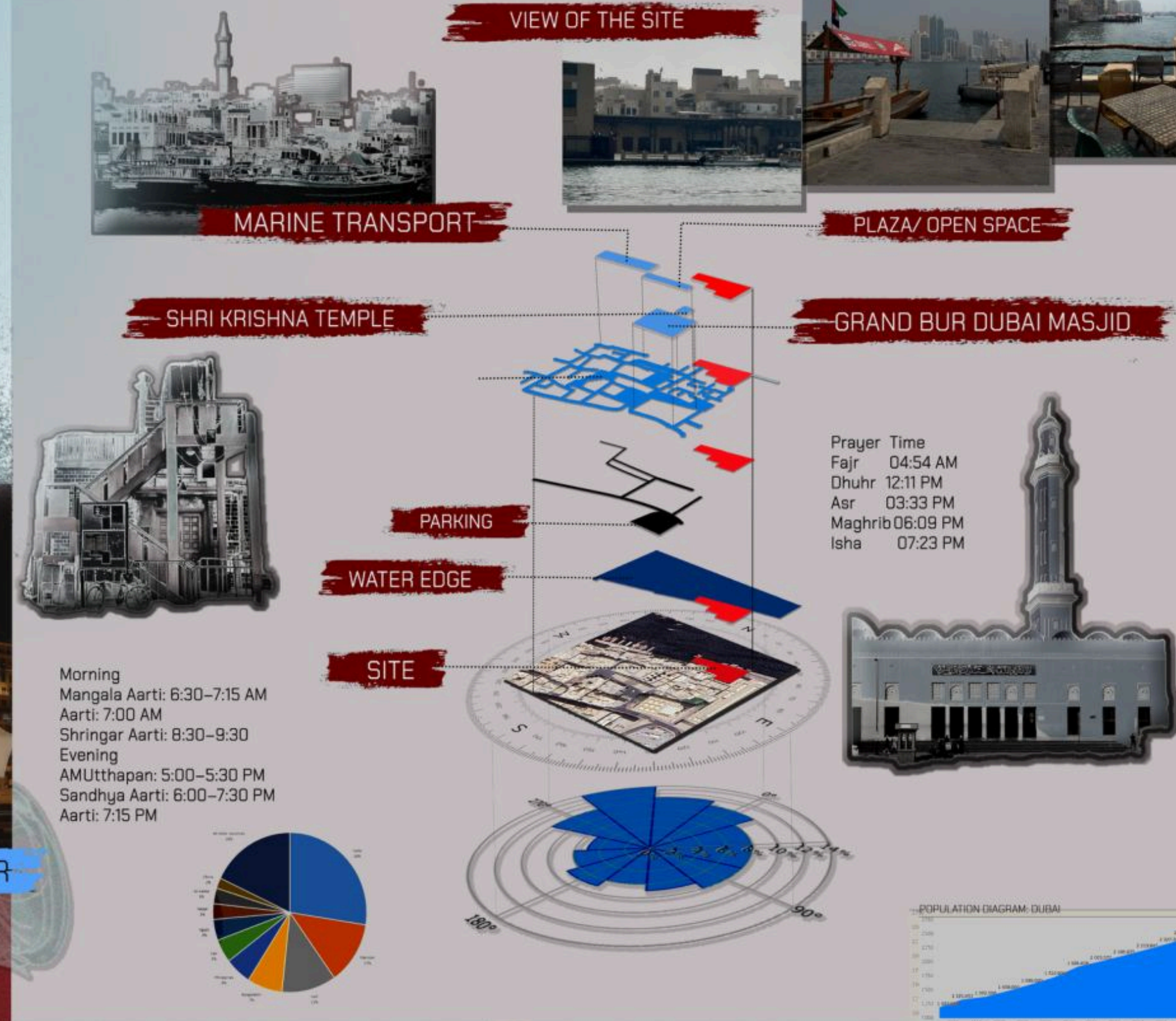
"WHISPERS BENEATH THE
SCALES OF AL HAFIDI"

1



NORTH- EAST VIEW

"INTRIGUING; INVITING; PLAYFULL"



JID

GRAND BUR DUBAI MASJID

Prayer Time
Fajr 04:54 AM
Dhuhr 12:11 PM
Asr 03:33 PM
Maghrib 06:09 PM
Isha 07:23 PM

+

RISING SEA LEVEL OVERCROWDED AREAS TRANSPORTATION NO SURROUNDING SHADING PARKING SITUATION	FLOATING DEVICE SPACECREATING DESIGN WATER MODULES REFLECTIVE DESIGN VOIDS CREATING SHADE PRIVATE PORT
--	--

REMORA FISH

SUCTION DISK
SPINE

SHERI FISH

SWIM BLADDER
SCALE

COMPOSITE LAMELLAE
MADE OF FLEXIBLE AND
CALCIFIED TISSUES

ATTACHES OF SHARK/WHALE THAT
SHE USES FOR TRANSPORTATION
AND FOOD SUPPLY

THEY ATTACH TO VARIOUS
PARTS OF SHARKS BODY

SPINE VERTEBRAS ALLOW
FISH TO MOVE IN VARIOUS
DIRECTIONS

3D-PRINTED SOFT POLYMER
SUCTION DISC

3D-PRINTED
COMPOSITE "LAMELLAE"
MADE OF VARIOUS
ARTIFICIAL COMPOUNDS

AIR BUBBLE SUPPORTING MODULE
ADJUST TO THE RISING SEA LEVEL
INSPIRED BY SWIM BLADDER

FLEXIBLE DESIGN
INSPIRED BY SPINE MOVEMENT

MOVEABLE MECHANISM FLOATS
ON THE WATER AND ADJUST TO THE
WATER DIRECTION

SCHOOL OF FISH

MODULES CONNECTING TO MAIN BODY
FLOATING ON THE WATER
CONNECTING WITH SUCTION DISC

REFLECTIVE FACADE
BIPV COLLECTING SOLAR ENERGY
CREATES PRIVACY AND CREATES
INTRIGUING VIEW
INSPIRED BY SHERI FISH SOLAE

INSPIRED BY VERTEBRA
HOLES CREATE VOID THAT
CAN BE TRANSFORMED
INTO SPACE

PART ABOVE
WATER, PART UNDER WATER
THAT OFFERS VIEW OF
UNDERWATER MARINE

SYMBIOT

AL FAHIDI: HISTORICAL HERITAGE
DUBAI CREEK, UAE
25.26216 N, 55.29496 E



REMORA



SUCTION CUP



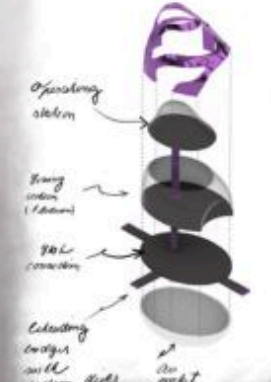
FISH SPINE



SWIM BLADDER



BIOARTEFACT



PARASITISM

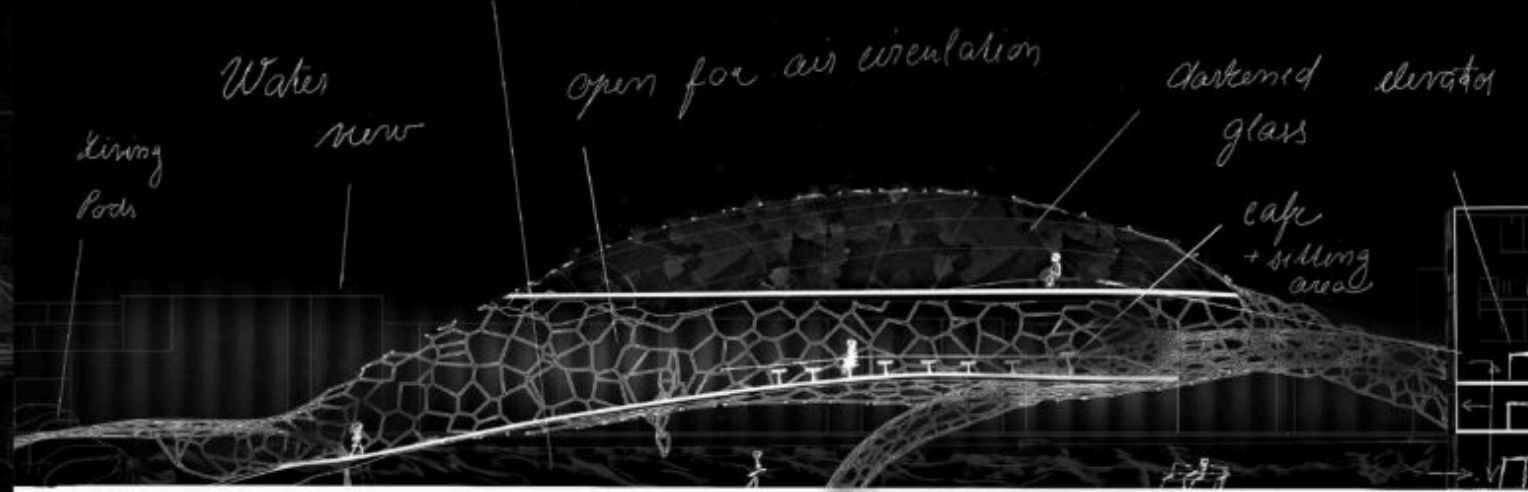
21WBI0212

POD UNIT FLOOR PLAN

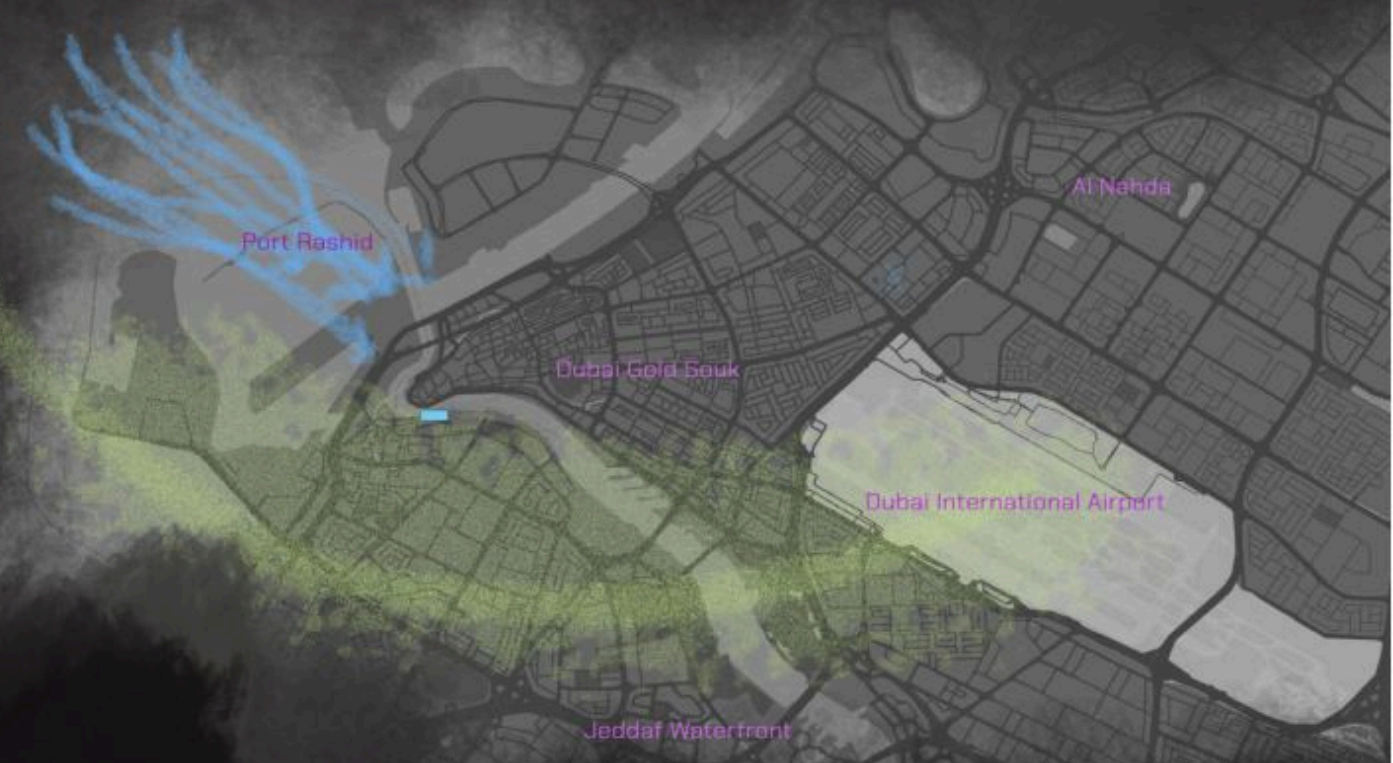


Section AA

minimum build up area for 6 floor



SCALE 1:250





1833 The Bani Yas tribe settles in the area that becomes Dubai. Deira emerges as one of the main settlements on the northern side of Dubai Creek.

1841 Deira is officially recognised as a separate community from Bur Dubai, located on the opposite side of Dubai Creek. It begins to develop as a trading hub due to its proximity to the creek.

1900s Deira becomes Dubai's commercial heart. Traditional markets (souks) such as the Gold Souk, Spice Souk, and Perfume Souk start to flourish.

1930s Pearl diving and fishing are major economic activities in Deira. The area continues to grow with merchants and traders settling there.

1950s Deira expands significantly as Dubai's economy grows. Modern roads and infrastructure are introduced. Some warehouses and trading houses are built along the creek.

1960s Construction of bridges over Dubai Creek begins, improving connectivity with Bur Dubai. Deira becomes a hub for retail and trade.

1970s Dubai Oil Company starts influencing economic growth, indirectly affecting Deira's development.

1980s Rapid urbanisation. New hotels, banks, and commercial buildings are constructed. Deira solidifies its status as Dubai's financial and commercial centre.

1985 Completion of Al Maktoum Bridge connecting Deira and Bur Dubai. This facilitates trade and commuter traffic.

1990s Deira sees modernisation but retains its traditional markets. Introduction of shopping malls alongside traditional souks.

2000s Waterfront developments along Dubai Creek. Deira faces competition from newer areas like Downtown Dubai and Dubai Marina but remains culturally and historically significant.

2010s Dubai Creek revitalisation projects begin, including Dubai Creek Harbour and tourist infrastructure upgrades.

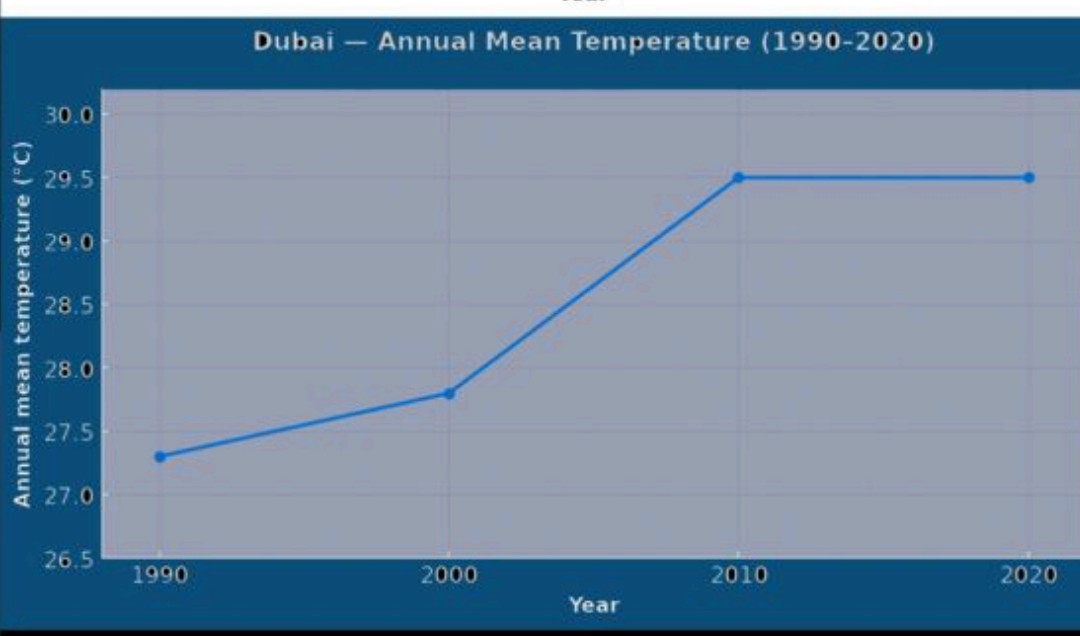
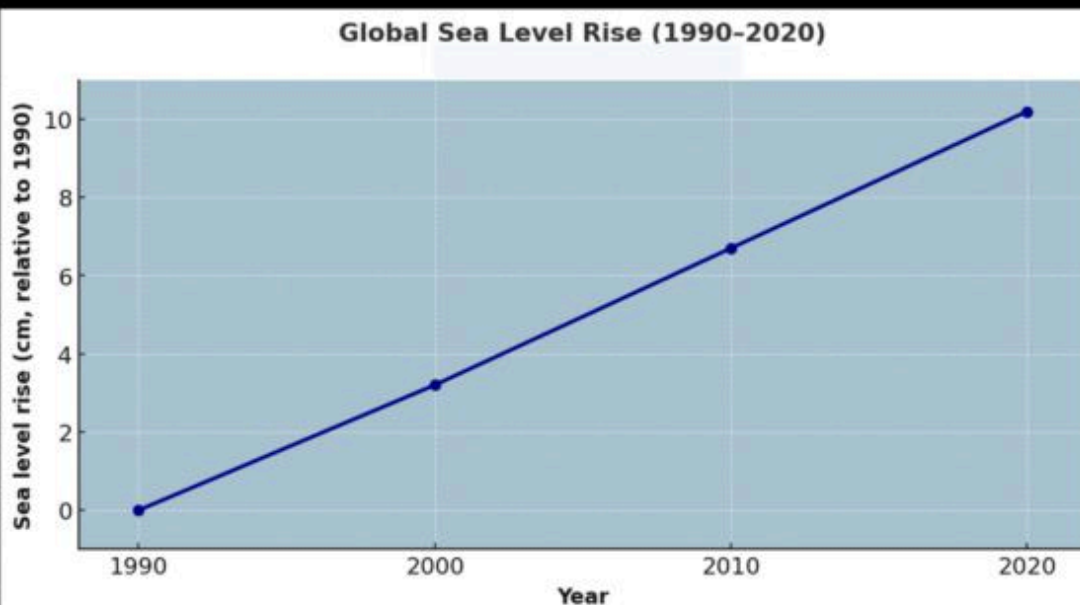
2020s Deira maintains its reputation as a cultural and commercial hub. Old souks and traditional buildings are preserved amid modern development.



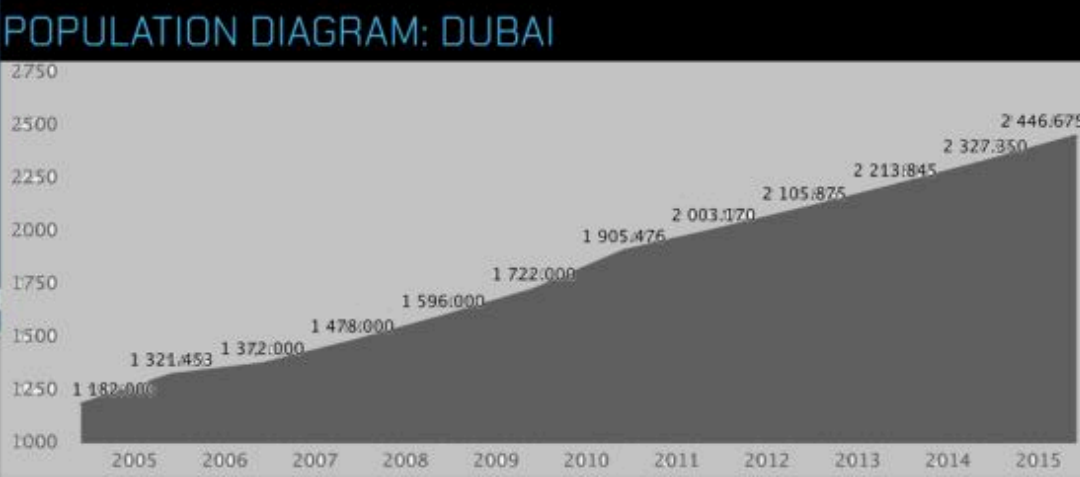
Rising sea levels caused by climate change threaten coastal cities worldwide, including Dubai. Strengthening climate literacy is essential to help communities and decision-makers understand these risks and plan resilient, sustainable responses for the future.

CLIMATE LITERACY // IMPACT ON DUBAI

Amphibious housing strategy for rising sea levels.
Harmonization with marine ecosystems & heritage fabric.
Using bio-architecture, water-sensitive urban design and resilient materials.



-	+
RIISING SEA LEVEL	FLOATING DEVICE
OVERCROWDED AREAS	SPACECREATING DESIGN
TRANSPORTATION	WATER MODULES
NO SURROUNDING SHADING	REFLECTIVE DESIGN
PARKING SITUATION	VOIDS CREATING SHADE
	PRIVATE PORT



Micro site analysis

AL FAHIDI; HISTORICAL HERRITAGE
DUBAI CREEK, UAE
25.26216 N , 55.29436 E

December 21ST

June 21ST



09:00 am.



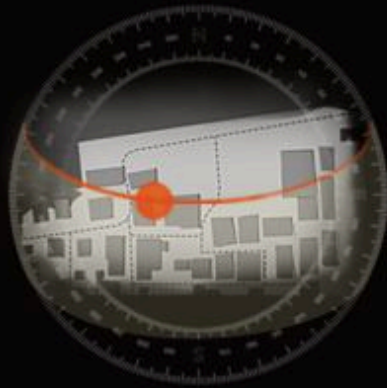
13:00 pm.



17:00 pm.



09:00 am.



13:00 pm.



17:00 pm.

ACCESS DIAGRAM

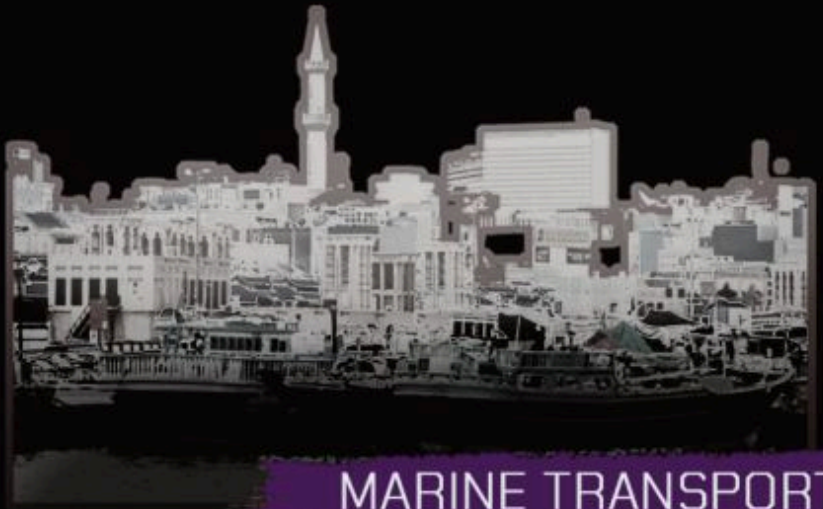
WEST VIEW

NORTH- EAST VIEW

VIEW OF THE SITE

SUN SHADOW DIAGRAM

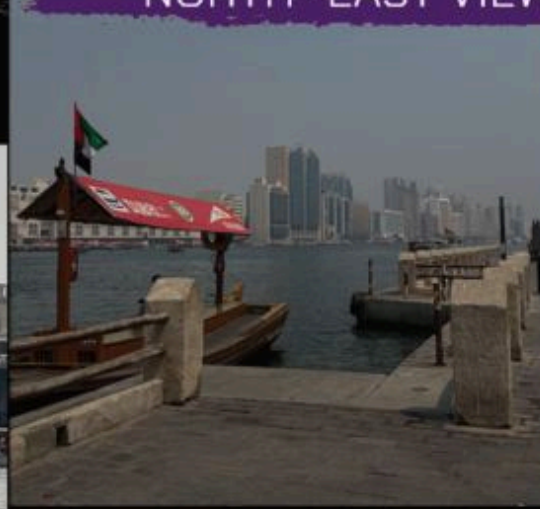
DATE	SUNRISE	SUNDOWN	DURATION
DECEMBER 21st	6:59 AM.	17:33 PM.	10H 43 MIN
JUNE 21st	5:29 AM.	19:12 PM.	13H 43MIN



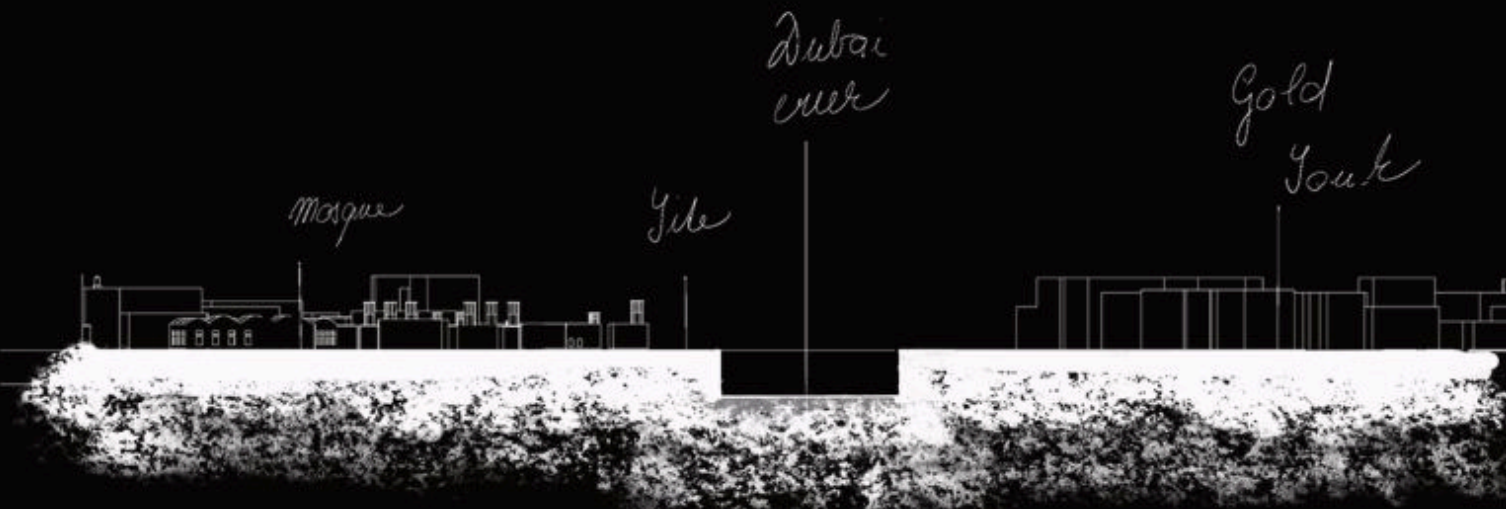
MARINE TRANSPORT



PLAZA/ OPEN SPACE



WATER EDGE DIAGRAM



SHRI KRISHNA TEMPLE

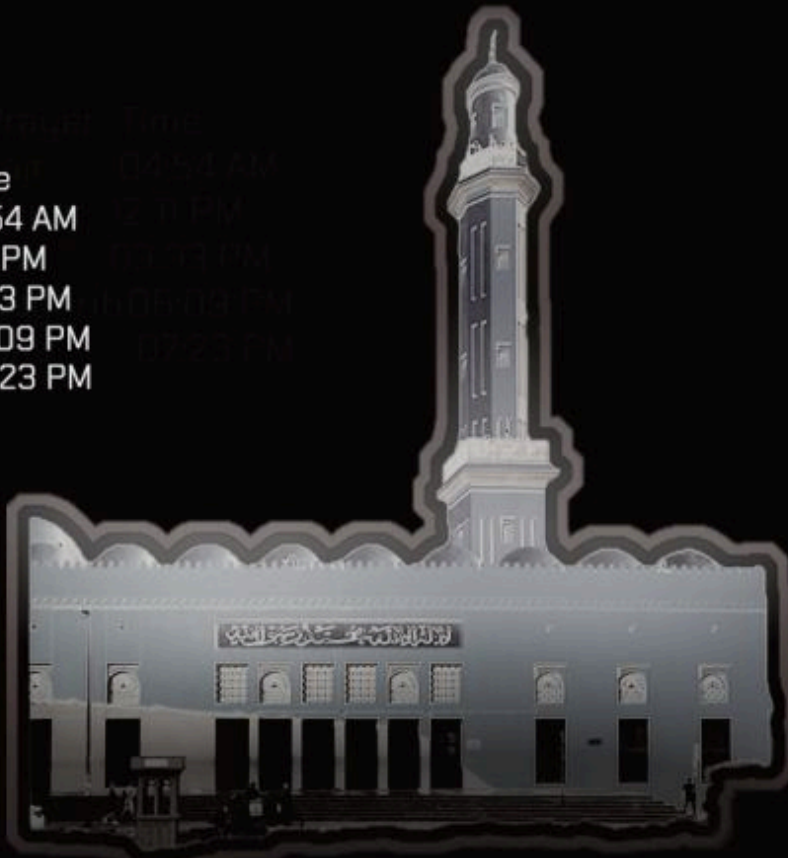
GRAND BUR DUBAI MASJID

PARKING

WATER EDGE

SITE

Prayer Time
Fajr 04:54 AM
Dhuhr 12:11 PM
Asr 03:33 PM
Maghrib 06:09 PM
Isha 07:23 PM

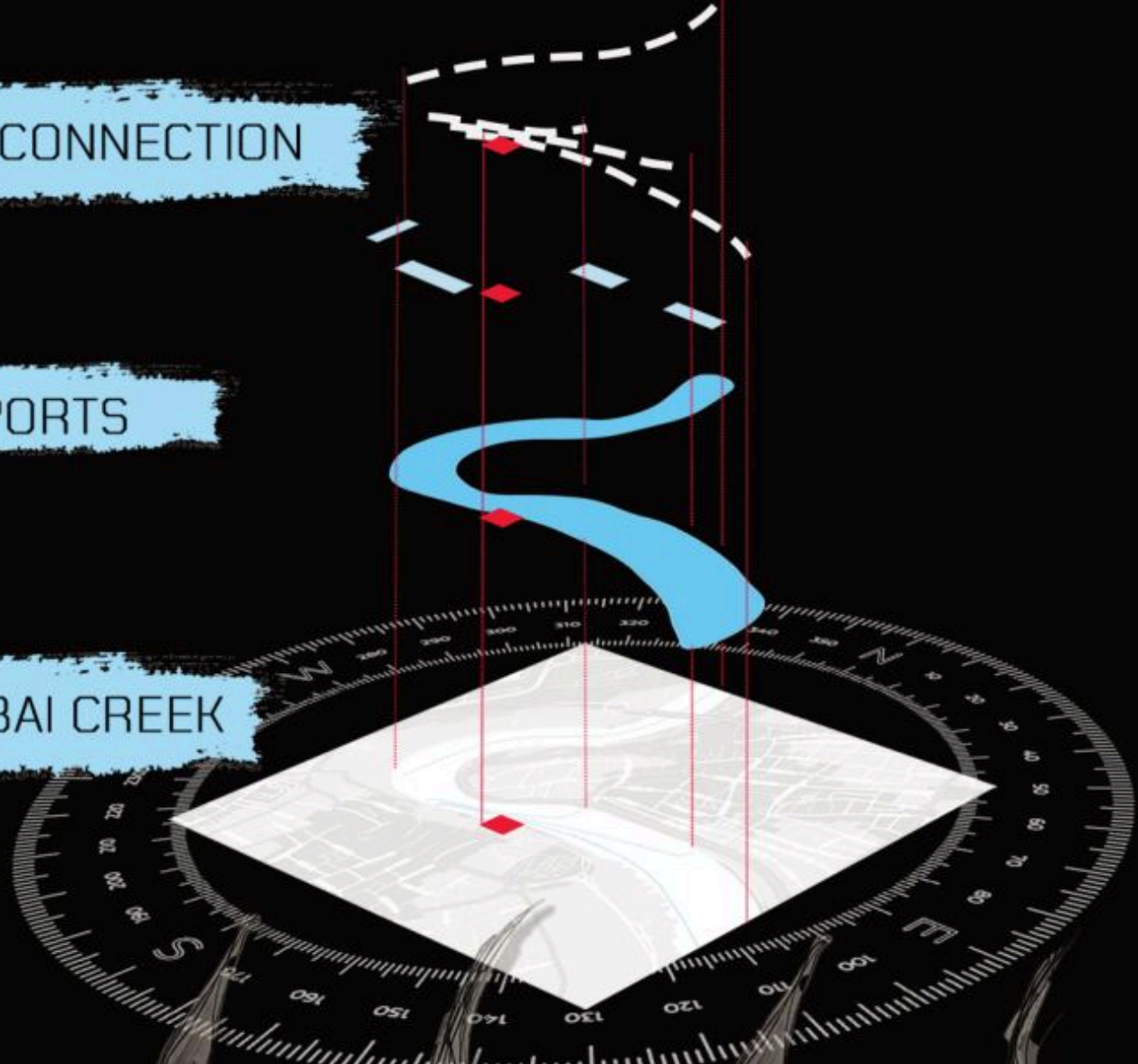


PORT CONNECTION

PORTS

DUBAI CREEK

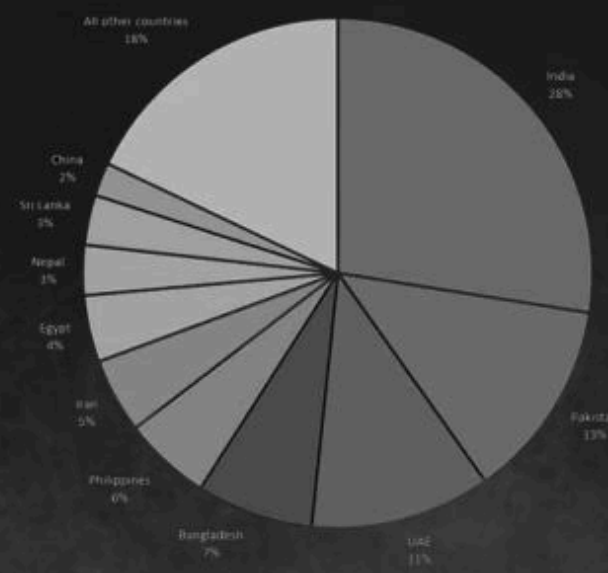
Morning
Mangala Aarti: 6:30-7:15 AM
Aarti: 7:00 AM
Shringar Aarti: 8:30-9:30
Evening
AMUtthapan: 5:00-5:30 PM
Sandhya Aarti: 6:00-7:30 PM
Aarti: 7:15 PM



Macro site analysis

AL FAHIDI; HISTORICAL HERRITAGE
DUBAI CREEK, UAE
25.26216 N , 55.29436 E

NATIONALITIES
DIAGRAM



Season	Time of Day	Typical Humidity	Comfort Level
Winter	Early morning & night	~75-85%	Cool, moderate humidity
	Midday (lowest)	~35-40%	Dry and comfortable
	Overall	Lower daytime humidity	Pleasant for outdoor activity
Summer	Early morning & night	~85-93%	Very humid, sticky
	Midday (lowest)	~47-55%	Hot and humid
	Overall	High humidity throughout	Uncomfortable, especially nights

1: PORT RASHID

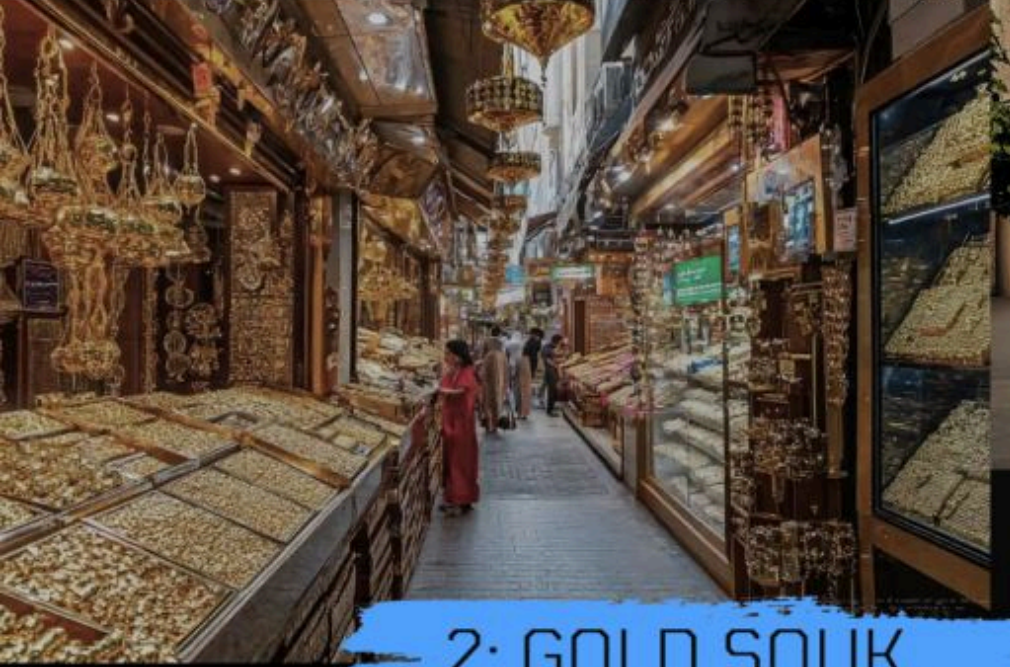


3: DUBAI OLD VILLAGE



DEIRA CLOCKTOWER

2: GOLD SOUK



Port Rashid

Dubai Gold Souk

Al Nahda

Dubai International Airport

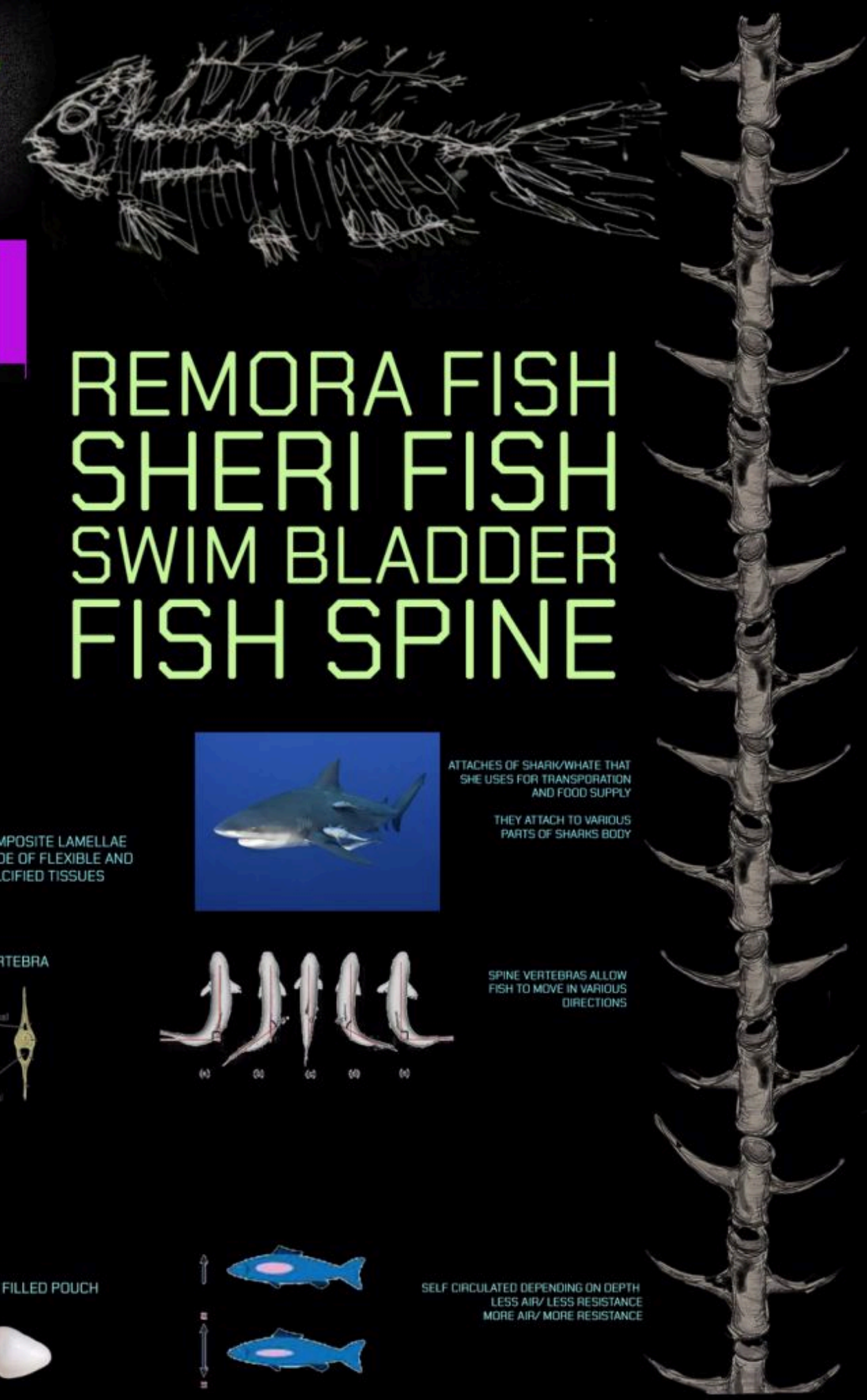
Jeddaf Waterfront

Season	Avg Wind Speed	Main Direction	Wind Type	Comfort Effect
Winter	8-18 km/h	N-NW	Shamal	Cooling, comfortable
Summer	10-22 km/h	NW	Sea breeze	Humid, limited relief

BIOARTEFACT

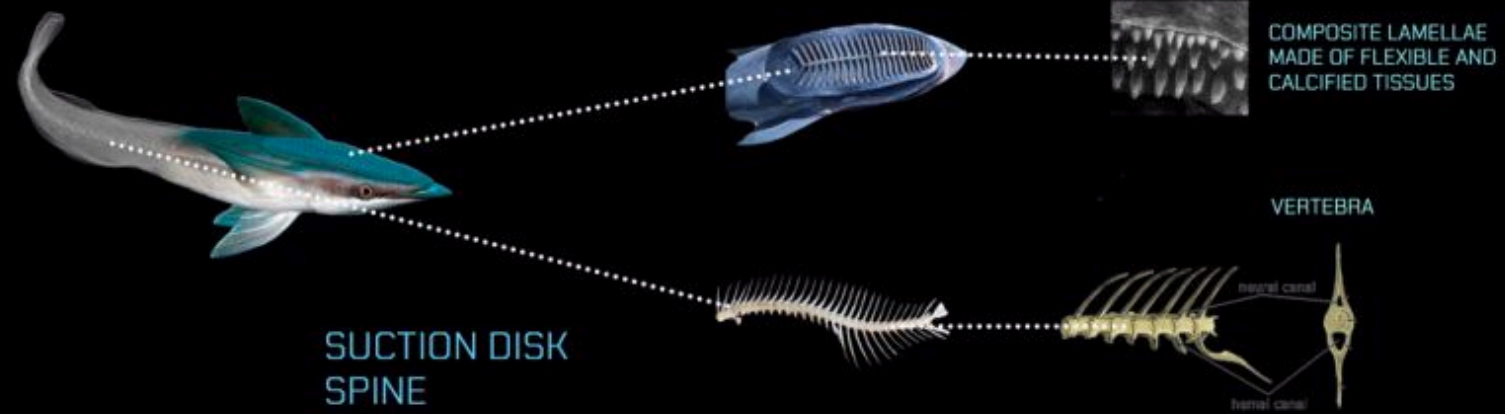
PARASITISM

21W BIO 212



REMORA FISH SHERI FISH SWIM BLADDER FISH SPINE

REMORA FISH

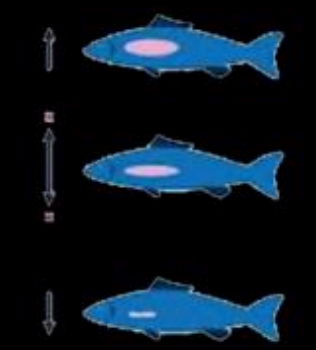
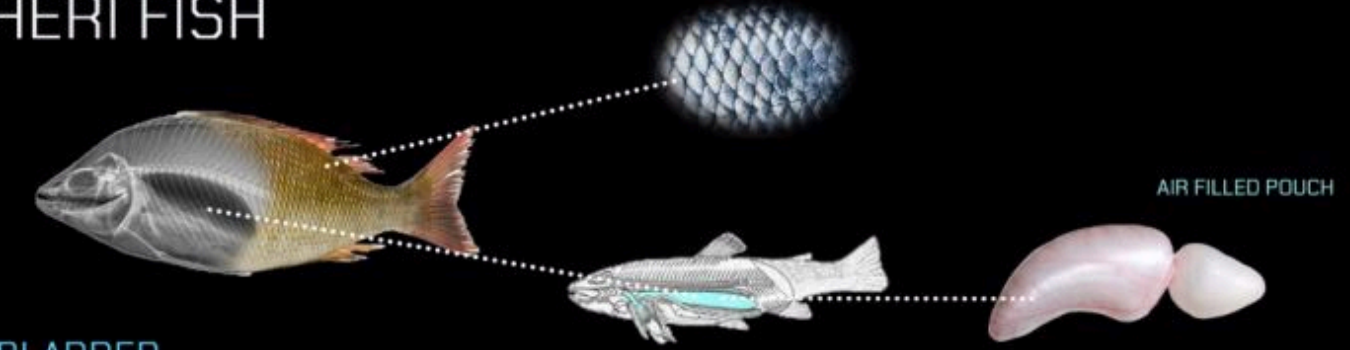


ATTACHES OF SHARK/WHALE THAT SHE USES FOR TRANSPORTATION AND FOOD SUPPLY
THEY ATTACH TO VARIOUS PARTS OF SHARKS BODY



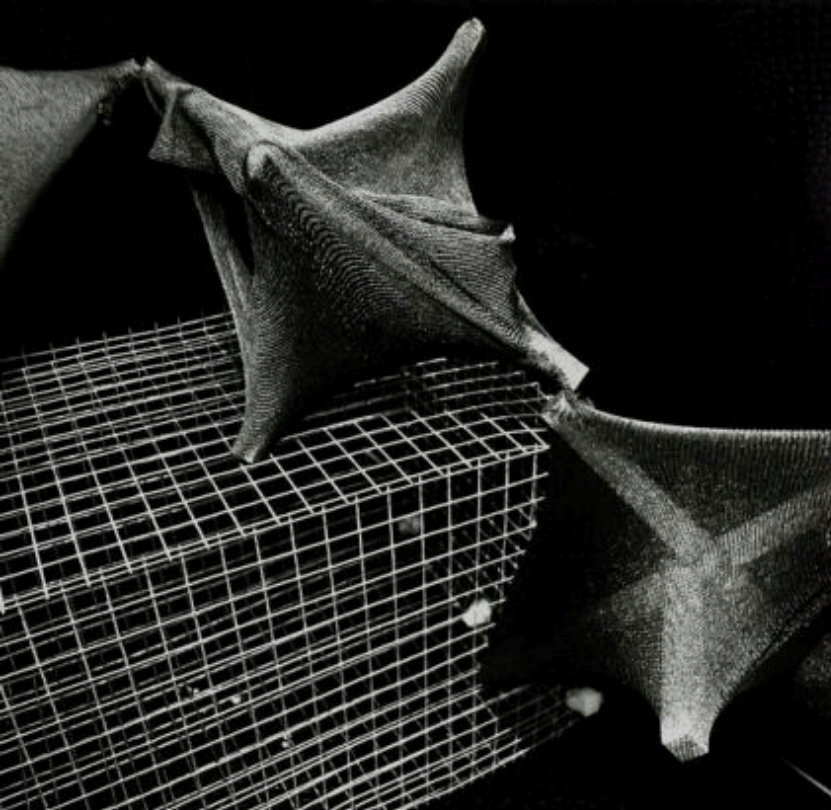
SPINE VERTEBRAS ALLOW FISH TO MOVE IN VARIOUS DIRECTIONS

SHERI FISH

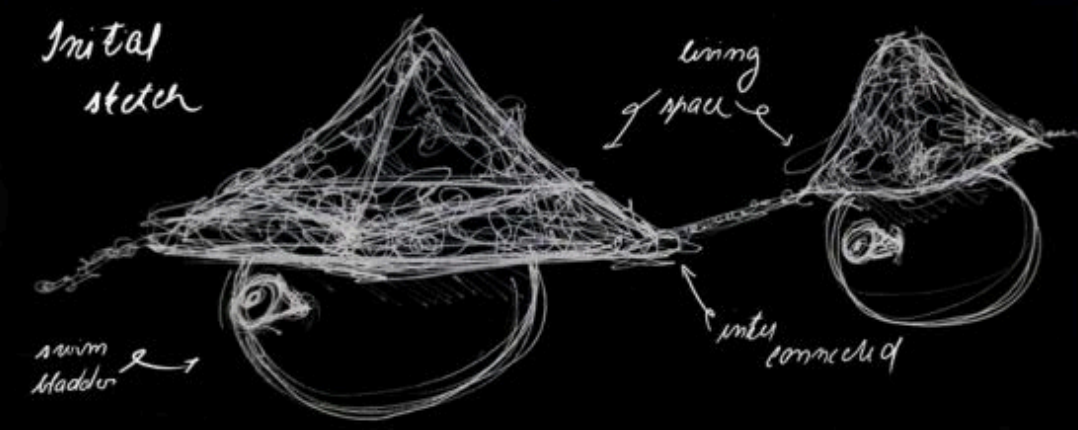


SELF CIRCULATED DEPENDING ON DEPTH
LESS AIR/ LESS RESISTANCE
MORE AIR/ MORE RESISTANCE

PHYSICA MODEL/ REMORA INSPIRED ATTACHMENT/ PARASITISM



PODS SUPPORTED BY "SWIM BLADDER"



BENDING INSPIRED BY FISH SPINE
ALLOWS MOVEMENT ON WATER
ANGEL IS KEPT MY RESISTING WATER PRESSURE FROM BOTH SIDES

PHYSICA MODEL/ FISH SPINE



SUPPORTED BY A SERIES OF AIR-PRESSURE CONTROLLED SOFT CYLINDRICAL ROTARY JOINTS



3D-PRINTED SOFT POLYMER SUCTION DISC
3D-PRINTED COMPOSITE "LAMELLAE" MADE OF VARIOUS ARTIFICIAL COMPOUNDS

MODULES CONNECTING TO MAIN BODY FLOATING ON THE WATER CONNECTING WITH SUCTION DISC



FLEXIBLE DESIGN INSPIRED BY SPINE MOVEMENT
MOVEABLE MECHANISM FLOATS ON THE WATER AND ADJUST TO THE WATER DIRECTION

SCHOOL OF FISH

BOTH REMORA AND SHERI FISH SWIM WITH COMPANY



AIR BUBBLE SUPPORTING MODULE ADJUST TO THE RISING SEA LEVEL INSPIRED BY SWIM BLADDER



REFLECTIVE FACADE
BIPV COLLECTING SOLAR ENERGY CREATES PRIVACY AND CREATES INTRIGUING VIEW
INSPIRED BY SHERI FISH SCLAE

SINTERIOR FACING
INSPIRED BY VERTEBRA HOLES CREATE VOID THAT CAN BE TRANSFORMED INTO SPACE

PART ABOVE WATER, PART UNDER WATER THAT OFFERS VIEW OF UNDERWATER MARINE LIFE

PHYSICA MODEL/ SWIM BLADDER



Site inferences

360° sun

MINIMAL FOOTPRINT ON THE BASE
MAXIMIZES G FLOOR OUTDOOR AREA

GIVE TO PEOPLE

METAL PLATES REPRESENT
RESTING AREA ALONG ONE PATH



Wind direction

Water edge

Existing buildings

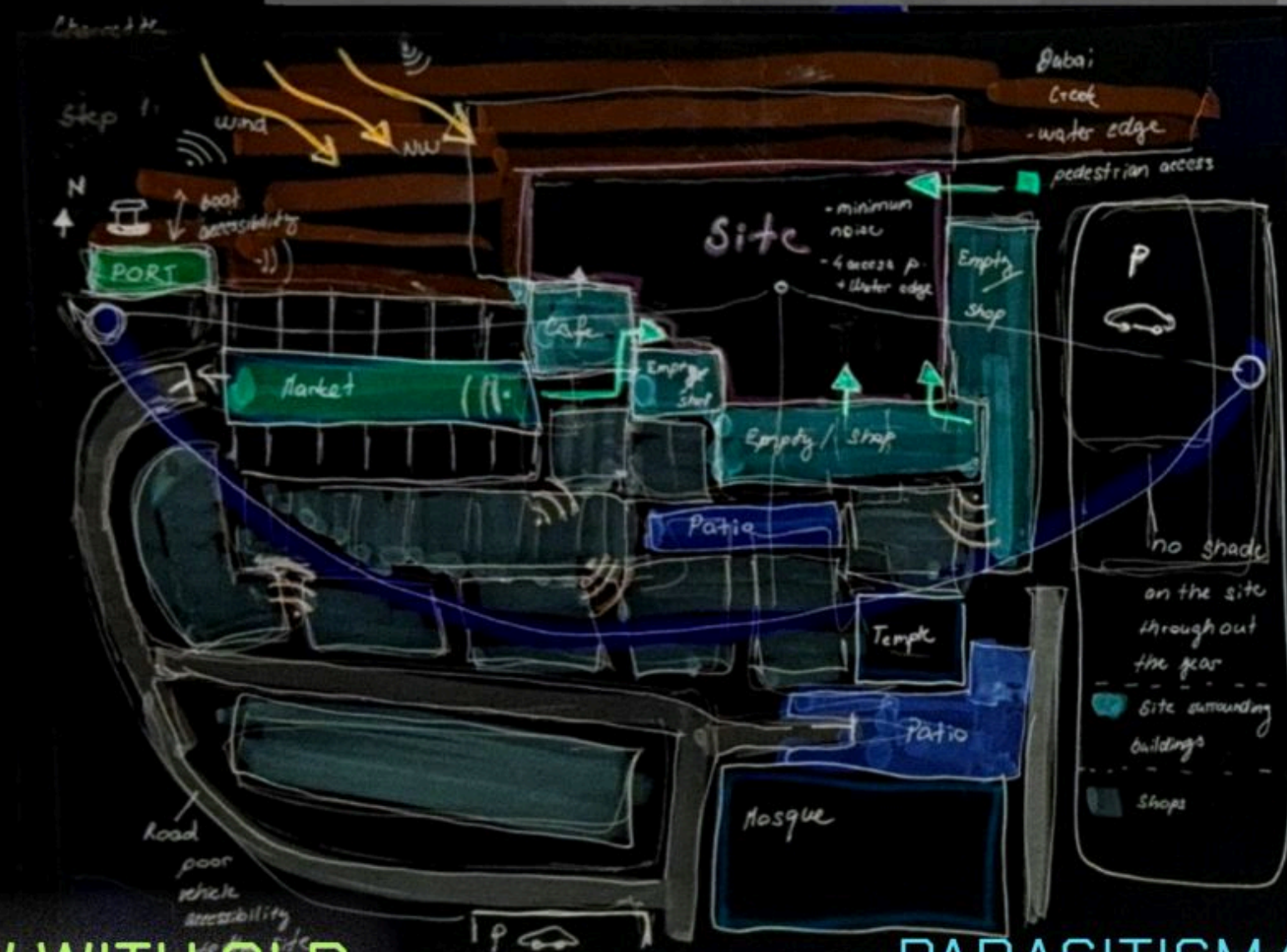
Pedestrian's inclination

SITE INFERENCES

SHADED/ OPEN AREA

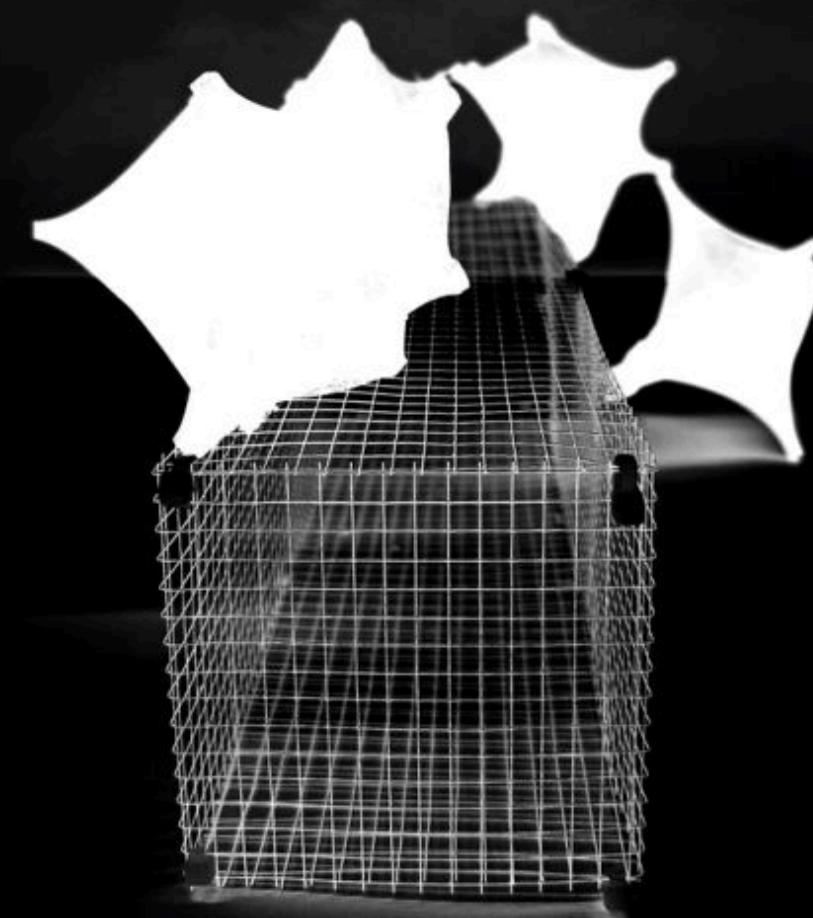
SITE INFERENCES
ABSTRACT MODES

CONNECT NEW WITH OLD



PARASITISM/
/CONNECTION/
/DISCONNECTION

ELEVATED
SPACIOUS
REST AREAS
INTERCONNECTED
PARASITIC
SEMI-OPEN
SPACIOUS

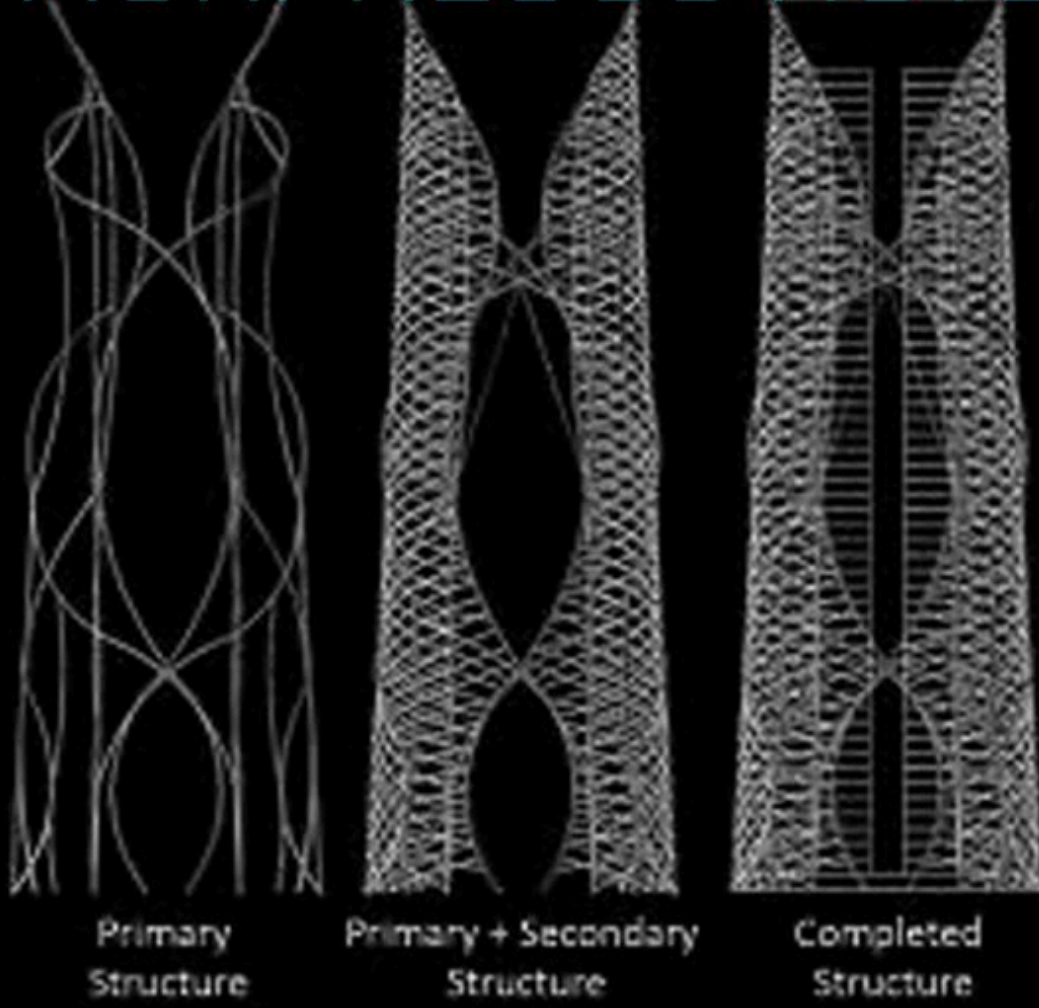


PRECEDENCE STUDIES

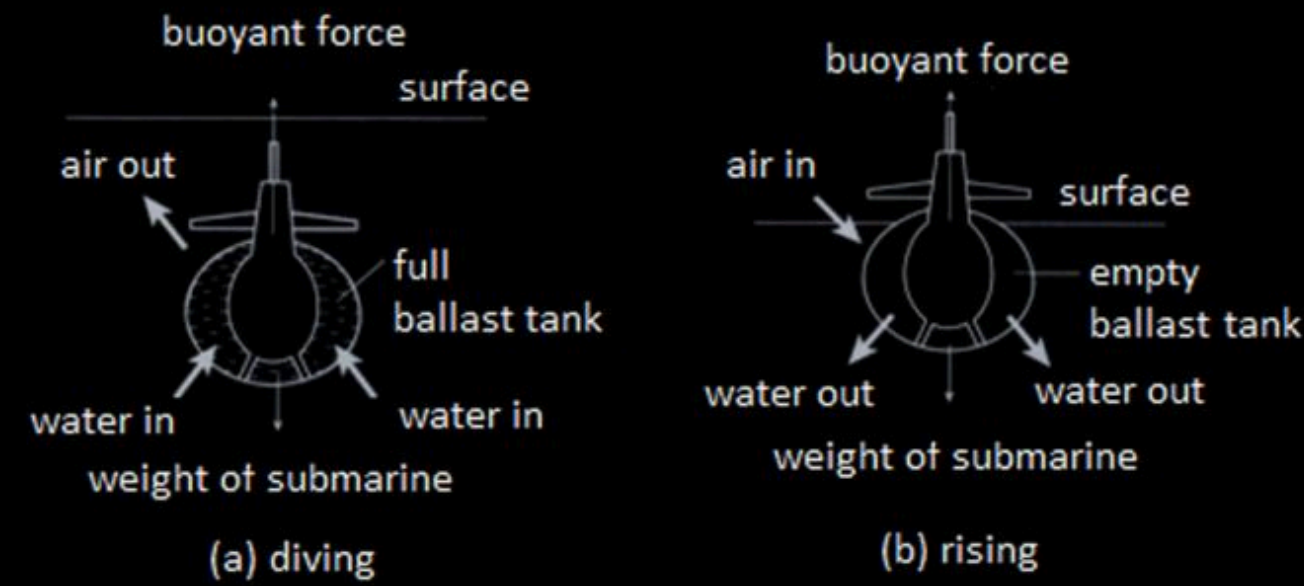
ATHÉNEA

ZAHA HADID MORPHEOUS HOTEL

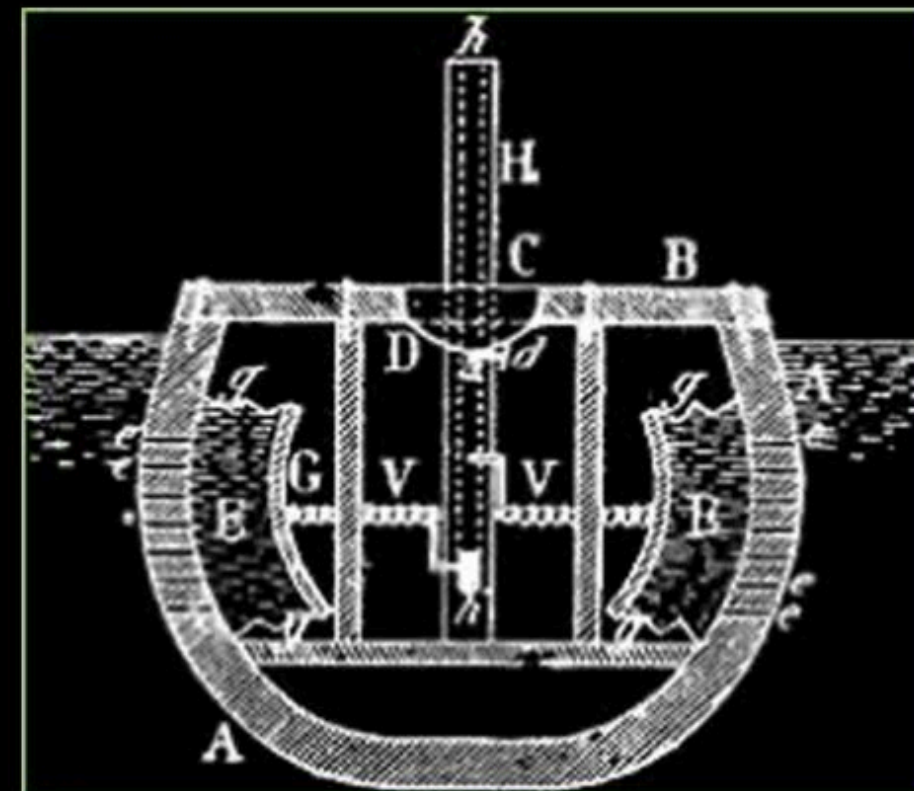
1. SUBMARINE



SUBMARINE DIVING MECHANISM



DECOMPRESSED CHAMBER



EXOSKELETON STRUCTURE

BOAT FLOATING PHYSICS

WATER FLOWING POD

360 VIEW OF UNDERWATER LIFE

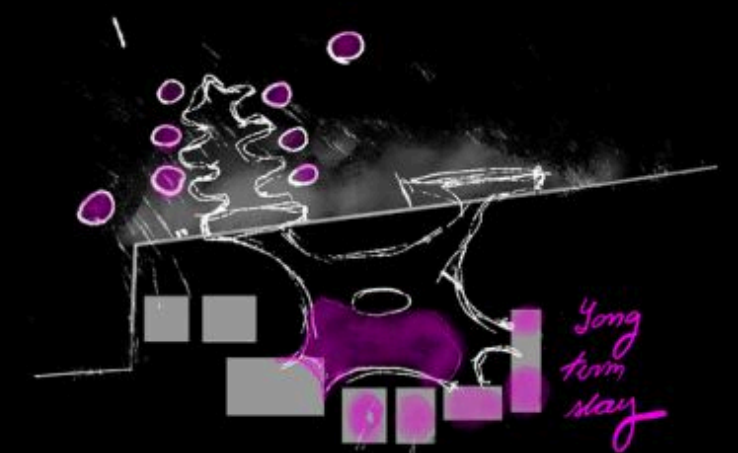
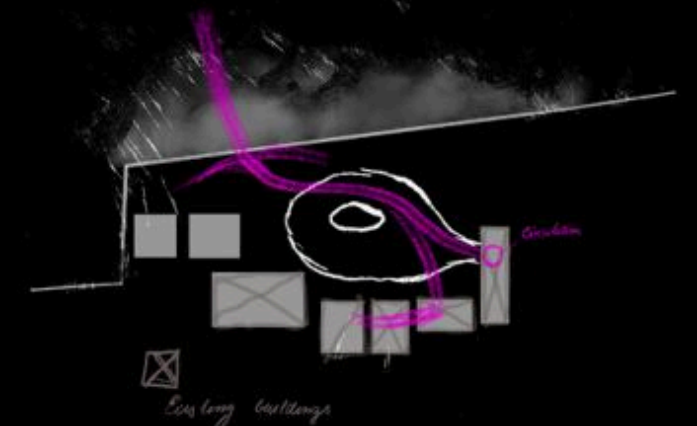
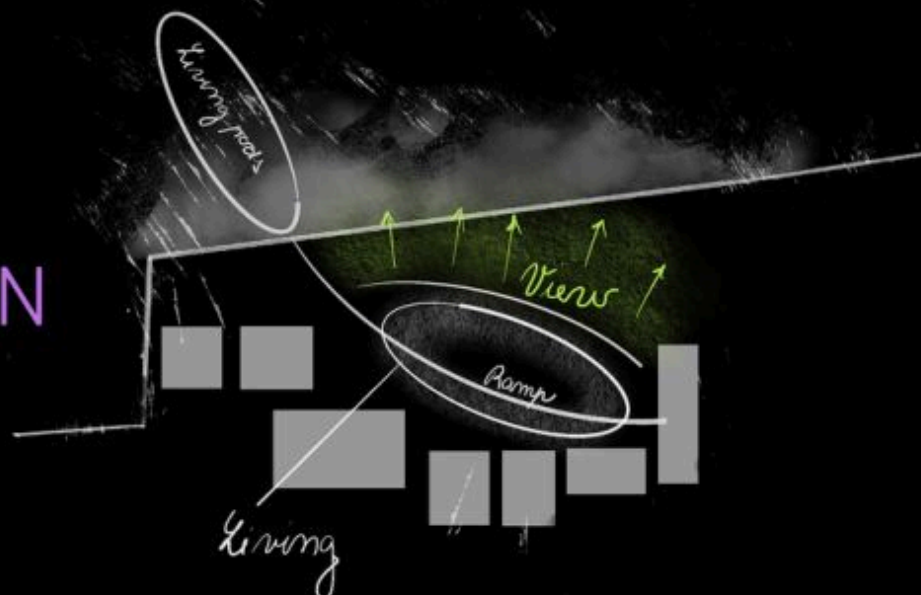
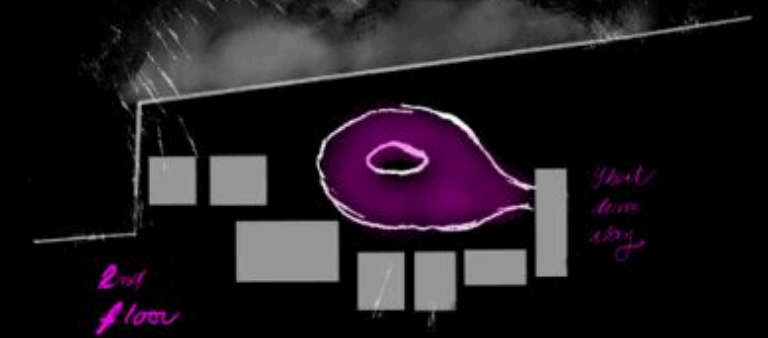


ZONING

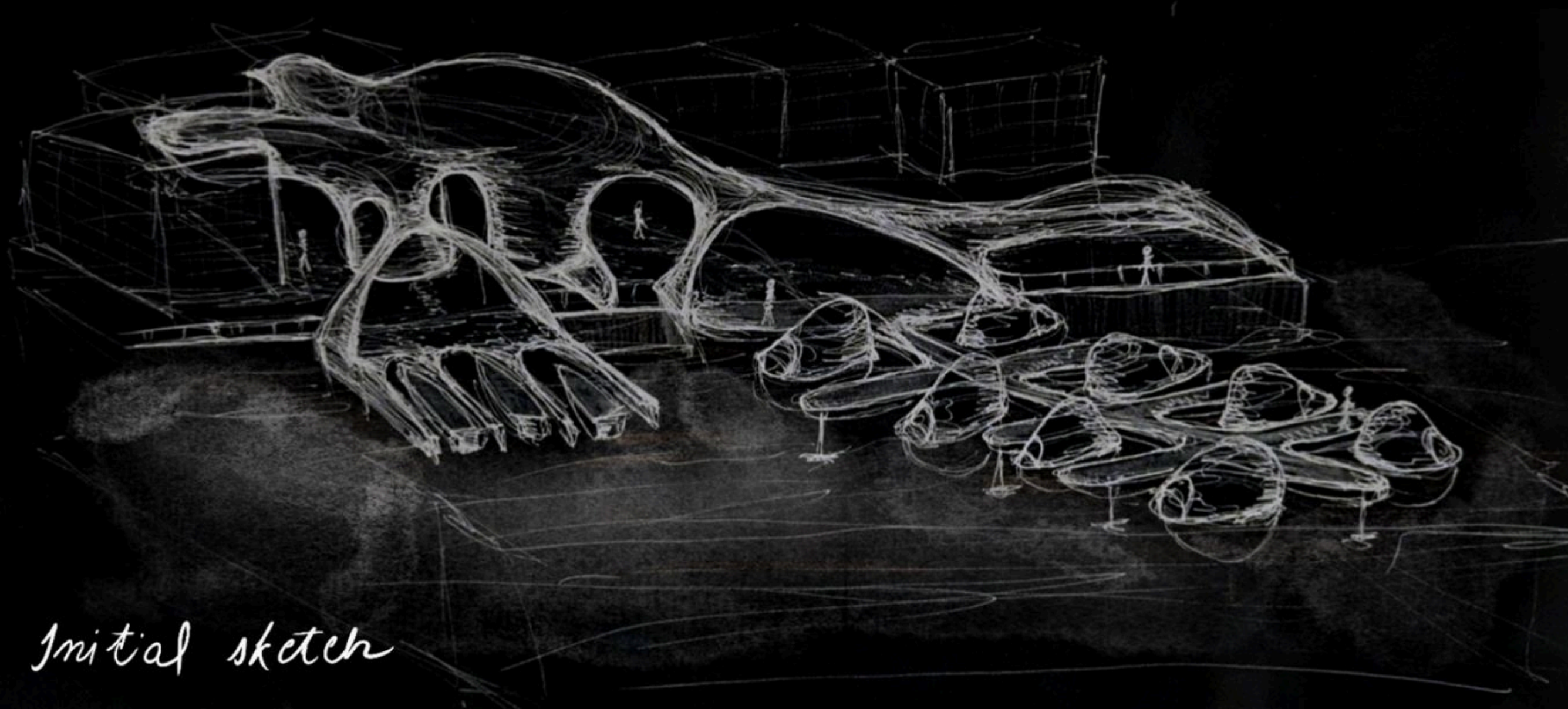
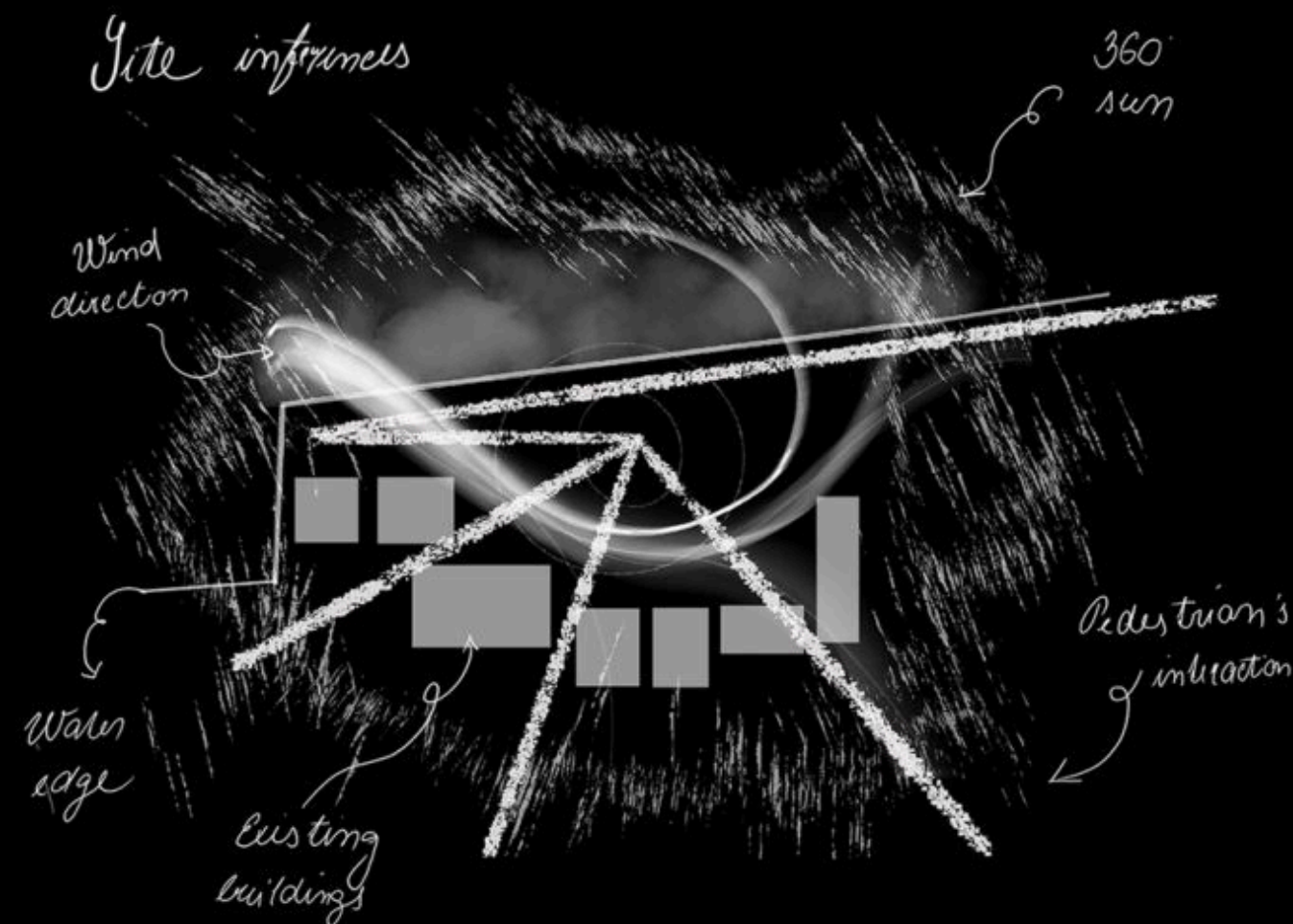
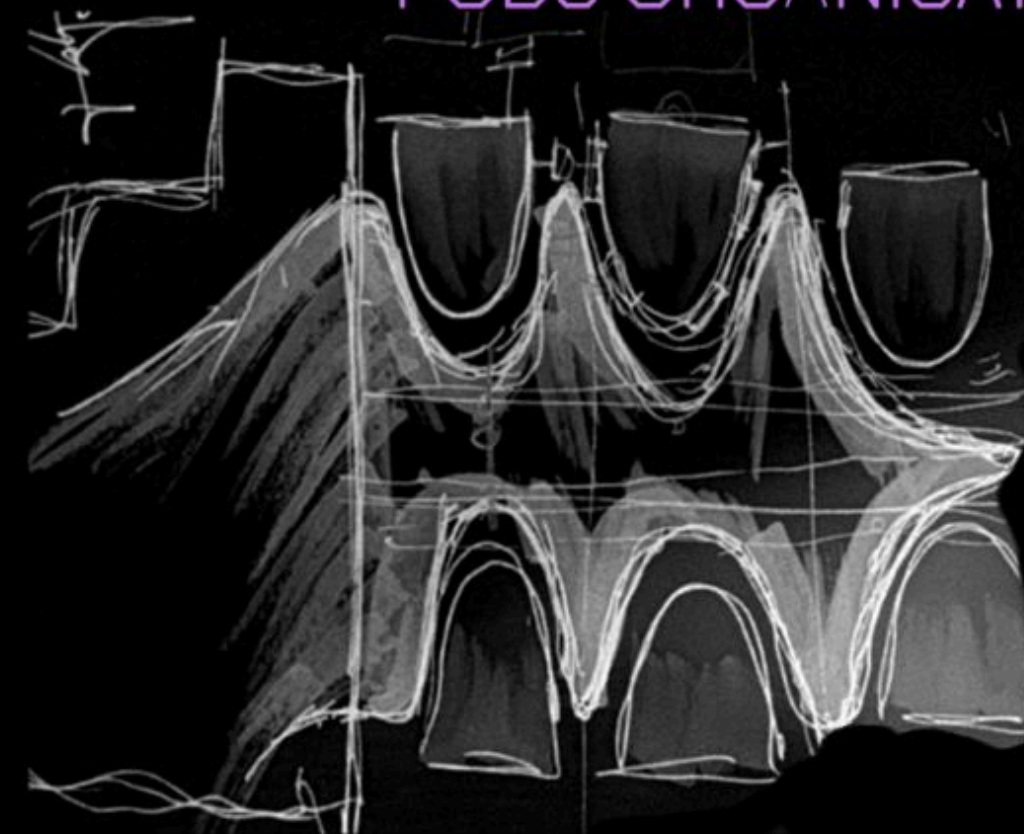
SHAPE DEVELOPMENT

REMORA FISH
FISH SPINE
ELEVATED

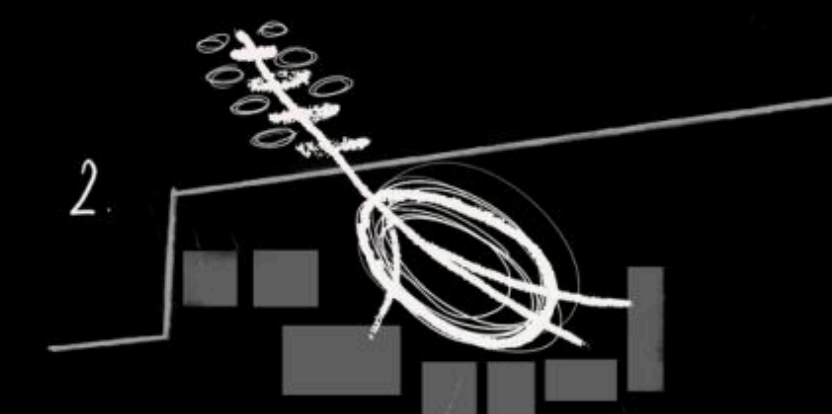
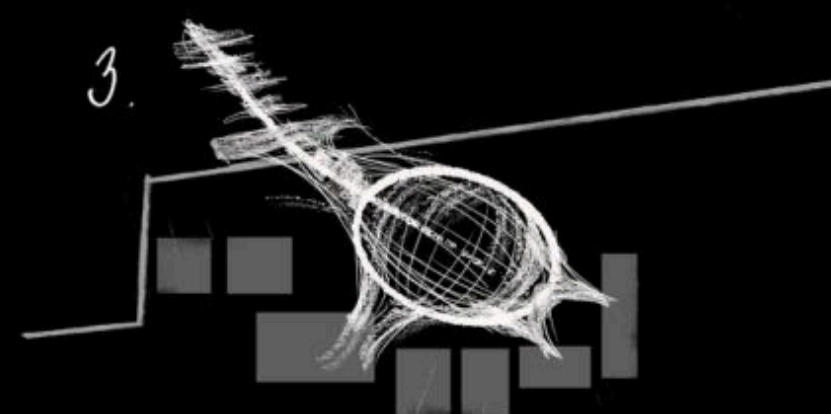
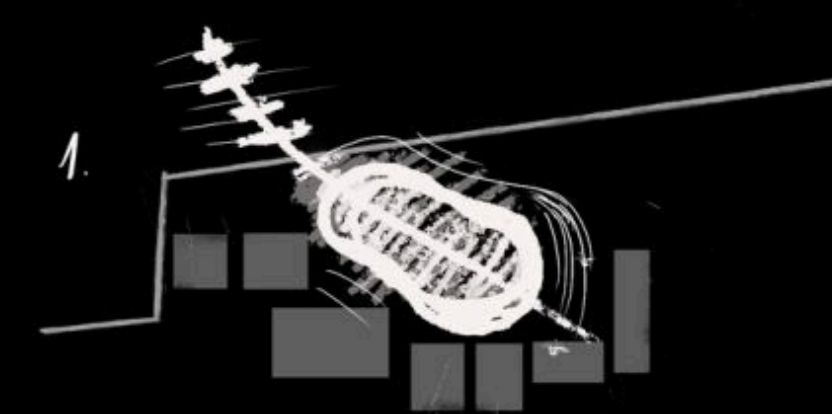
“INTRIGUING; INVITING; PLAYFULL”



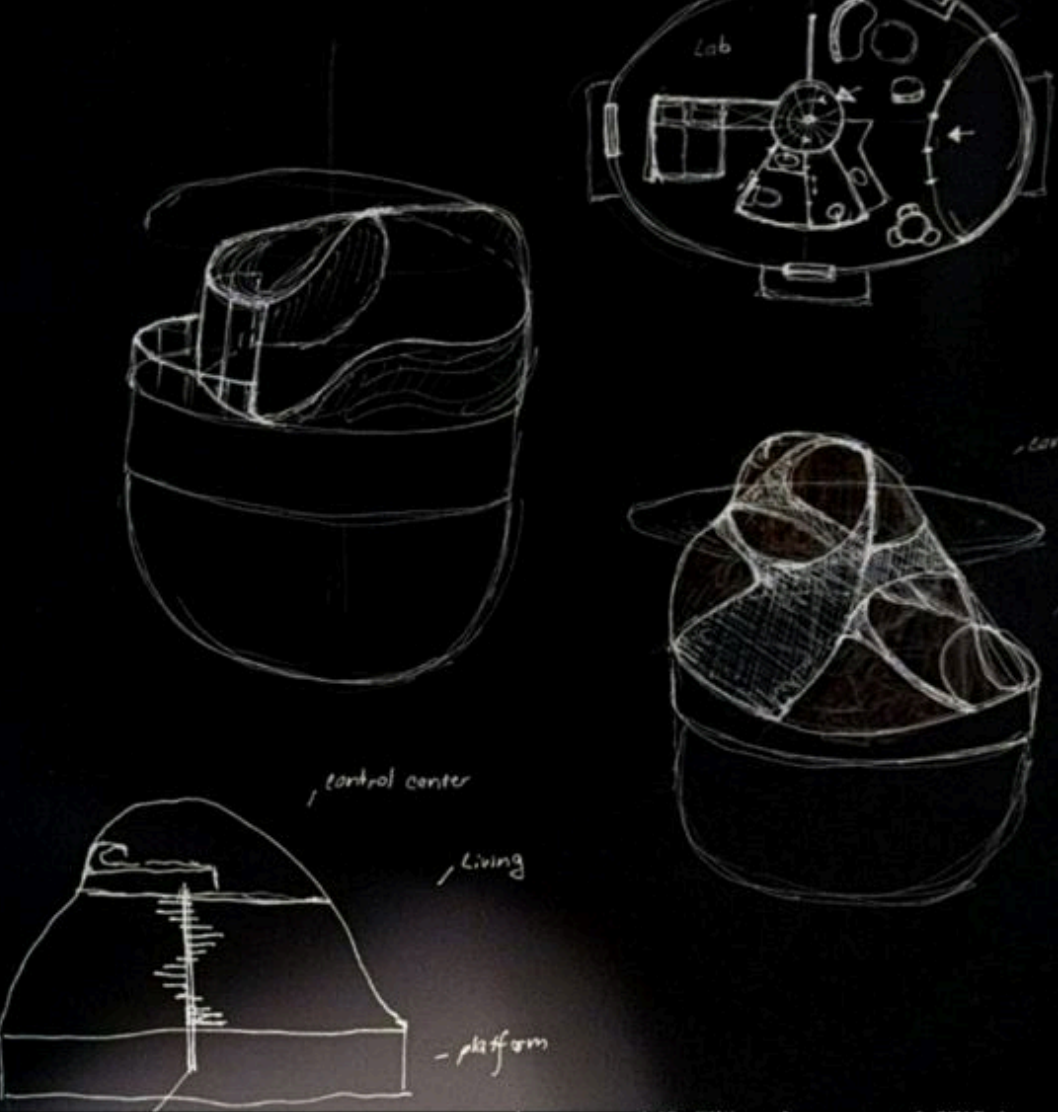
PODS ORGANISATION



Initial sketch

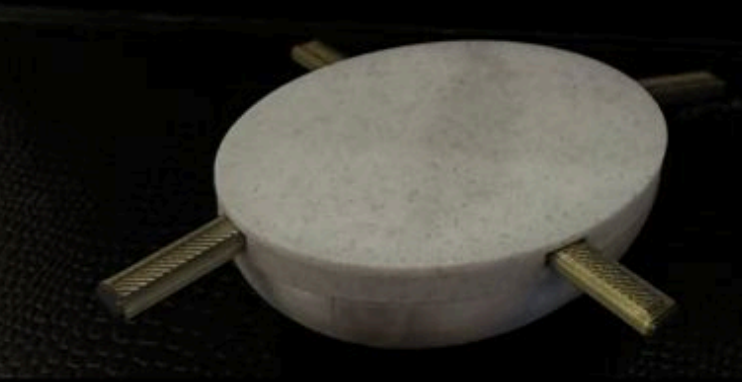


FLOATING//SINKING
POD

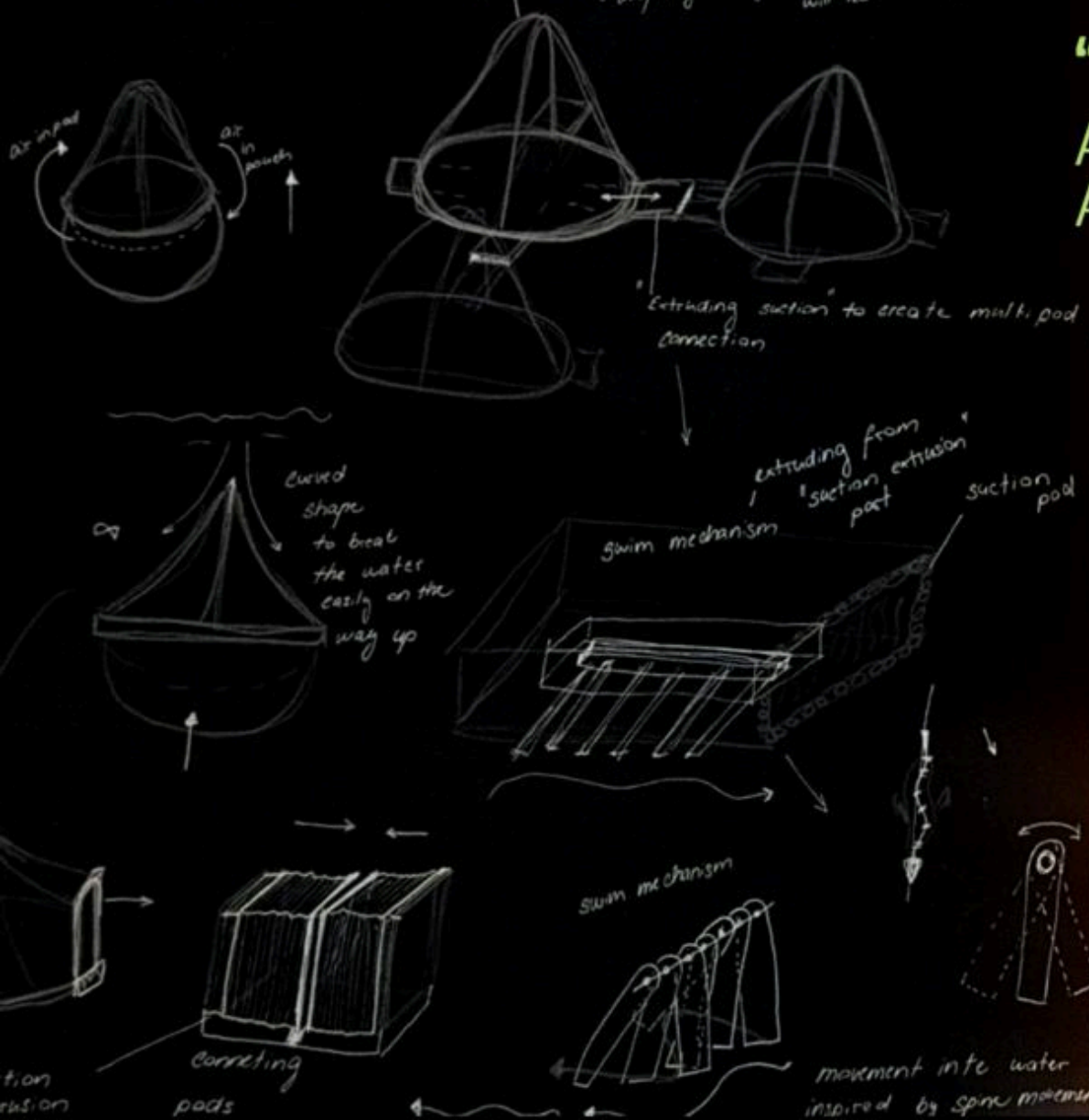


POD SHAPE
WEB COVERS PRIVATE ROOMS

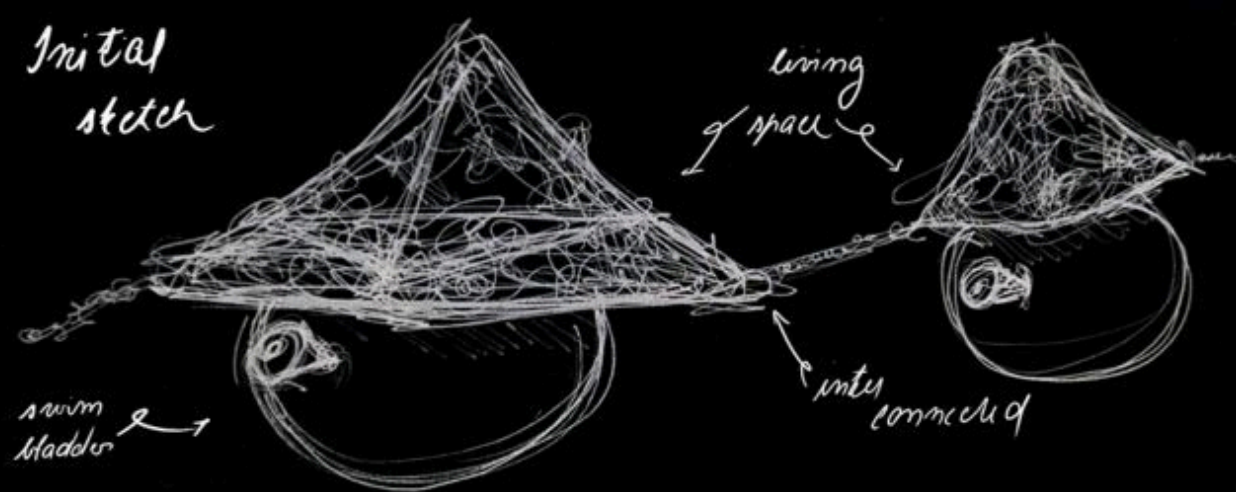
EXTRUDING
ARMS/BRIDGES
SUCTION CUP AT THE END
ALLOW CONNECTION
ACTS AS ANKER



"SWIM BLADDER"
ALLOWS SINKING/FLOWING
AIR BANK IN FLOOR SLAB



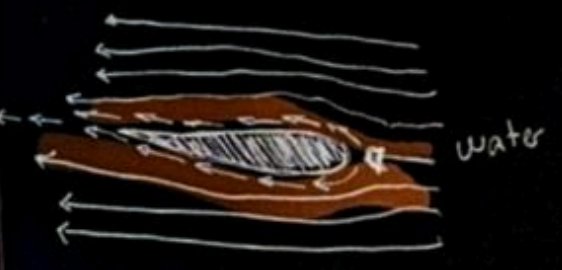
GLASS
LAB
FLOATING
SINKING
DEPRESSURISE
MARINE
FISH
INTERLOCKING
ANKER
MOVEMENT
TRAVEL
360



hydrofoils
- auto-align w flow
- lateral oscillation



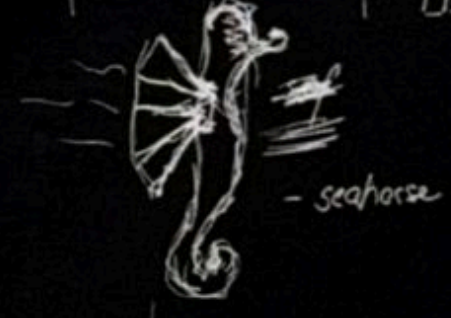
how it moves?



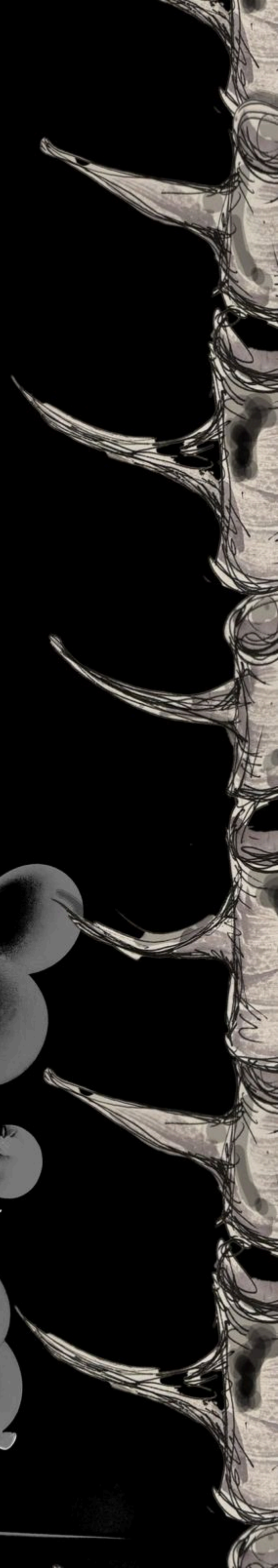
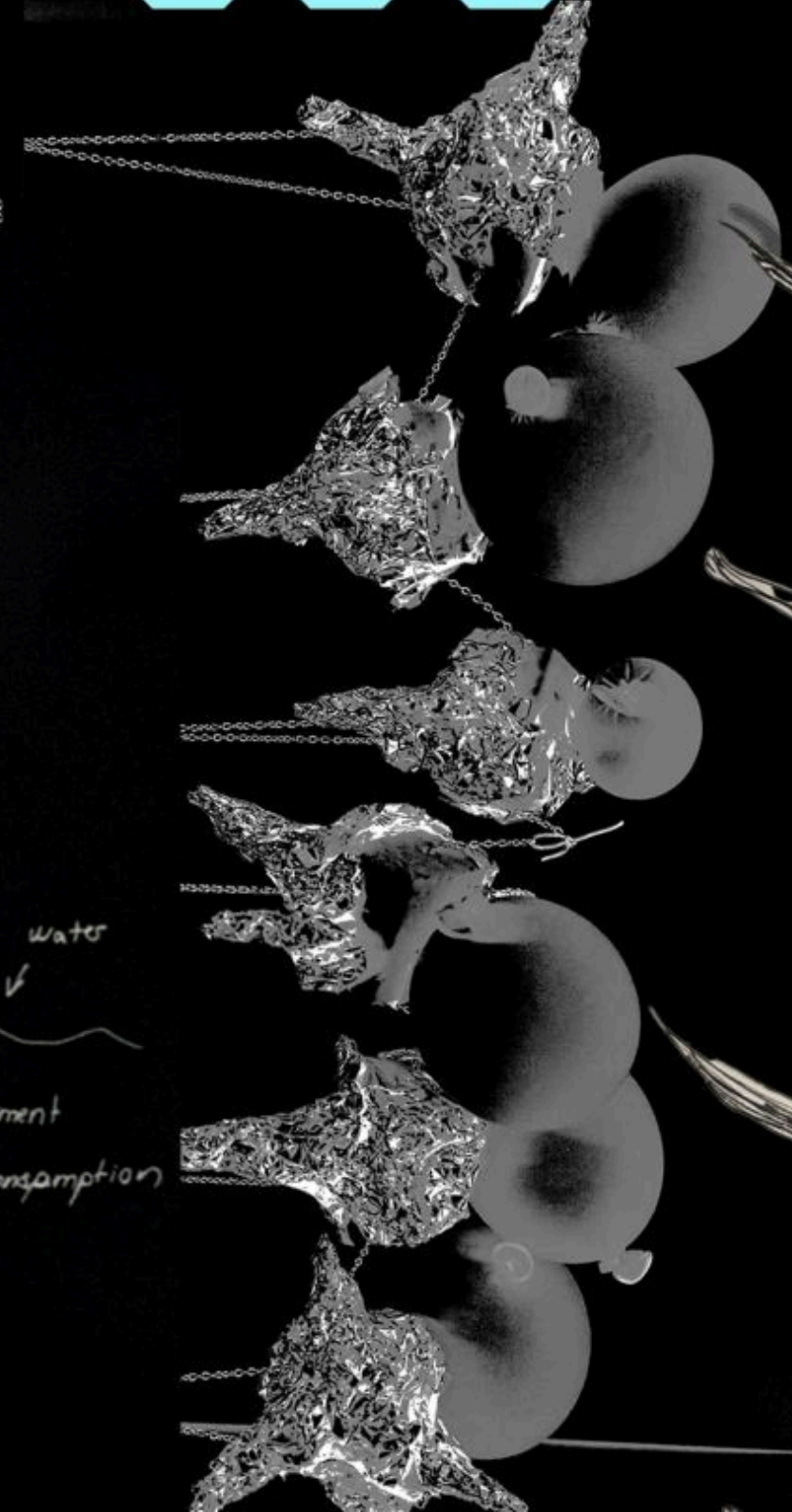
glides through water under angle which accelerates water (if angle upwards, water downwards)

- Newton's 3rd law
(every action has equal reaction)
water f. lifting

oscillating mechanism (propulsion)
- angled



- effective movement
- low energy consumption



G FLOOR PLAN

Under water pods

floating pods

Port station

water feature

Main circulation

site shaded by the building

Entrance from existing building

Existing buildings



SCALE 1:500



1ST FLOOR PLAN



Living
pods
(under/on water)

Long-term
living

park

Exoskeleton
structure

cafe

Public park
ramp

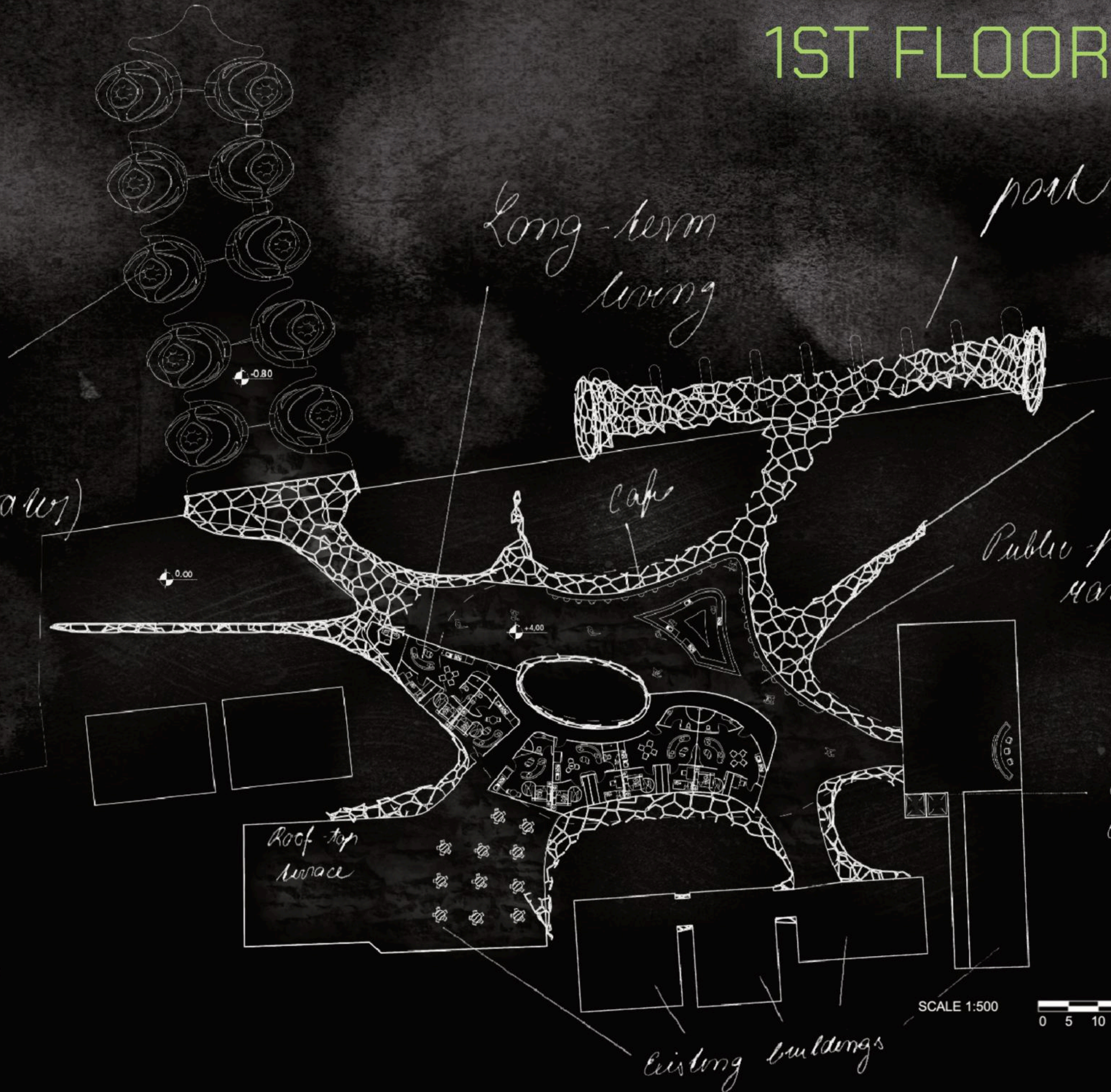
Elevator
circulation

Roof-top
terrace

water
edge

Existing buildings

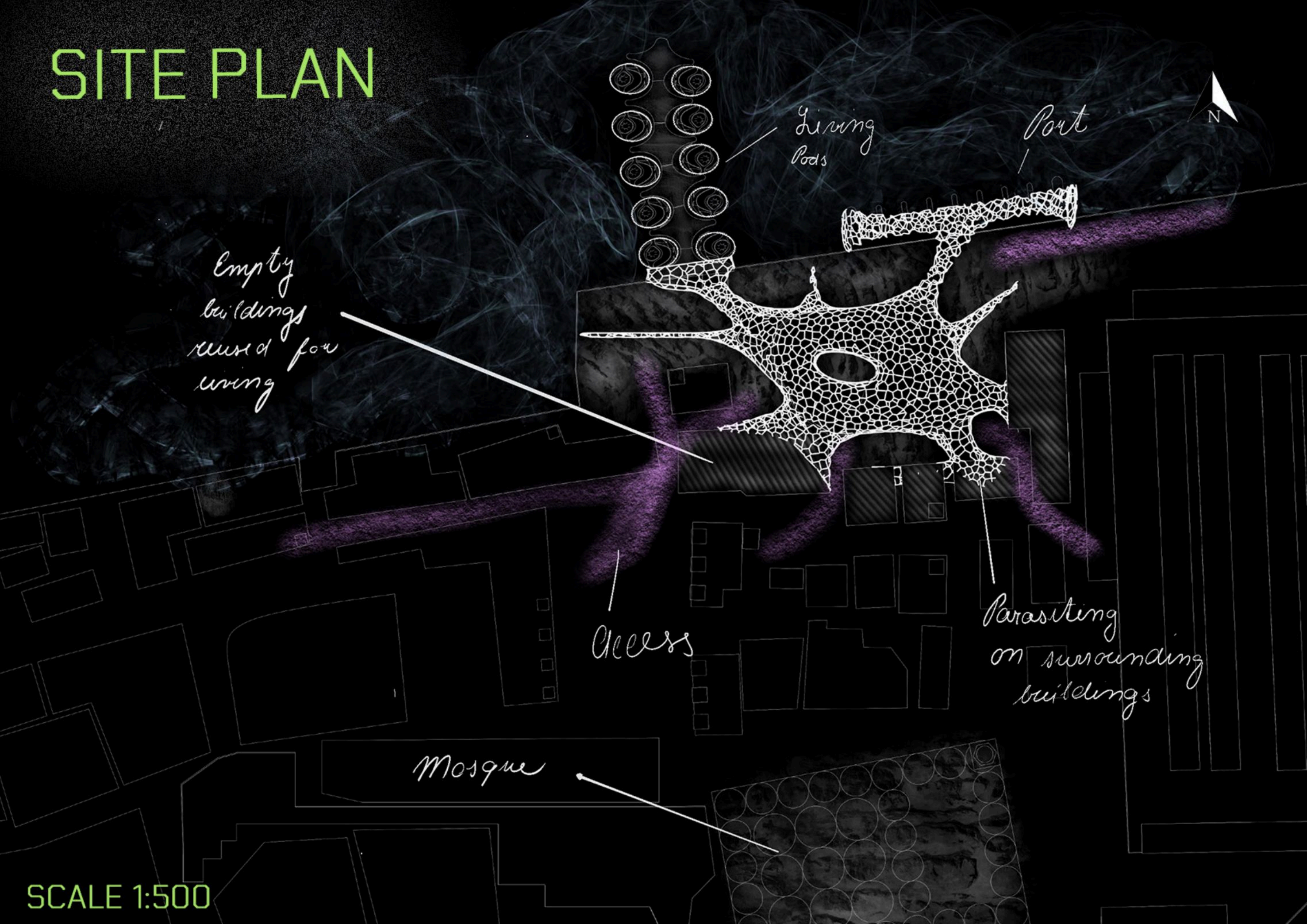
SCALE 1:500



2ST FLOOR PLAN

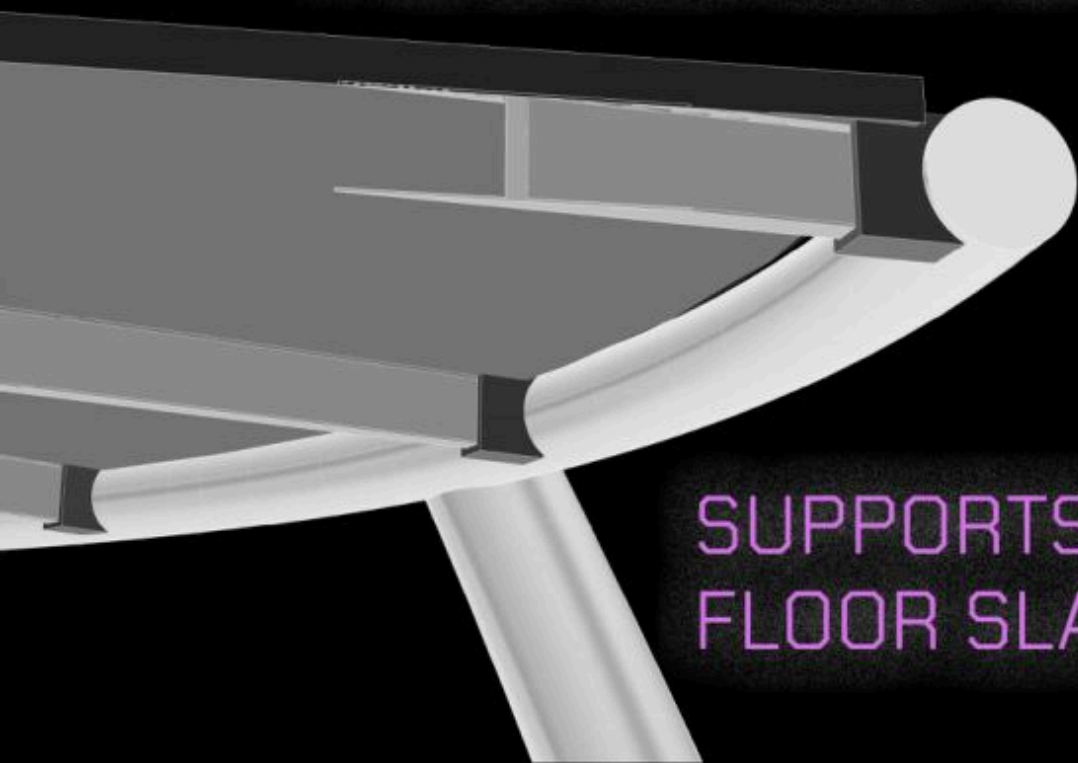


SITE PLAN



SCALE 1:500

H-SHAPED BEAMS INTERLOCKED INTO THE EXOSKELETON

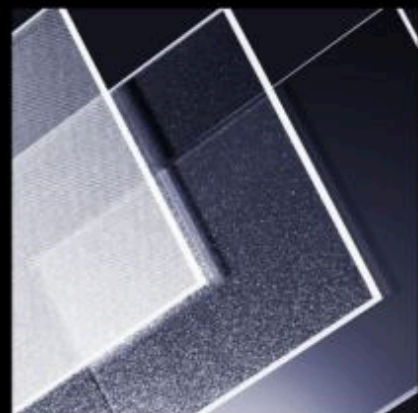


SUPPORTS CONCRETE FLOOR SLABS



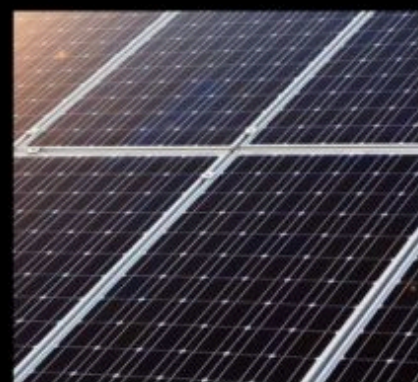
Aluminum Alloy
strength-to-weight ratio
Highly corrosion resistant
Lightweight → reduces structural load
Easy to fabricate, extrude, and anodize
Good thermal performance

METAL FRAME



Double-Glazed Low-E Solar Control Glass
Reflects infrared heat
Slight tint (blue/grey/green)
Reflects heat back outside
Improves insulation

SHAMED GLASS

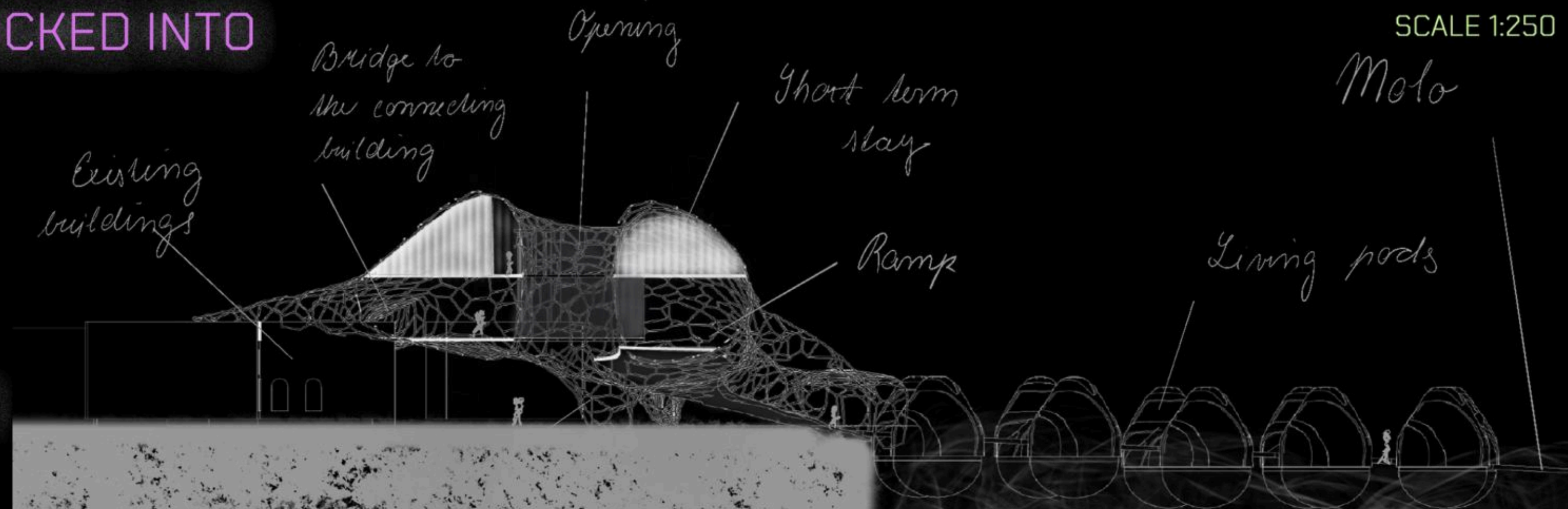


BIPV
Durable in high heat
UV-stable colours
Works well in Dubai climate

COLORED SOLAR PANELS

MATERIALITY

SCALE 1:250



SECTION AA

Yee Nio

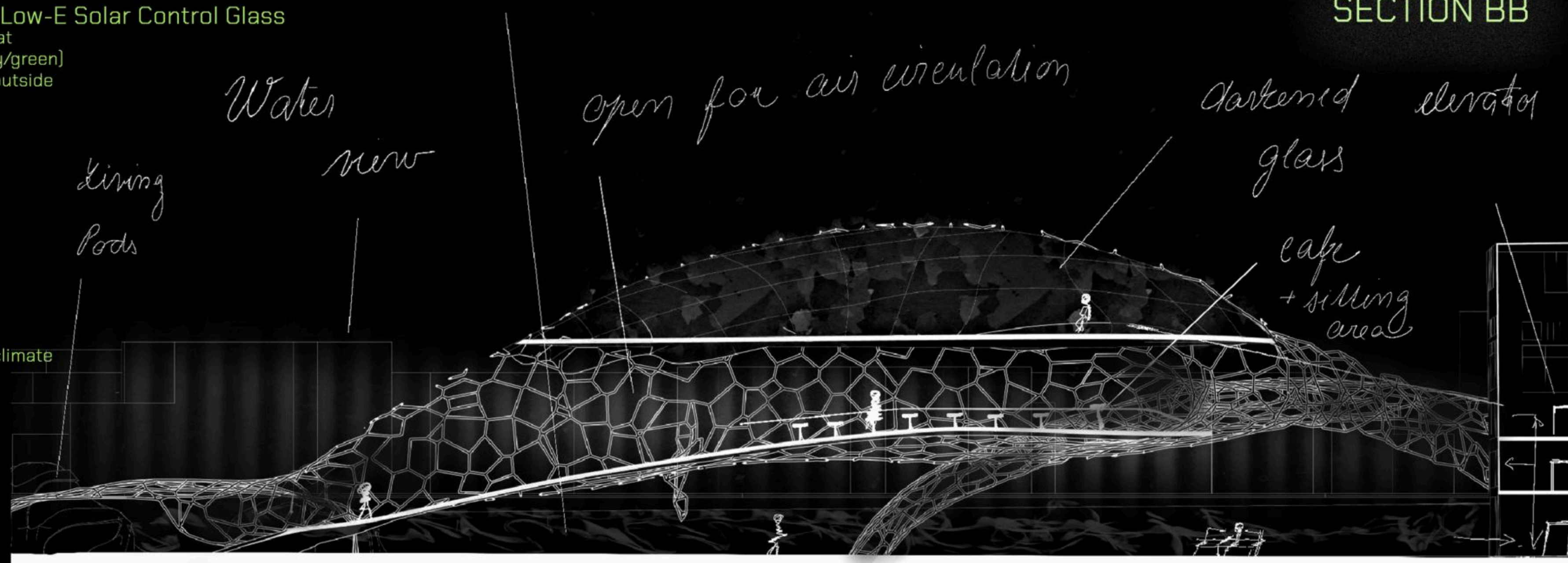
Support "legs"

Water edge

Water

minimum build up area for 6 floor

SECTION BB

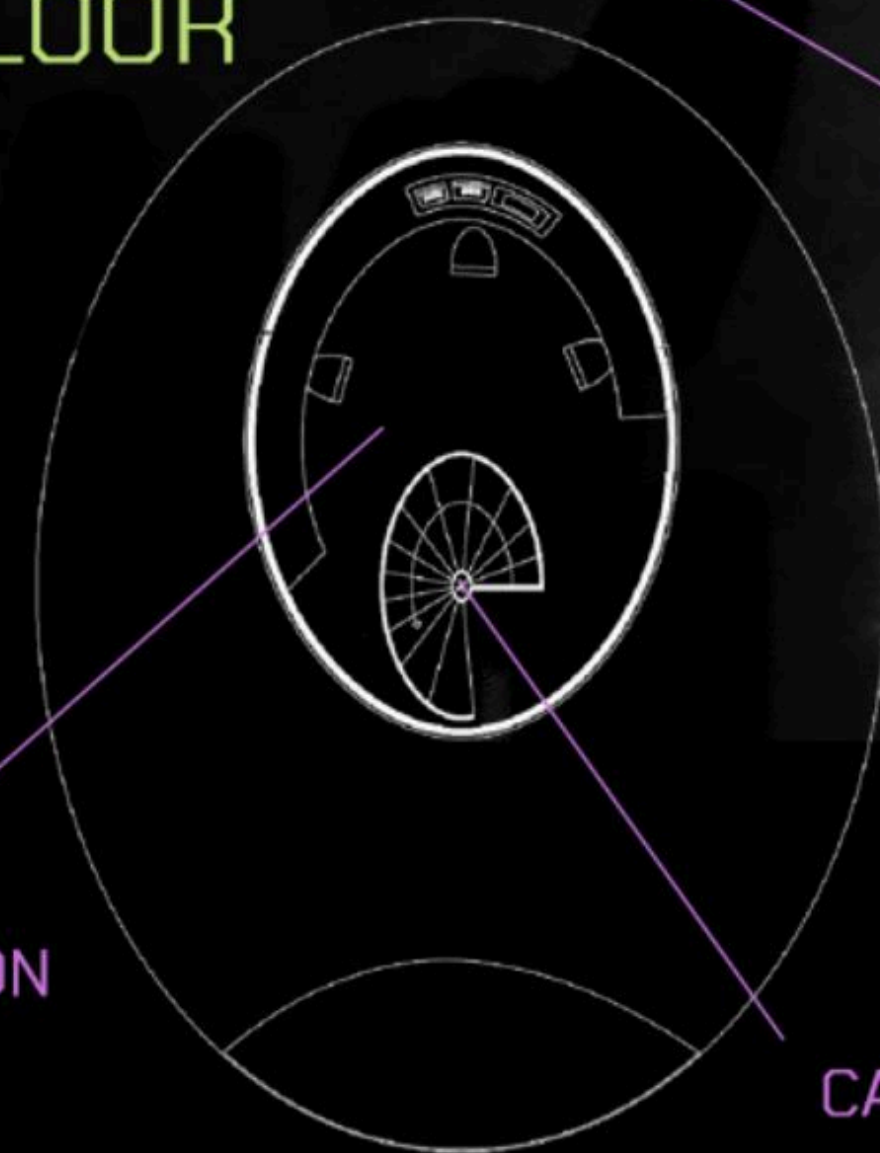


POD UNIT PLAN

G FLOOR



1ST FLOOR



LAB

TERRACE

OPERATION ROOM

CABLE PIPE

EXOSKELETON

GLASS

2ND FLOOR
FLOOR SLAB

1ST FLOOR
FLOOR SLAB

RAMP

EXOSKELETON

PRESSURE
EQUALIZATION
DUCTS

PORT

LIVING PODS

AXONOMETRIC
BUILDING

Operating
station

Living
section
(1 Bedroom)

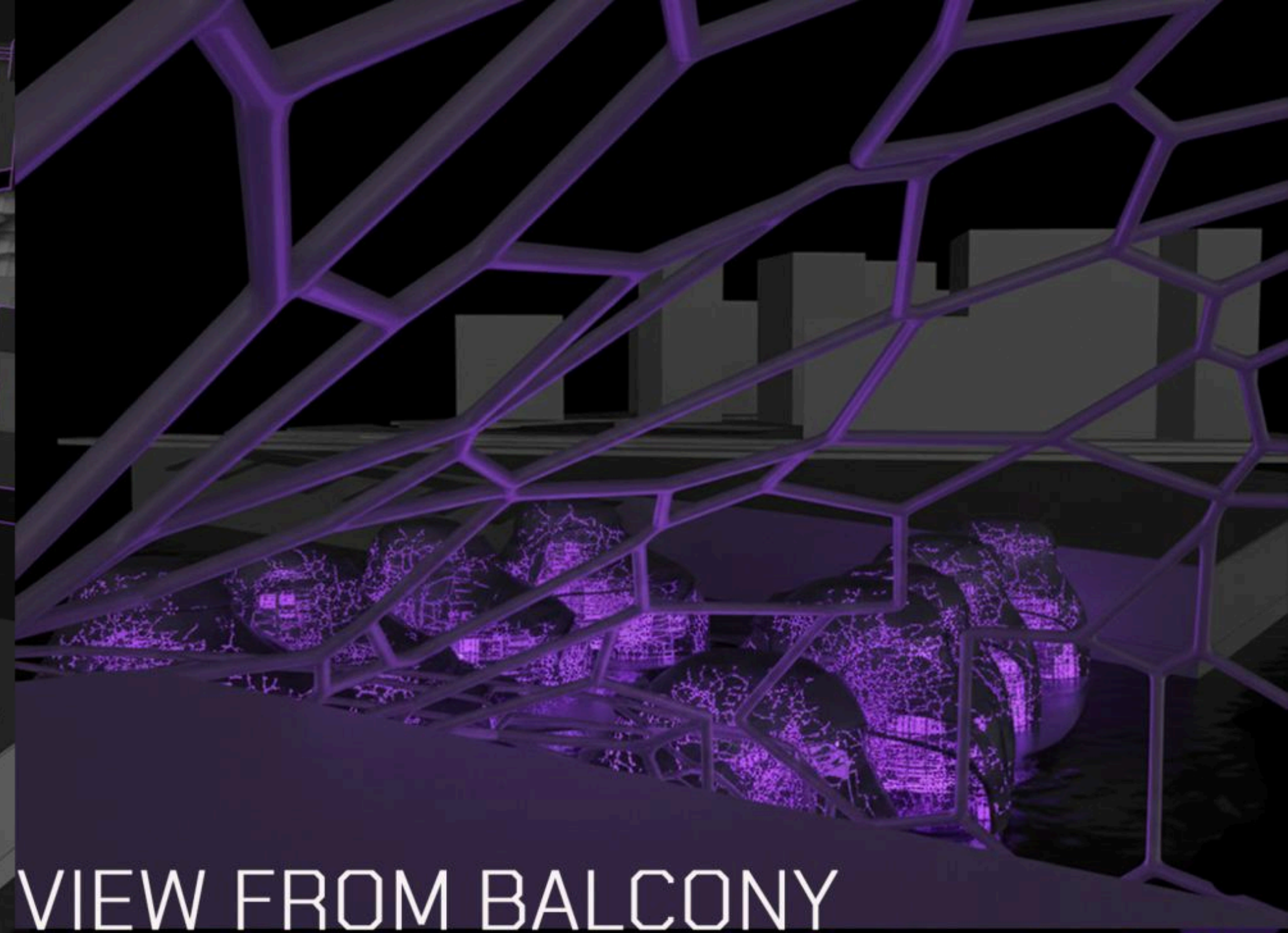
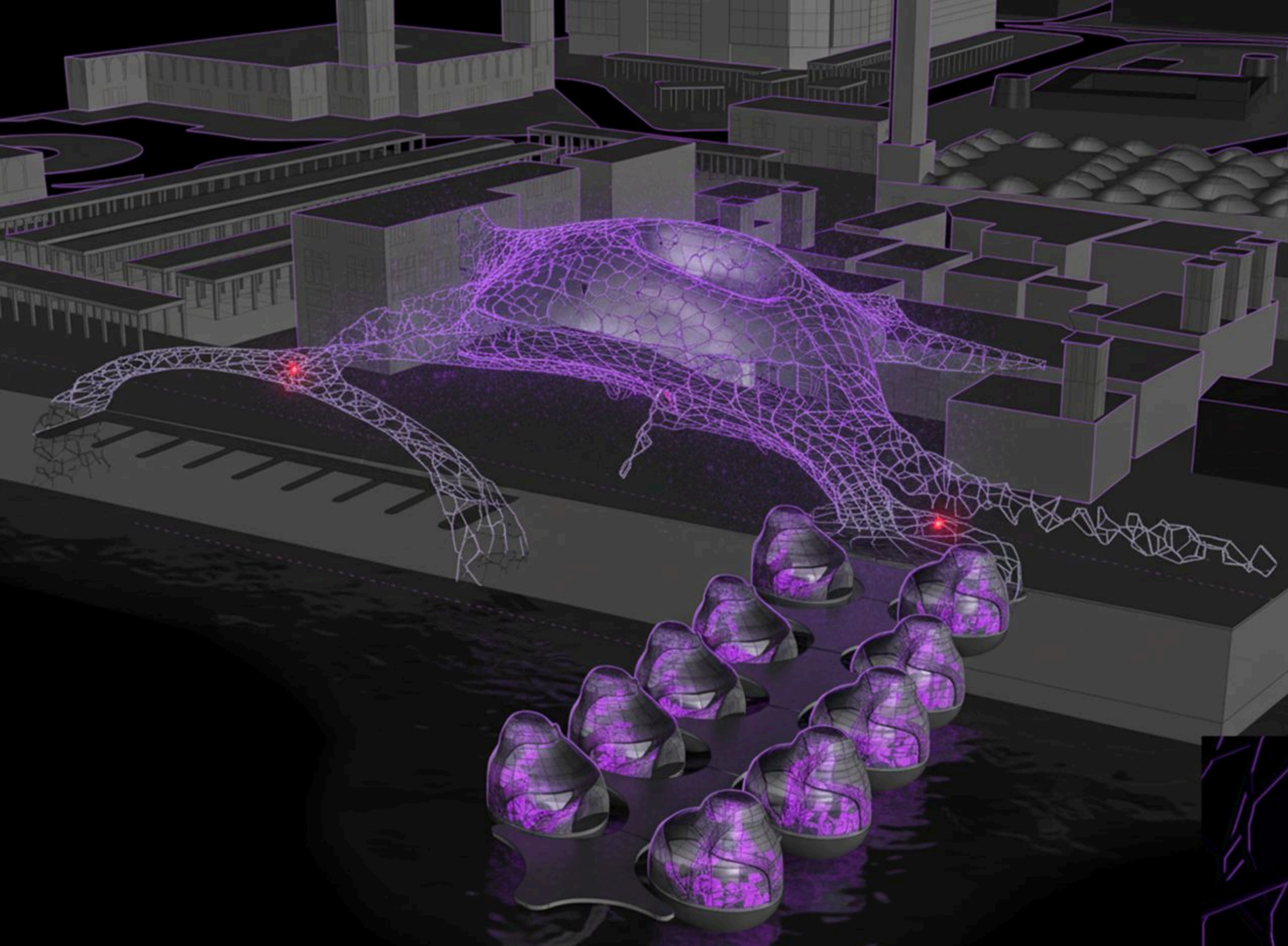
Glob
connection

Extruding
bridges
with
suction
disks

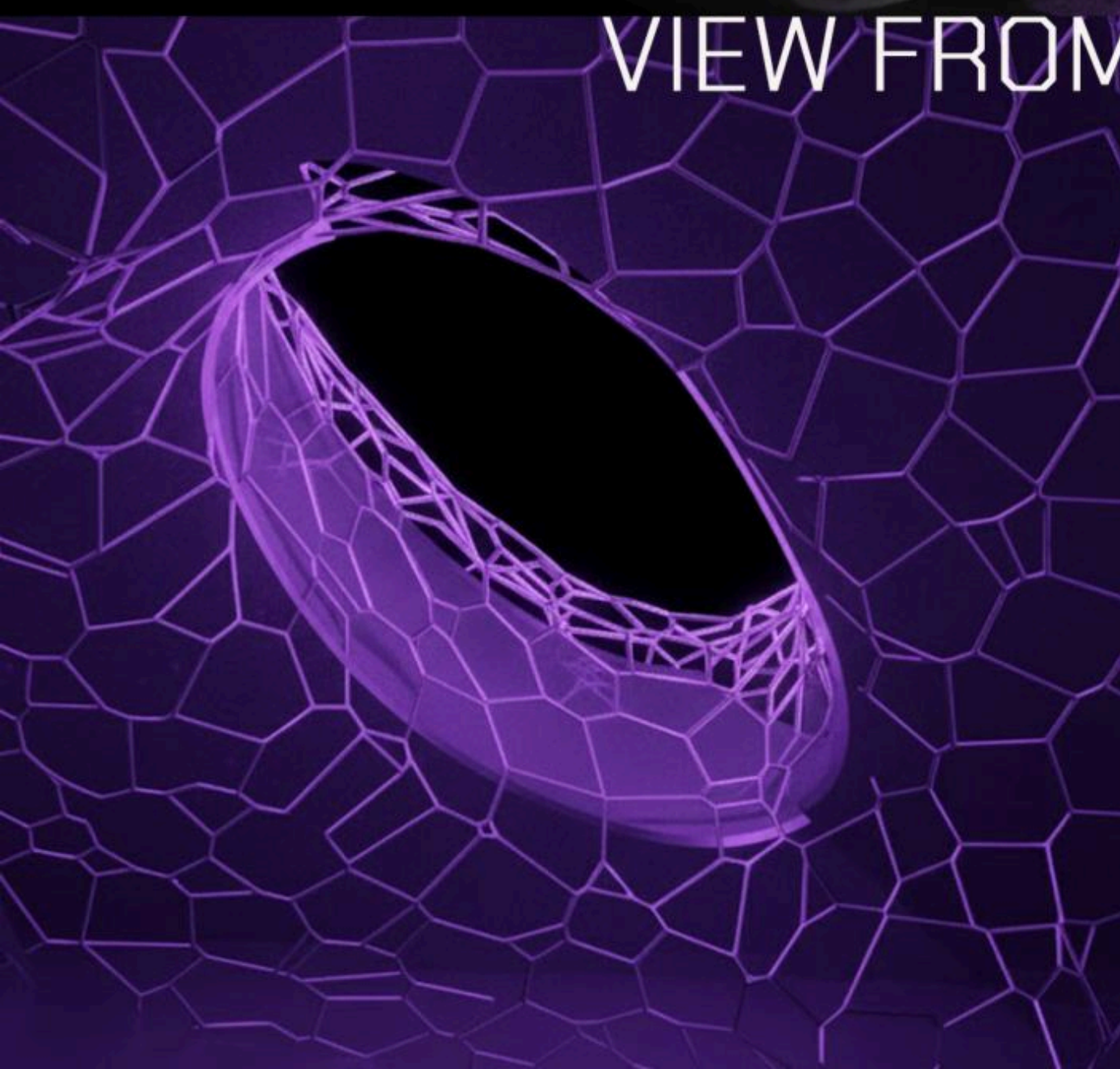
air
pocket

AXONOMETRIC
POD

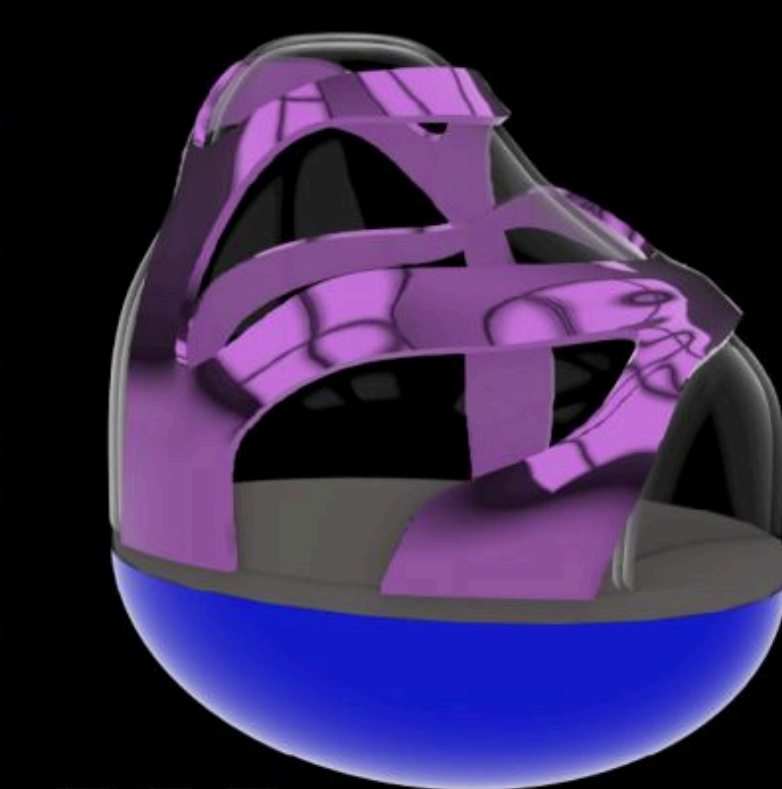
SCALE 1: 50



VIEW FROM BALCONY



VIEW FROM GROUND UP

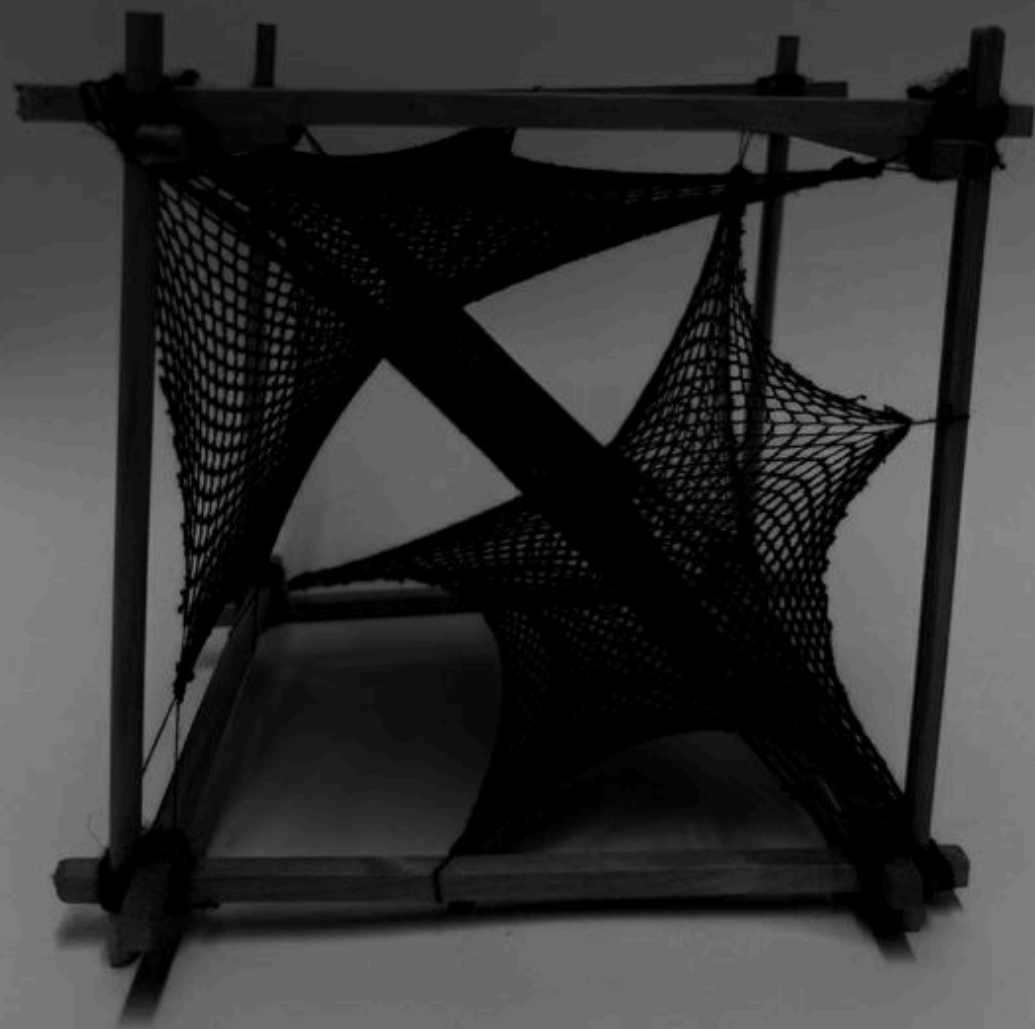


POD



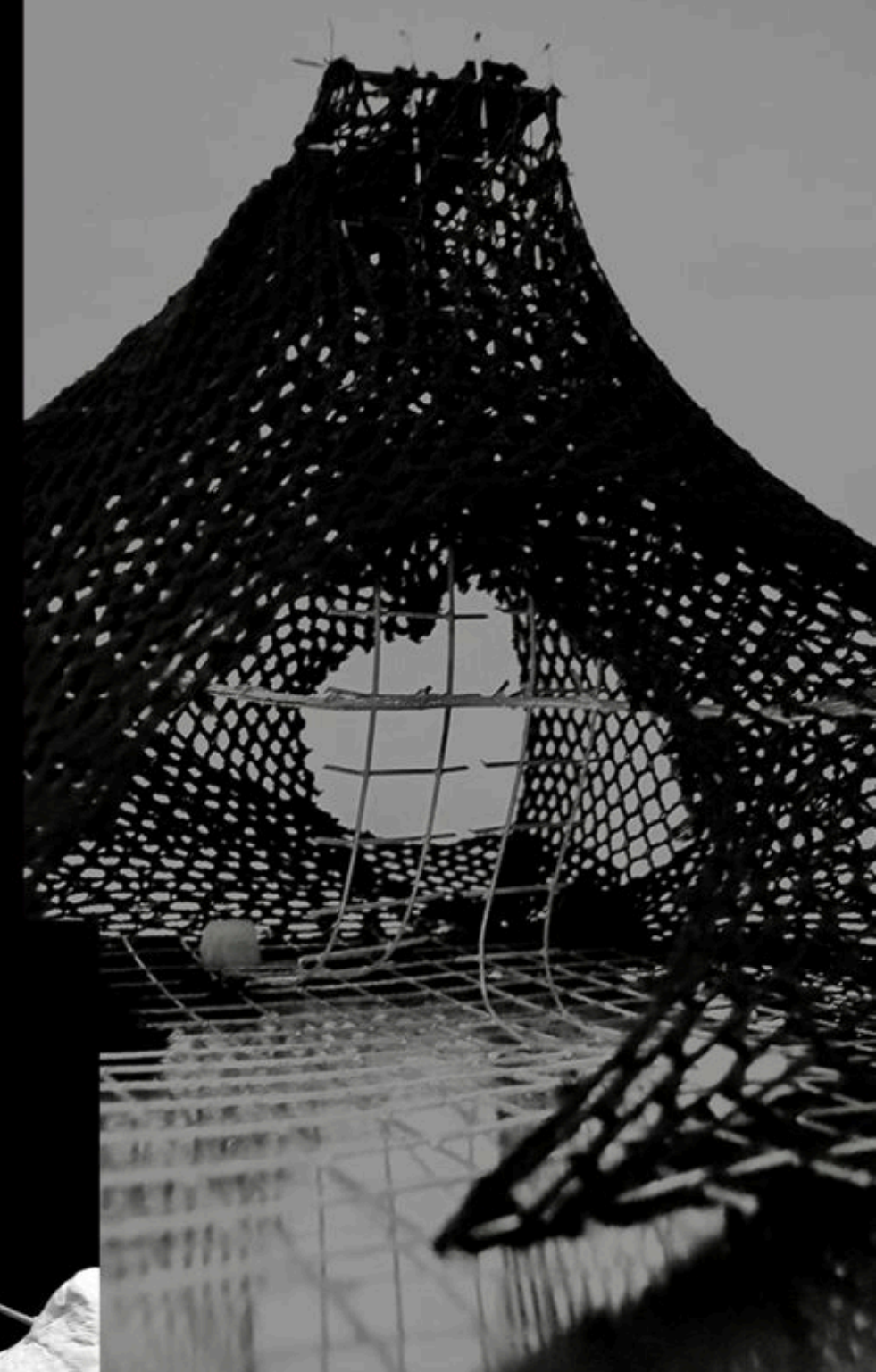
UP THE RAMP

SUPPORT

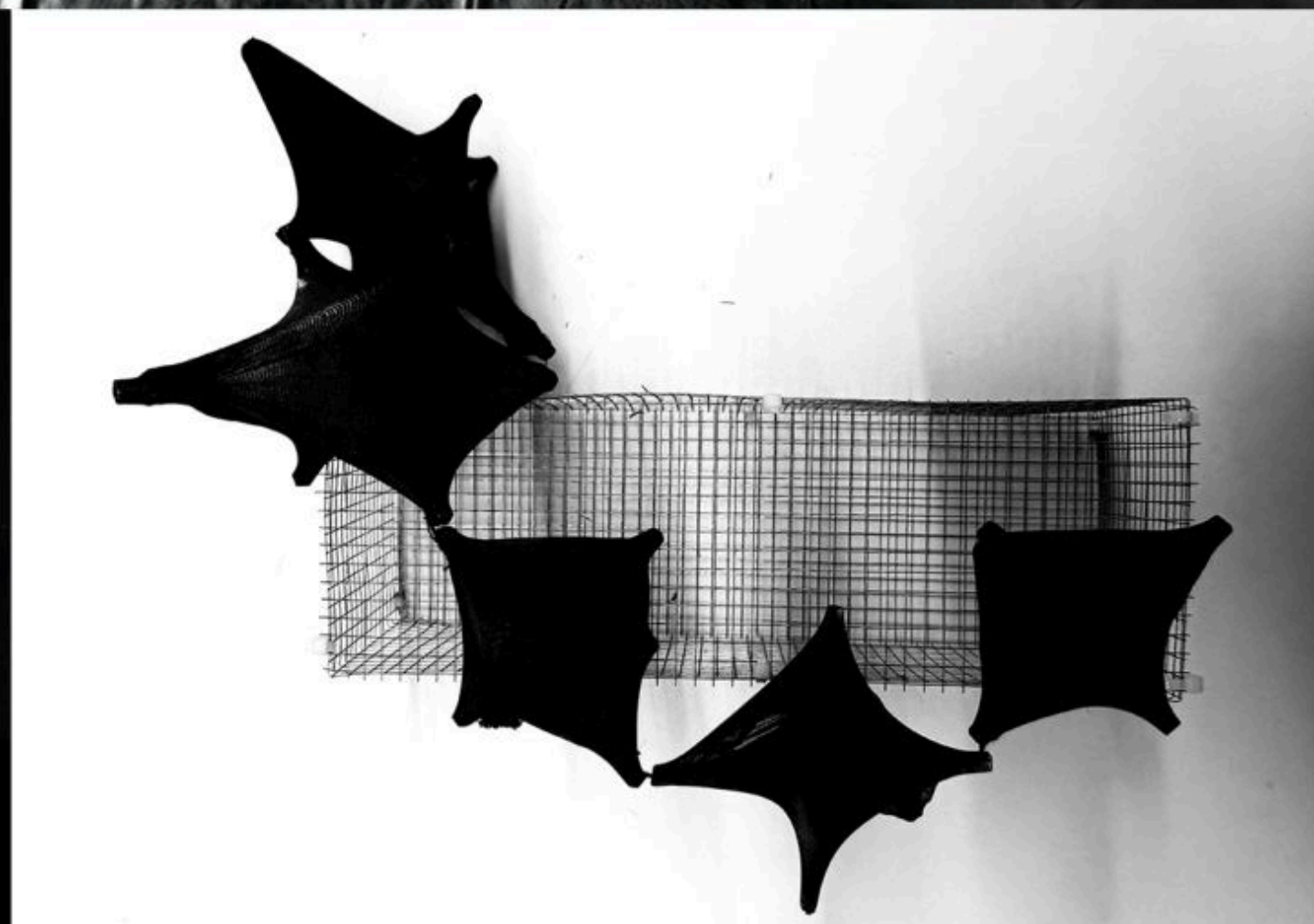


PHYSICAL MODELS

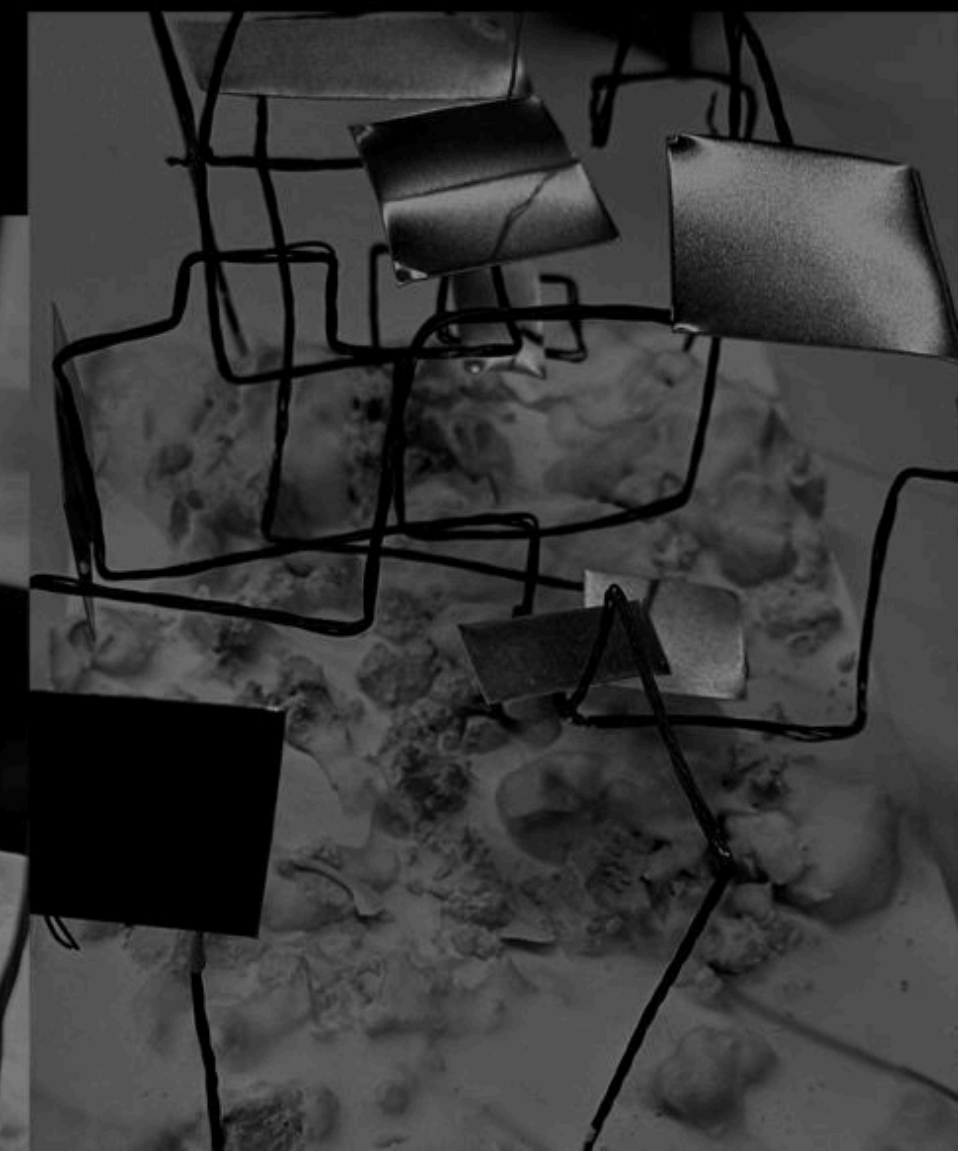
OPENED



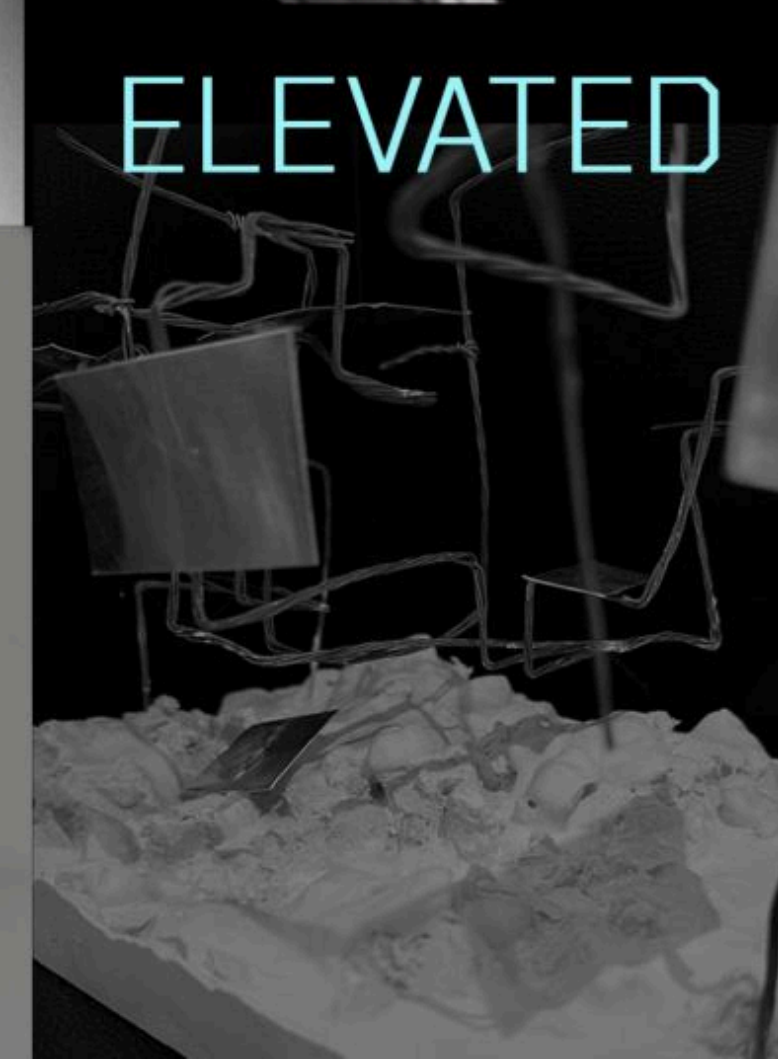
INTERLOCKED



PARASITE



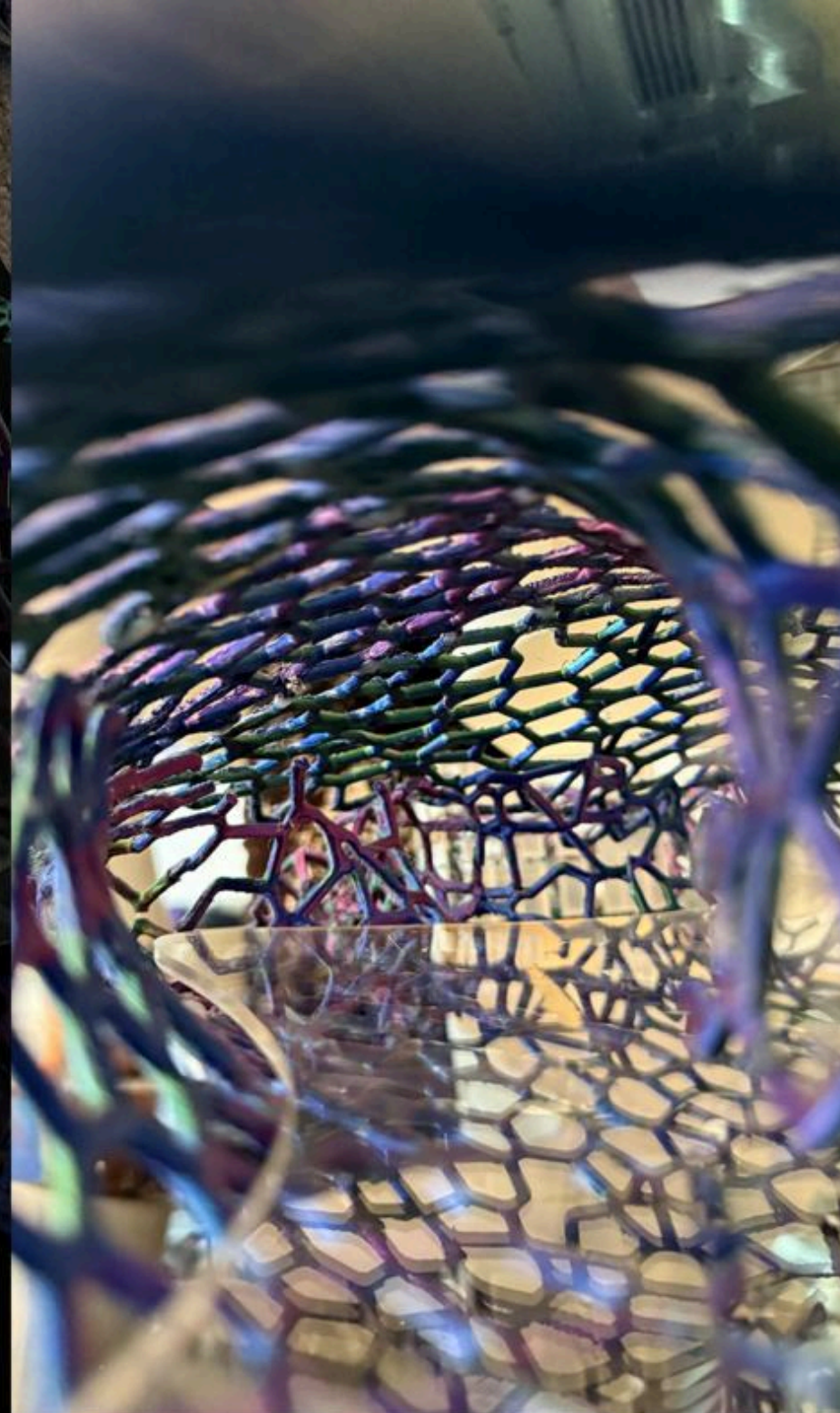
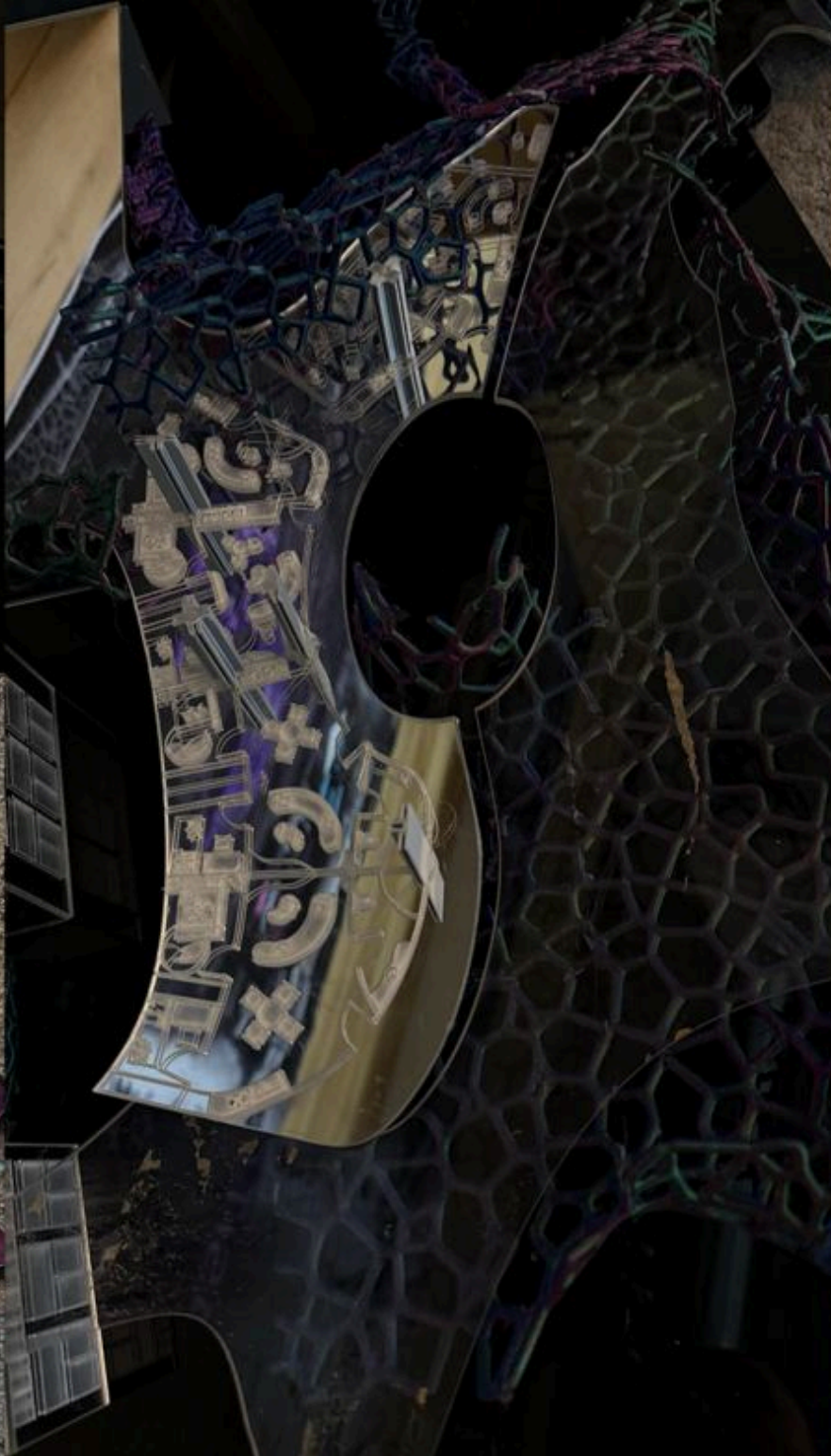
ELEVATED



MODEL 1:500

MOVEMENT



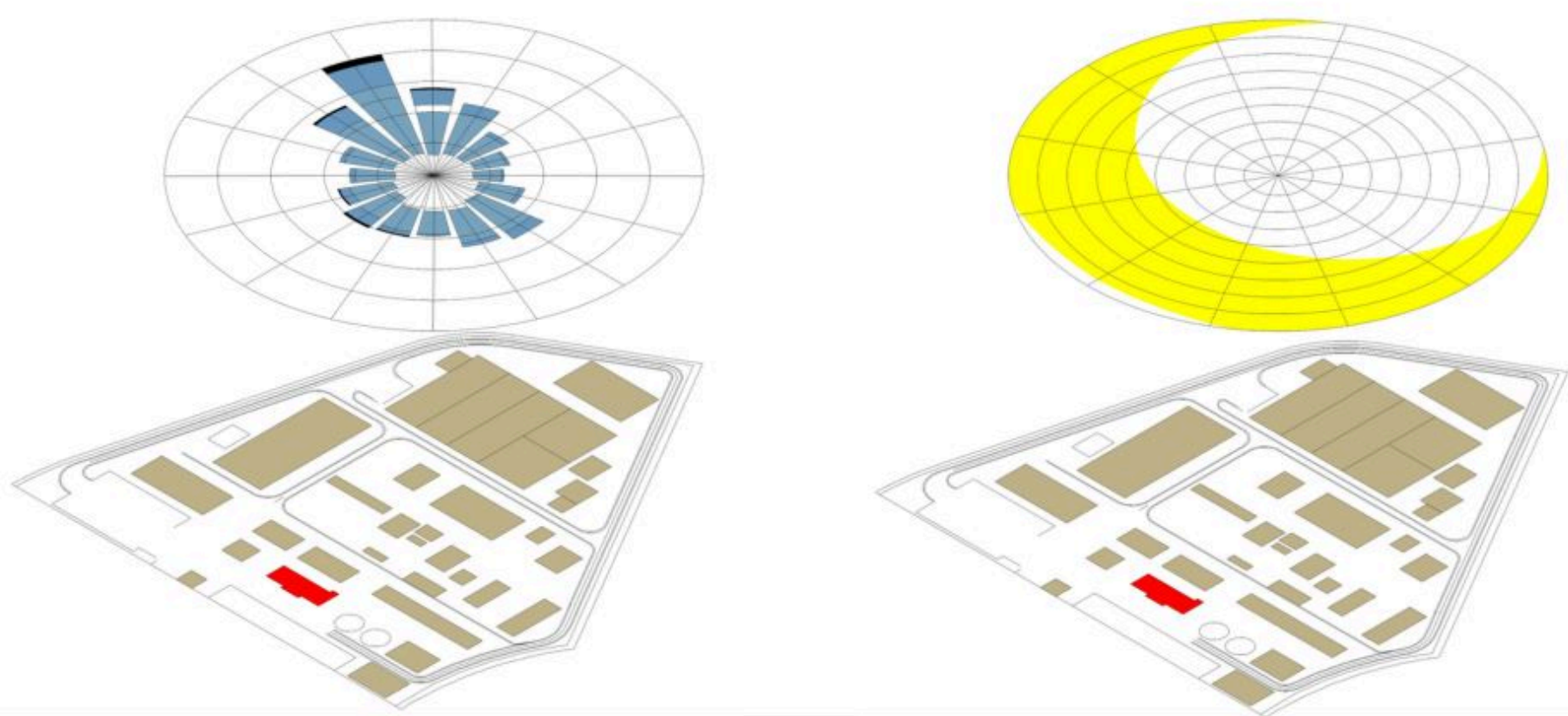


FINAL PHYSICAL MODEL



REGENA HEADQUARTERS

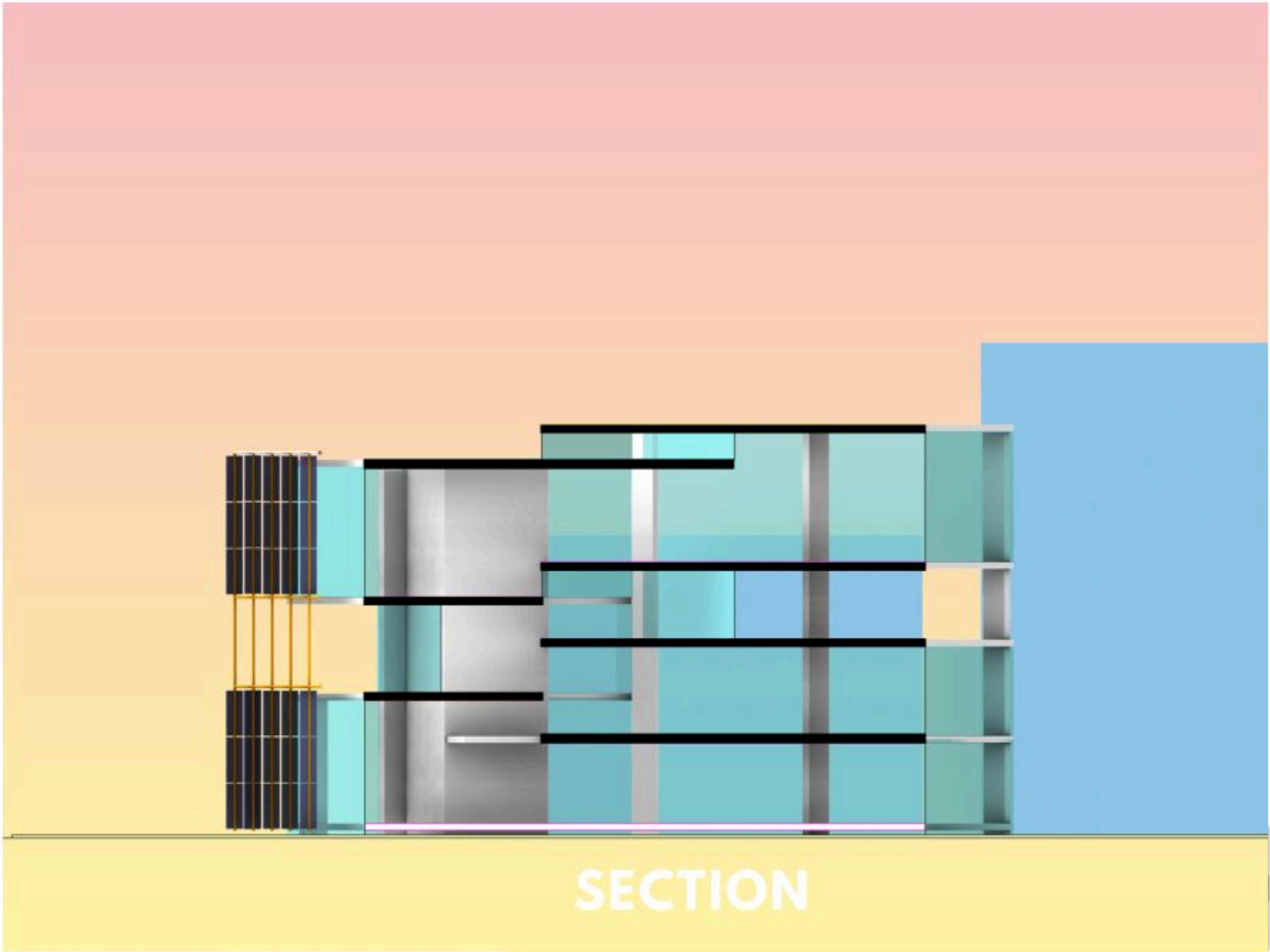
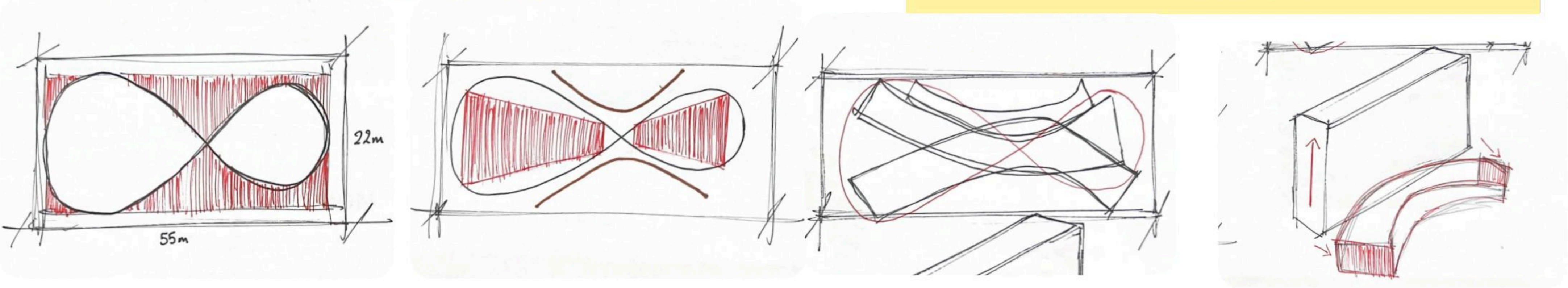
GROUP 6 MEMBERS: ARCH. ENG.- MILAN SWEMMER, CIVIL ENG.- ARAD PARVIN, KEBBIEL RISEN MEMBREBE, JOHN LUIGI PLASUELO, ARCH.- ZAYAN MAQSOOD MECCAWALA, TAMARA MIKLEOVA, AHMED MOHAMED, TANUSHRI UNNIKRI SHNAN



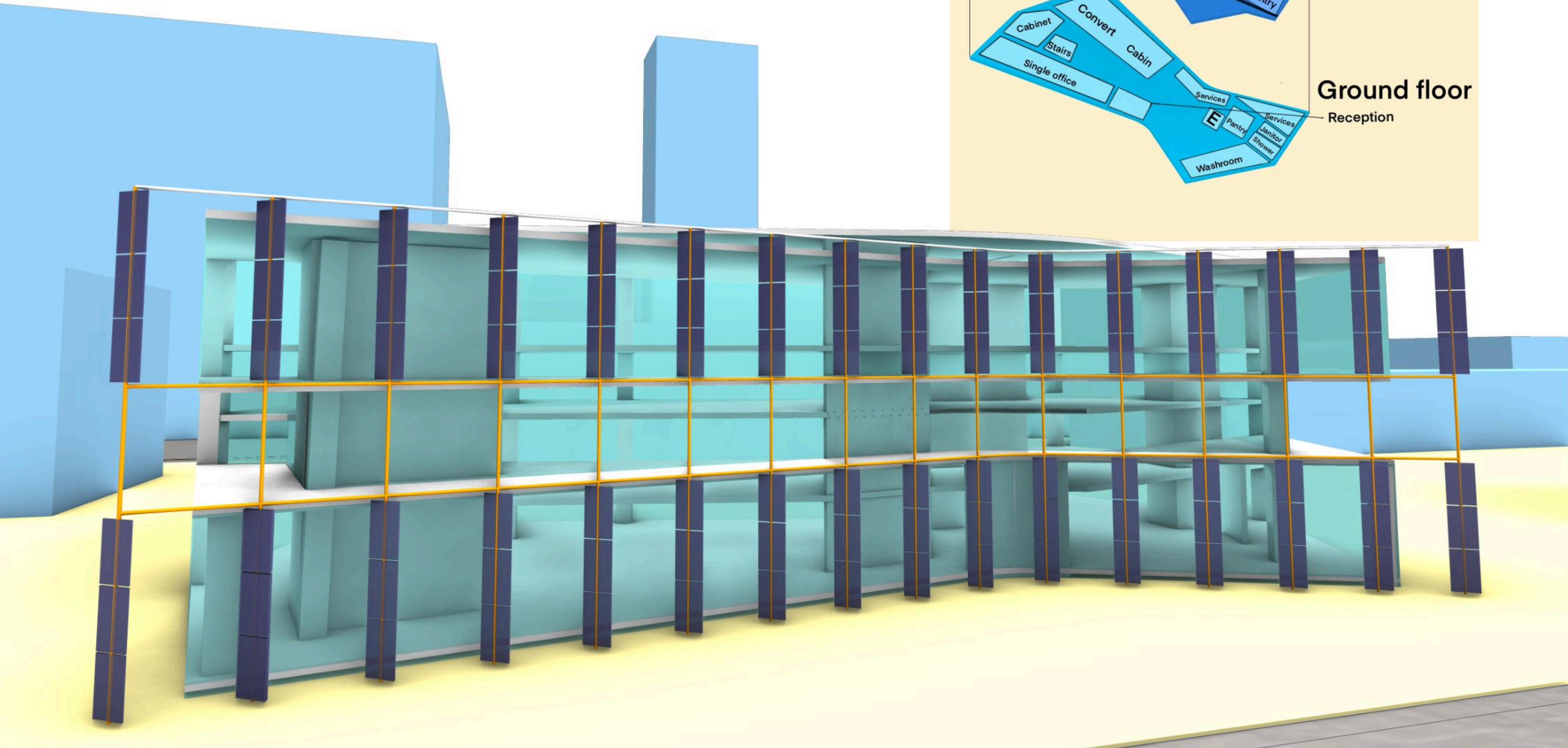
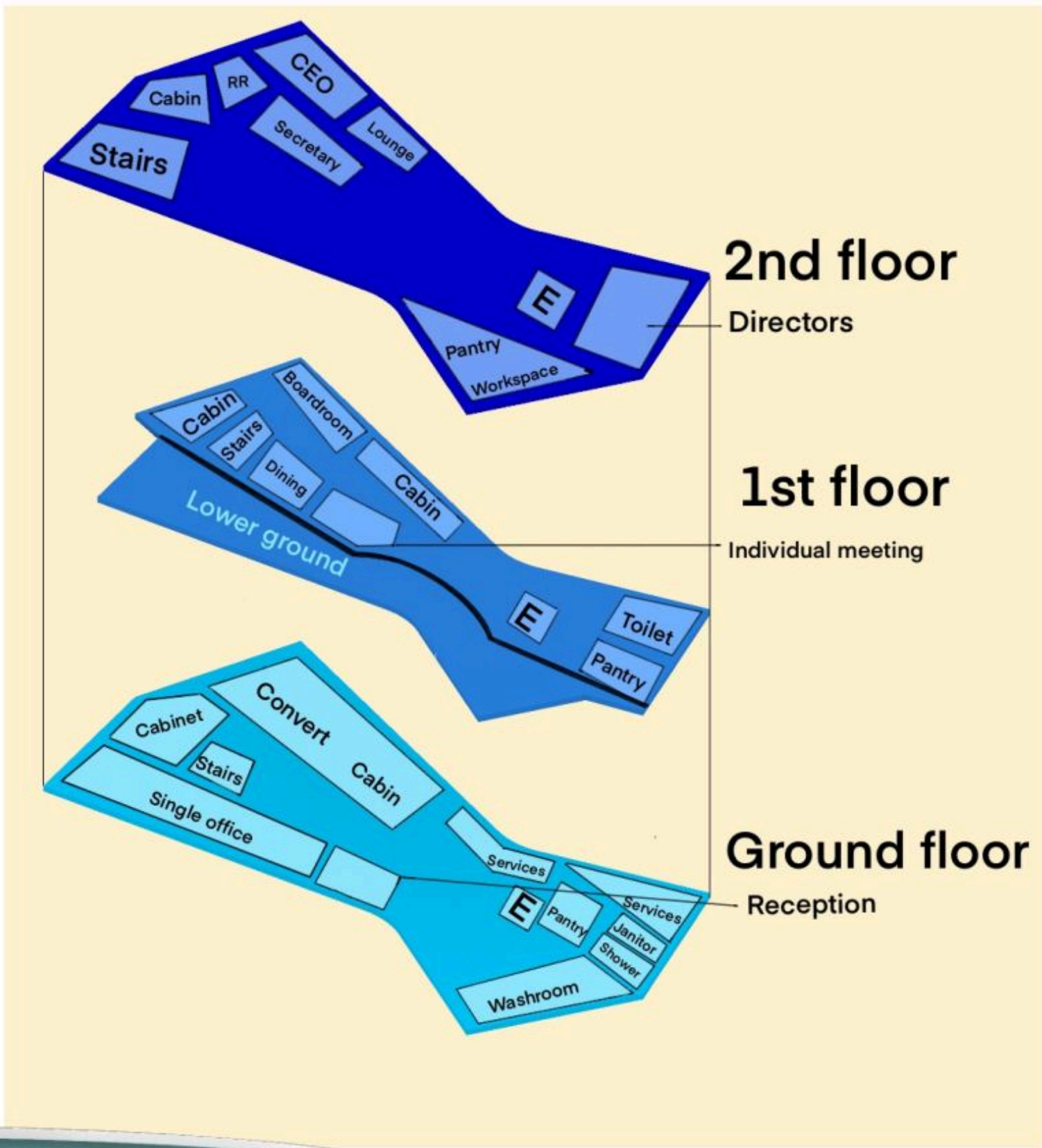
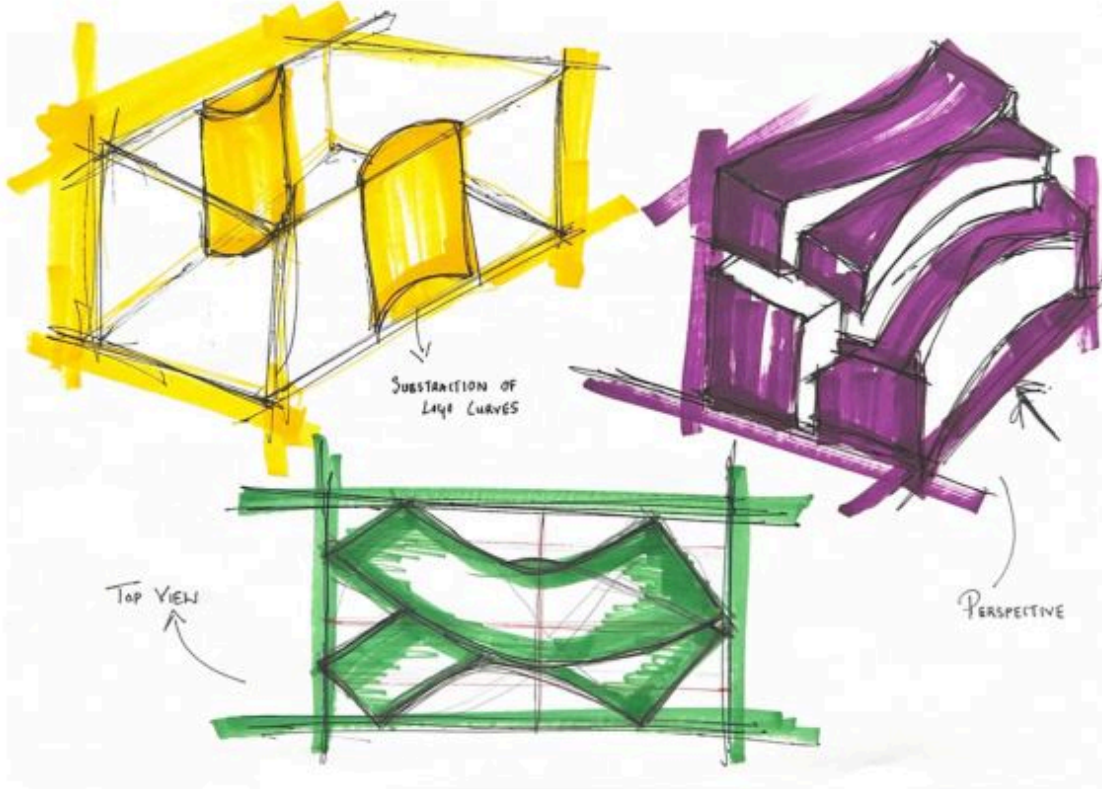
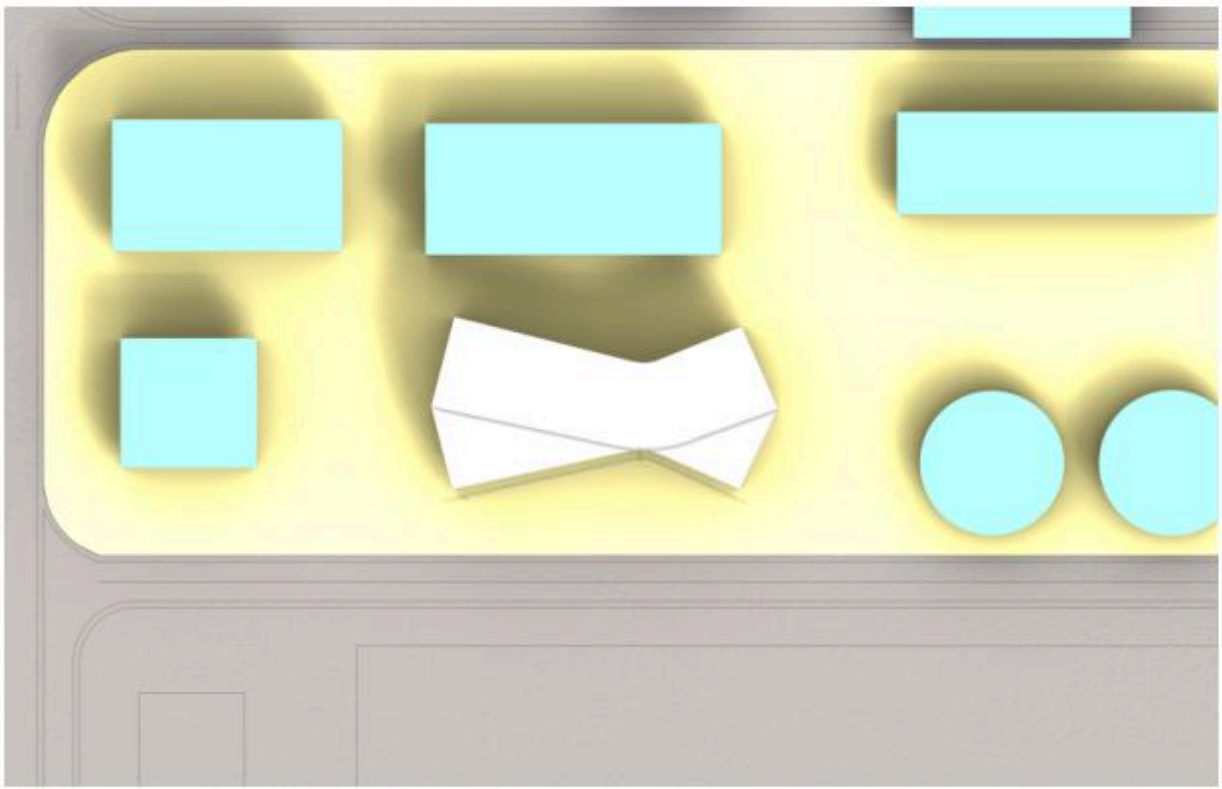
WIND PATH

SUN PATH

SITE ANALYSIS



SECTION

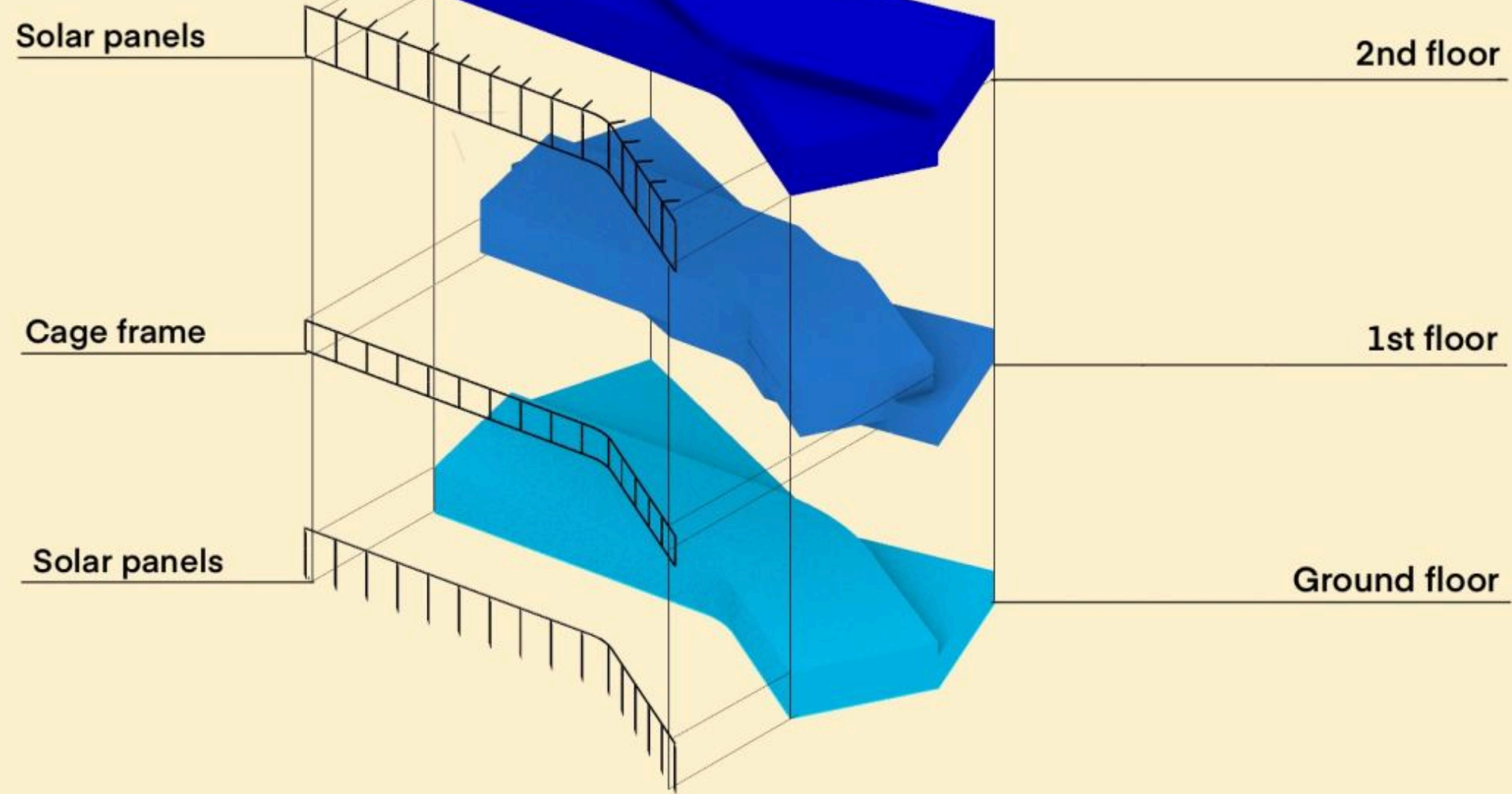




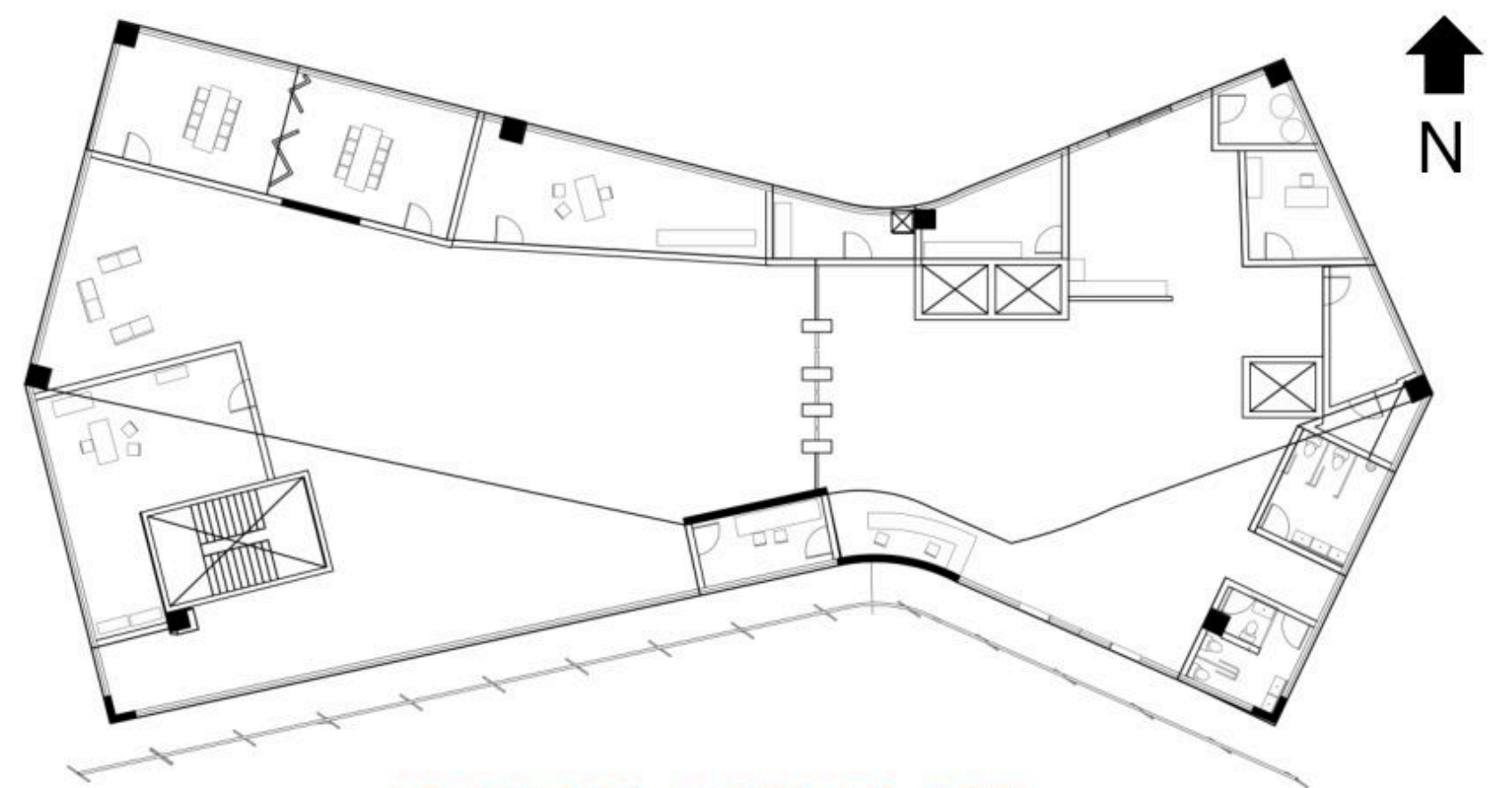
EXPLODED DIAGRAM

Double-faced
facade

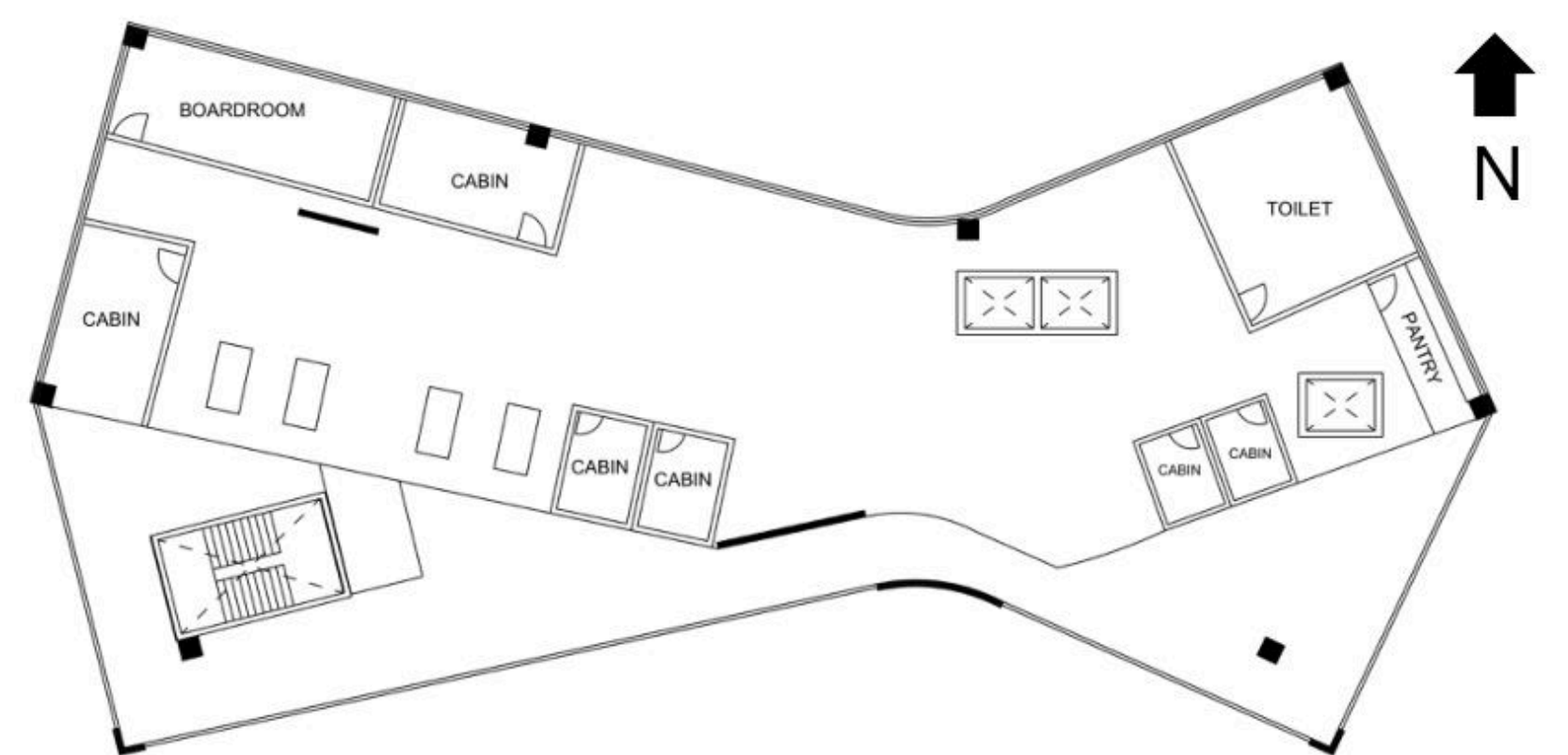
Sun-tracking
Photovoltaic
Kinetic



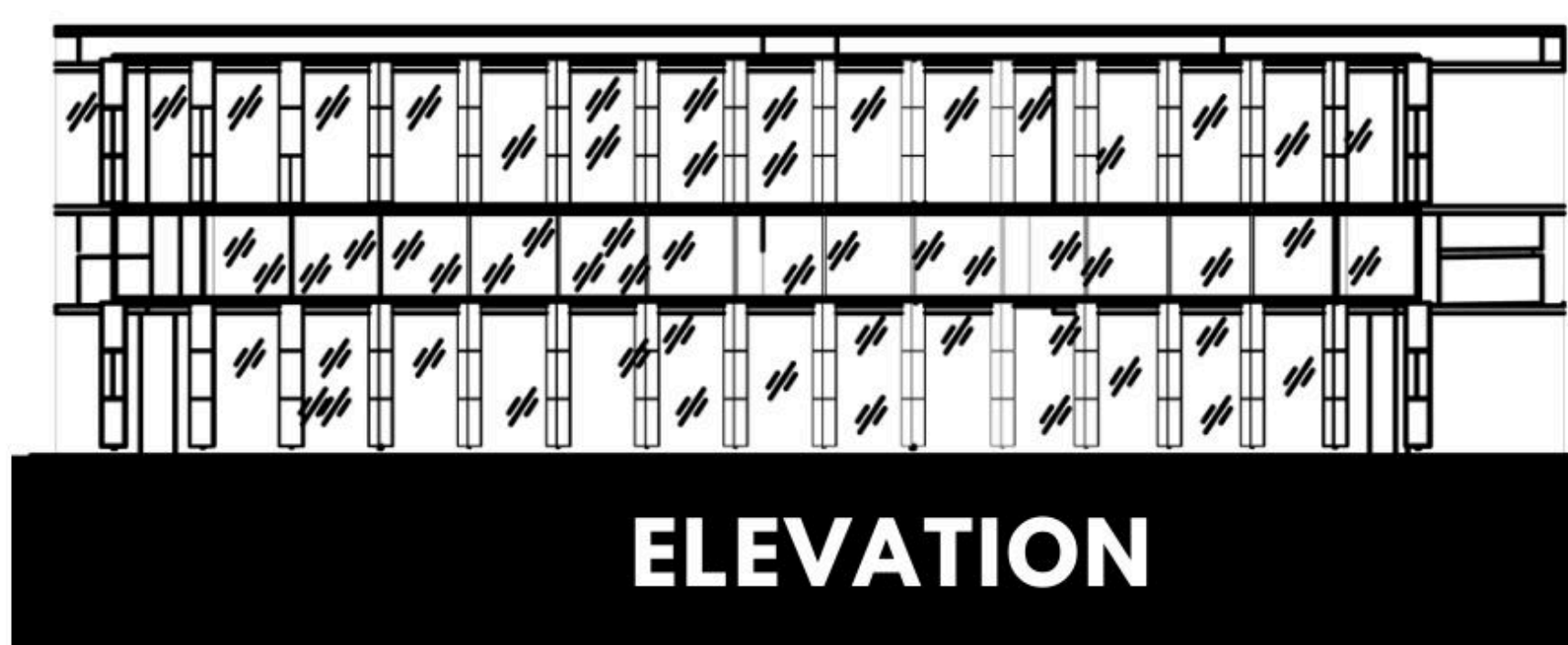
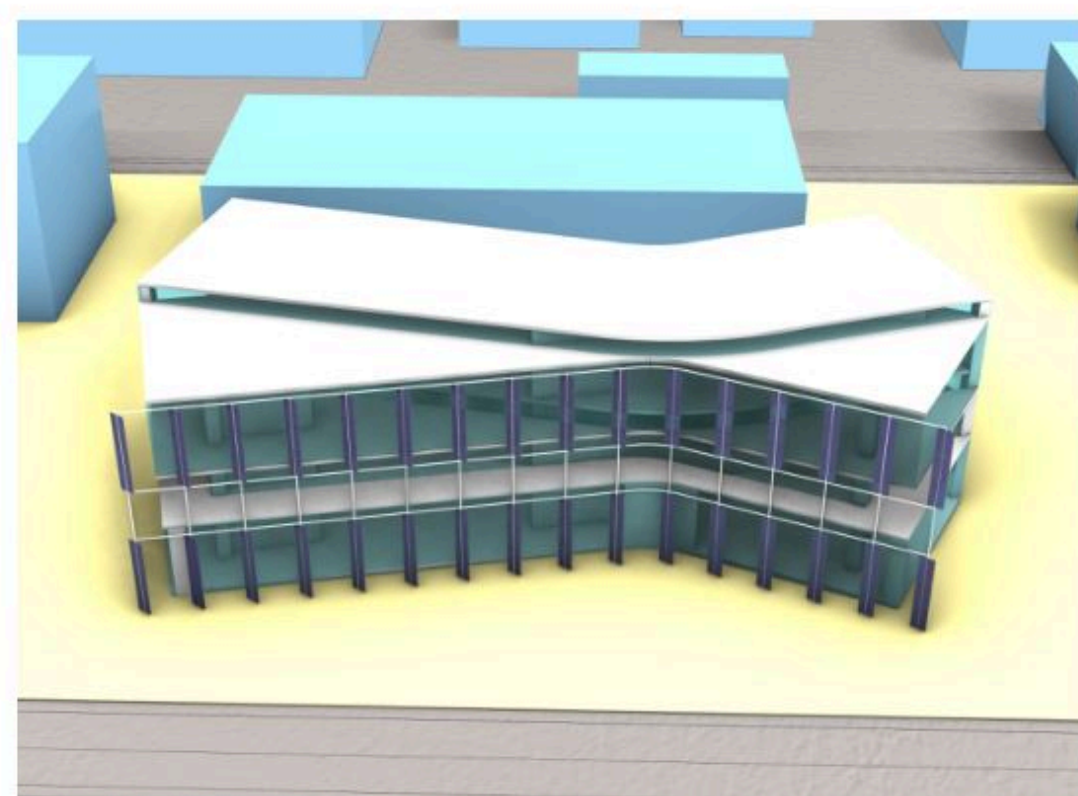
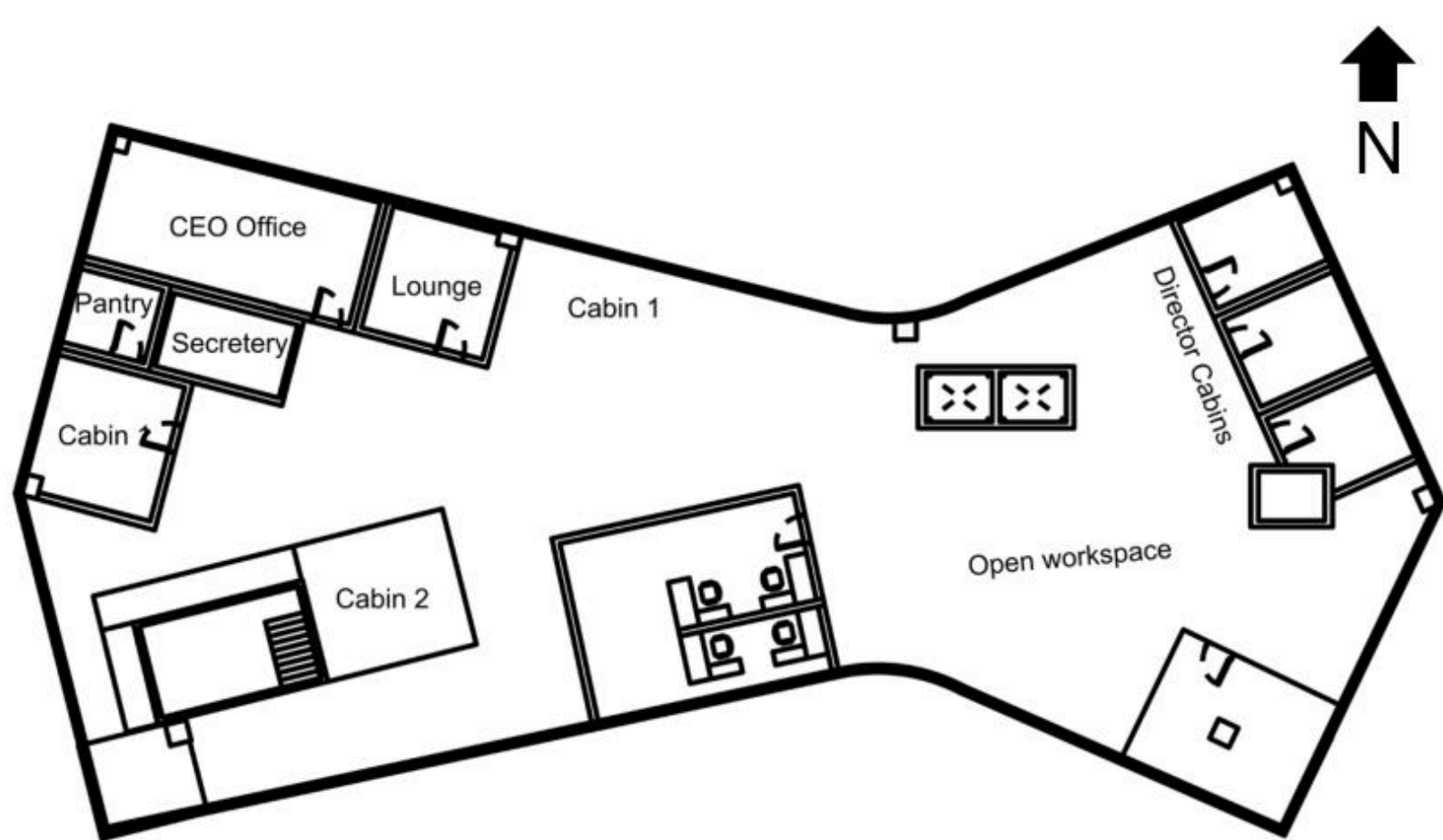
GROUND FLOOR PLAN



FIRST FLOOR PLAN

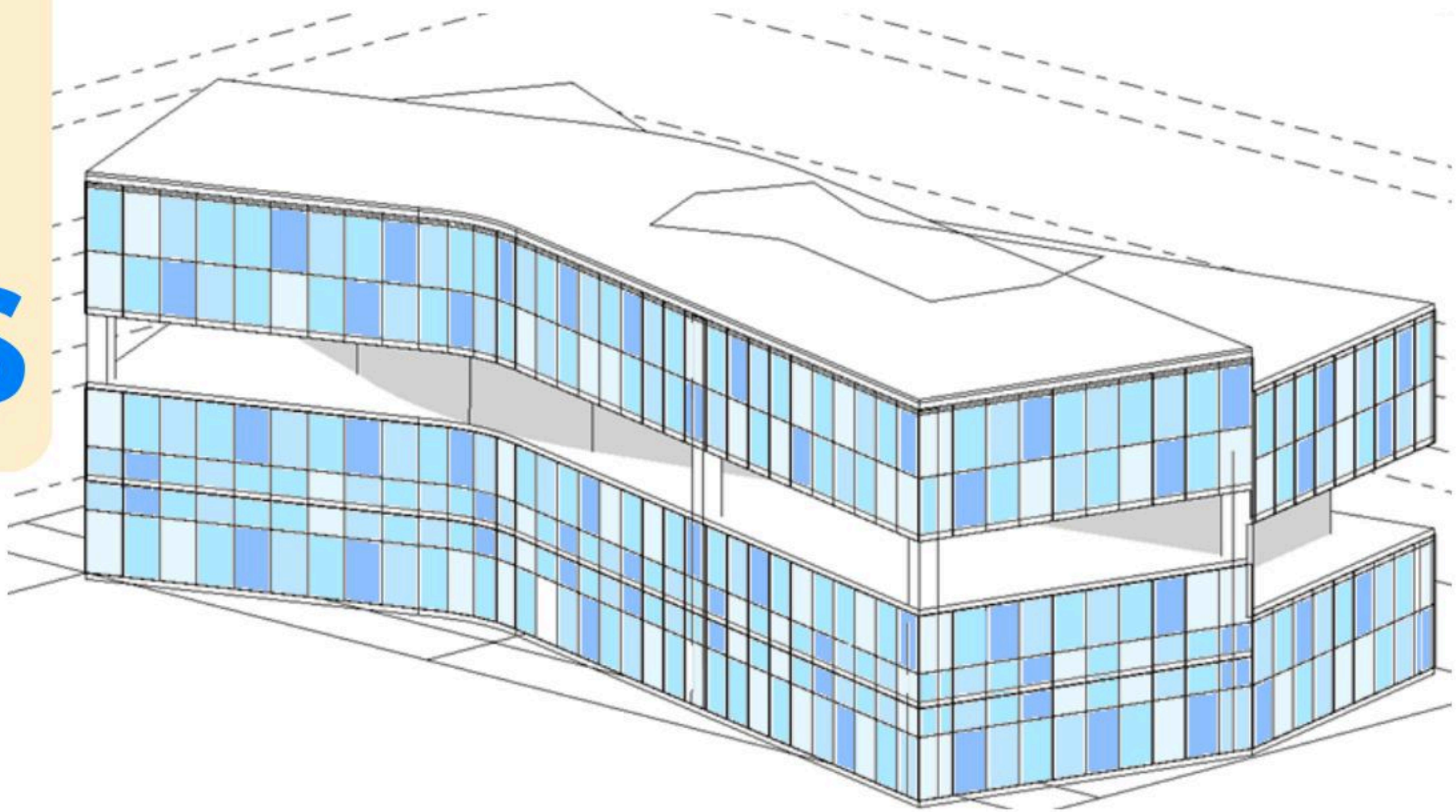


SECOND FLOOR PLAN



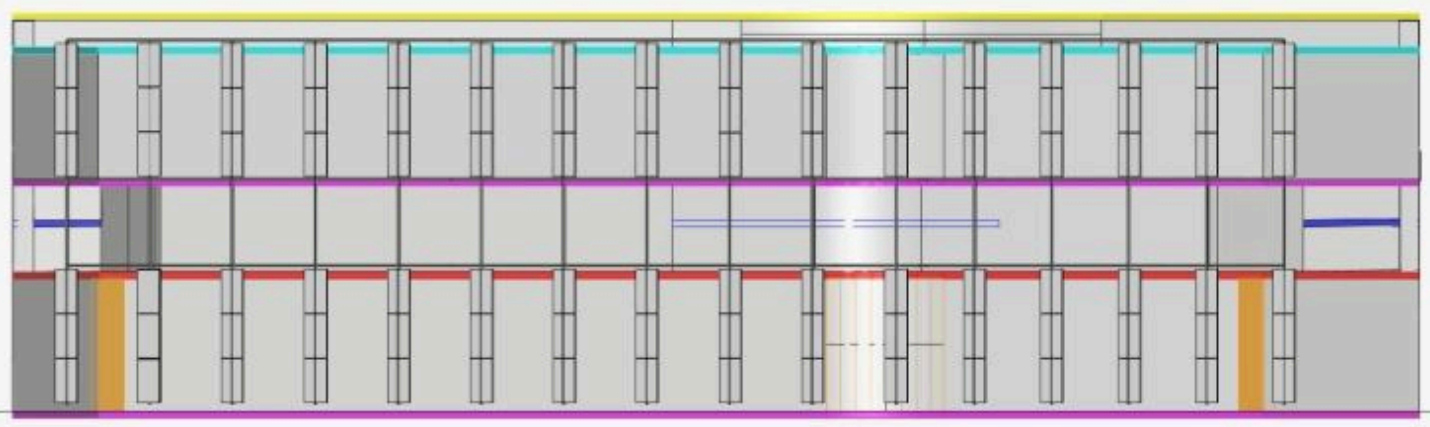
ELEVATION

SUSTAINABILITY + BUILDING SYSTEMS



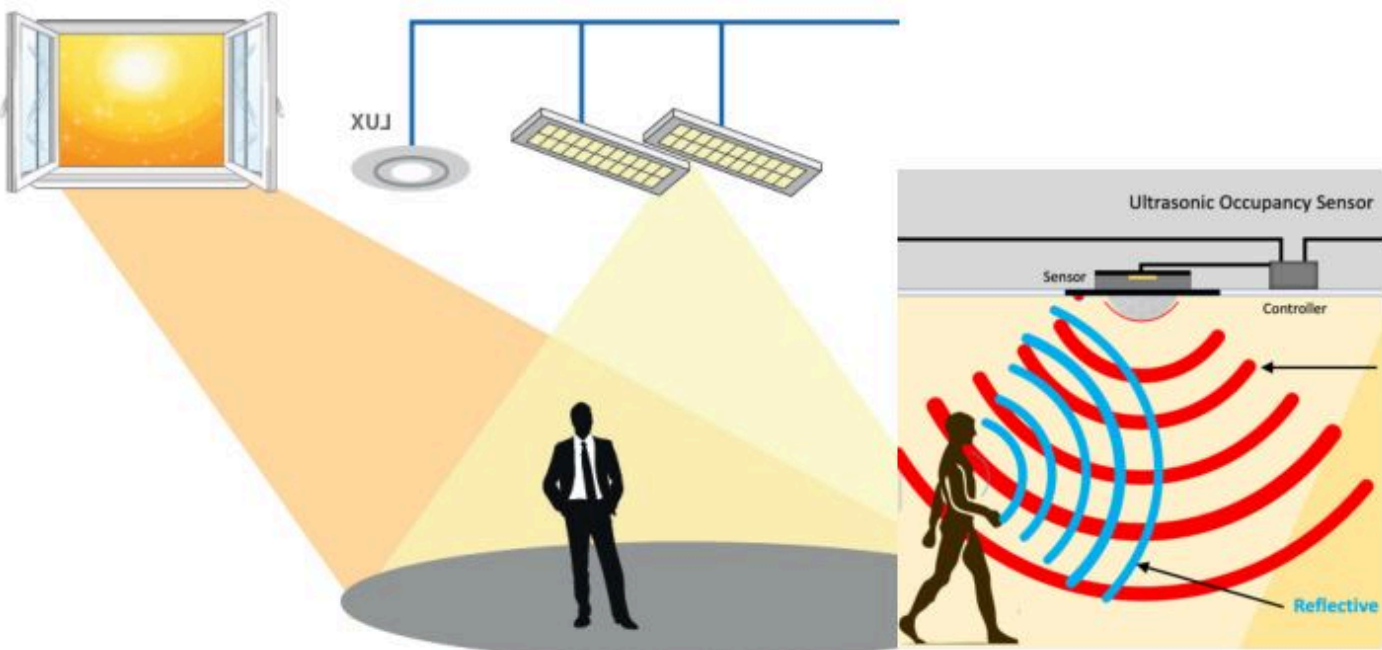
INTEGRATED
DEVELOPMENT
PROCESS

Building tomorrow, together!
Following Estidama.

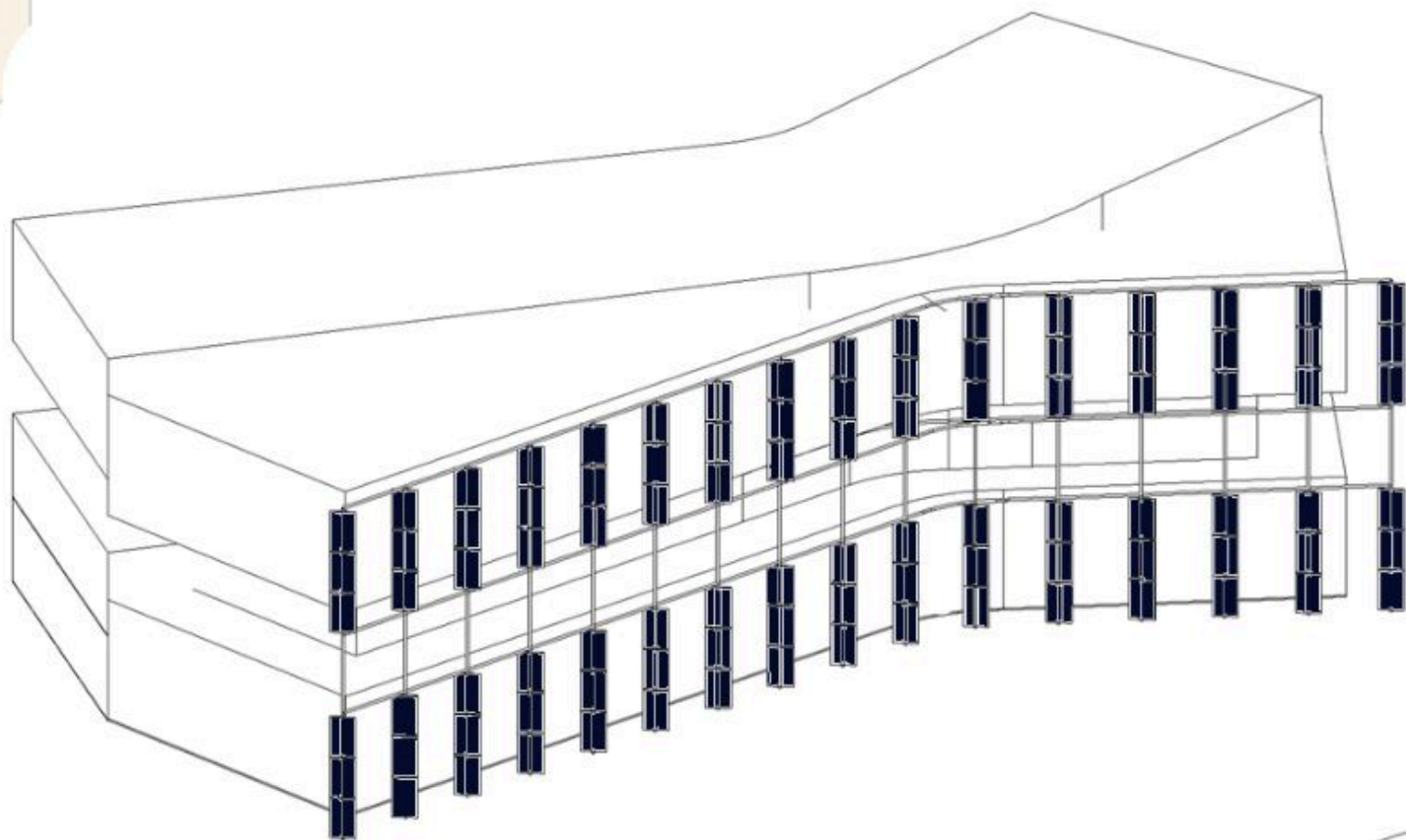


RESOURCEFUL
ENERGY

Daylight harvesting
Solar panels
occupancy sensors
BMS



Onyx Solar



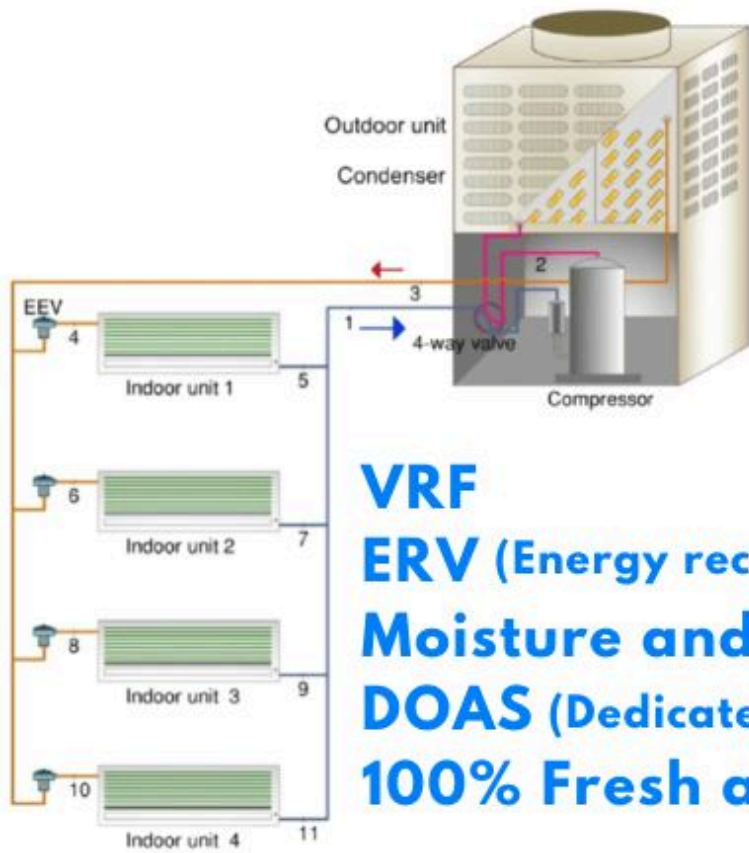
NATURAL
SYSTEMS



Green spaces incorporation



LIVEABLE
BUILDINGS

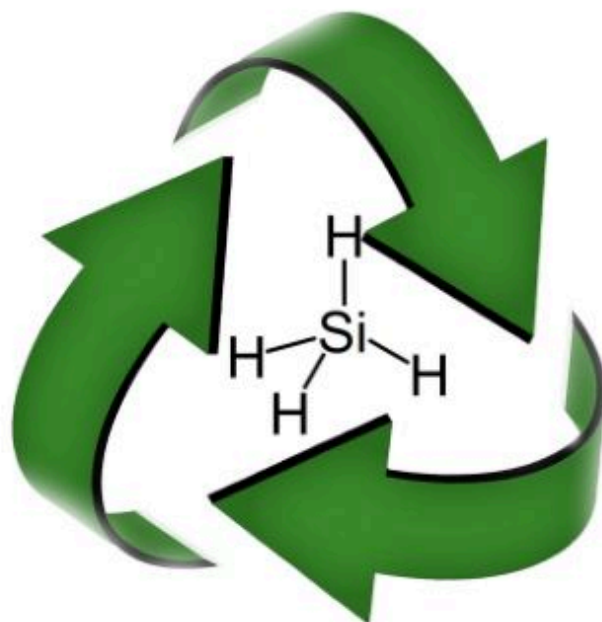


VRF
ERV (Energy recovery ventilator)
Moisture and heat removal.
DOAS (Dedicated outdoor air system)
100% Fresh air supply



STEWARING
MATERIALS

Silane recycling

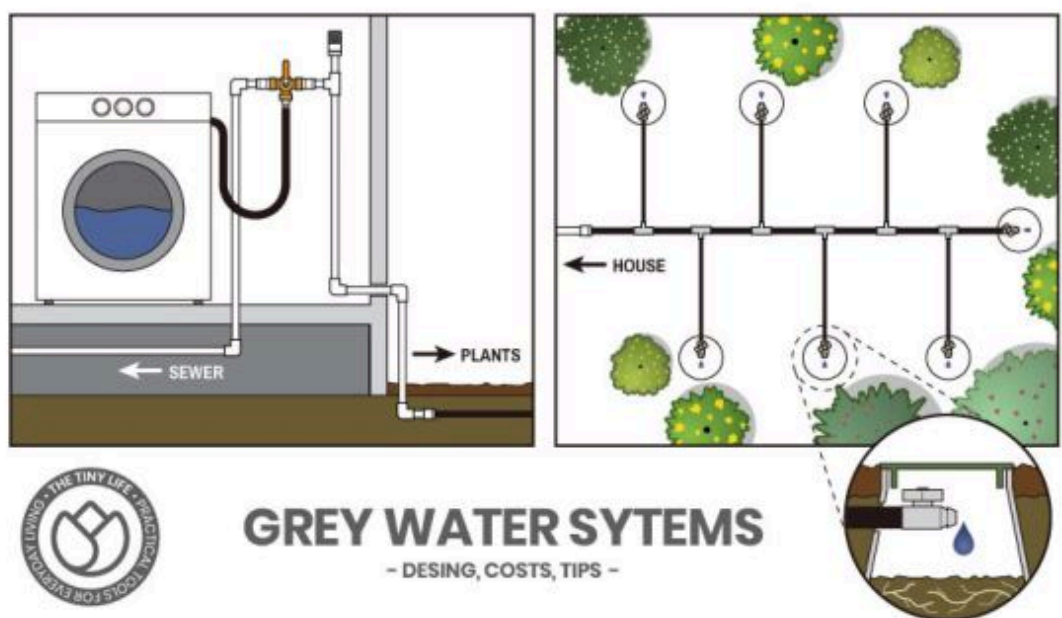


Saving between 4-16k CO2/year for 1kW
modules produced a year



PRECIOUS
WATER

Reduction in water use through
low-flow fixtures



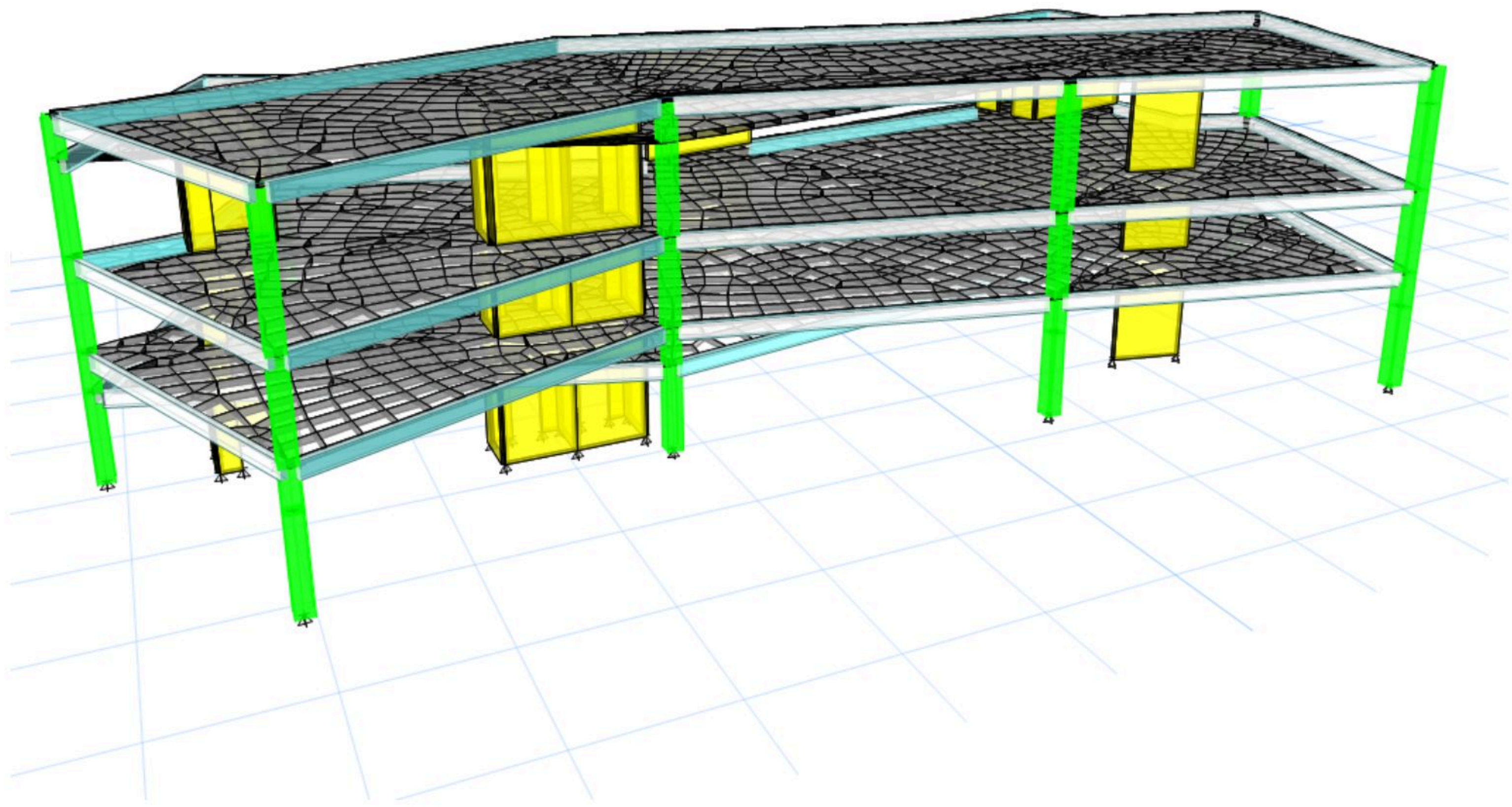
GREY WATER SYTEMS
-DESIGN, COSTS, TIPS-

HVAC system condensate recycling.



INNOVATING
PRACTICE





creativity meets practicality.

Post tension slabs

Shear and load-bearing walls

rebars and rebars and rebars!!!

codes?

every material and design judgement is up to code

